WORKBOOK

BY

DOUGLAS QUIGG

RESEARCH AND THEORY
EVERY CLAIM, WHAT KILLED IT, AND WHAT STILL STANDS

ORIOR
2026

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For Megan, my loving wife, the only person that really understands what I mean when I speak.

For Autumn and Opal, without you this thought would have never occurred.

To everyone else, I wish you well.

ABSTRACT

This is my workbook as it relates to general algorithmic implementation and testing. Hypotheses and refutations are kept.

HOW THIS WAS FOUND

This section records the order in which things happened, because the order bears on how the results should be read.

THE AUTHOR'S POSITION

I have no classical training in mathematics, linguistics, crystallography, cryptography or information theory. I am self-taught across all of them, arrived at each one sideways, and did not come to any of them through a curriculum, a supervisor or a department. I also have difficulty communicating, and a tool checks the prose in these research papers instead of my trusting it.

That is stated first because it explains the shape of what follows. Somebody trained in a field reads its literature before building, and meets a precedent before writing a line. I built first and read afterward. Every precedent named in these research papers was found that way, and finding one late is not the same as finding one early.

WHY THE REFUTED ENTRIES ARE KEPT

Honesty is the stated reason and it is not the working one.

I do not limit myself to conventional thought, or to conventional mathematics, and I do not accept a result because it arrived looking the way a result is supposed to look. That is how several things here were found. It is also how somebody working this way gets a long way down a wrong road without anybody stopping them, because there is no field standing behind me to say the road is wrong.

So the workbook is the thing that stops me. Writing down what a measurement killed, beside the claim it killed, is how I keep scrutinizing what I already think I know. An entry that says what was believed, what was measured, and which of the two lost is a check I can run on myself later, and it is the only check available to somebody with no supervisor and no committee.

An entry here is not a confession and should not be read as one. Refuted entries are working notes. The document would be less useful without them and so would I.

THE TIMELINE IS IN THE REPOSITORY AND IT IS SIGNED

The first commit of this repository is dated 2026-09-04 and is signed with my GPG key. Its subject is human language signature detection research. Seven days separate it from the date on this page.

Most of what is cited in these research papers I had not read before that commit. That is checkable and not asserted: the history records when each result appeared, when each search for a precedent was run, and when the citation was entered. A reader who doubts the order can read it off the commits instead of taking this section's word for it.

The seven days are also why the precedents arrive where they do. There was no interval in which to read a field first. Each result was built, and then somebody went looking for who had already had it.

THE ORDER THE PRECEDENTS WERE FOUND IN

The construction was reached from one general question, about how far an arrangement sits from the most disordered arrangement of its own parts. Neither crystal, cipher nor corpus was in mind. Each domain here is that question with a different answer to what counts as a part.

Arriving from there at something a field already uses is the good outcome. It is evidence the general question points at something real, and arriving somewhere nobody had ever needed to go would have been the weaker result. It gives up priority. I claim none here.

Each precedent was found by asking, once a result existed, how such a thing could have gone unnoticed. That question produced the search, and the search produced the citation.

- **The cross-language ordering result.** Montemurro and Zanette, *Universal Entropy of Word Ordering Across Linguistic Families*, PLoS ONE 6(5):e19875, 2011. Two hundred times the sample here, reaching Old Egyptian and Sumerian, and fourteen years earlier. The conclusion is theirs. This work reached it independently and later.
- **The null permutation.** Fisher, 1935, and standard in statistics ever since.
- **The maximum entropy reference.** Jaynes, 1957, and standard in physics.
- **Filtering with a necessary condition.** Standard in string matching, with Navarro reporting it dominating a large range of parameters. A necessary condition used to discard candidates, with a mandatory verification pass, is the filtering paradigm and it has a literature. No claim of novelty for the construction survives that reading.
- **The injectivity of the Σ functions.** Handschuh and Gilbert, by a polynomial argument, reproduced here for all four functions instead of two.

WHAT I HAVE NOT READ

The crystallographic literature has not been worked through. An autocorrelation of density in a crystal is very unlikely to be new, a century of the field has worked on exactly that object, and the reasonable prior is that the construction has a name, a date and a literature behind it. Nobody here has gone and found out. Until somebody does, nothing in that research paper is a claim to a new method and nothing in it is an attribution either.

WHAT NOW CHECKS THIS

Novelty is a claim about other people's work. Nothing in this repository can check one.

`maint/citations/prior_works.py` reports every novelty claim in the research papers with no citation attached and no line saying I have not read the literature. I decide each one. No pattern separates my priority claim from my description of somebody else's work.

`maint/citations/citations.py` is the other half. A surname in a comment is not a citation. I register a source once I have it open and can enter an author, a year, a title and an identifier. An empty row means I have not looked it up yet.

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THE WORKBOOK

1.1 ORIOR RESEARCH WORKBOOK

Purpose: Hold the settled results apart from the open ones. Work picked up later then re-derives neither. **Scope:** `src/engine/nbody/orior/orior.{h,c}`, the four benches at `bench/bench_{sift,lattice,entropy,ab}.c`, the engine's files, `docs/steering.md`, `docs/findings-for-verification.md`, the delta null research paper at `theory/theory/delta_null/`, and the corpus tools under `maint/data/`

Numbers here come from two places and they are not interchangeable. The anchor, lattice, entropy and A/B rows reproduce from the four benches built through CMake in a clean tree. The corpus rows, meaning everything about languages, ciphers, clustering and the selection rules compared in Python, come from the scripts under `maint/data/` run against texts fetched into `build/corpora`, which is a model of the selection rule and not the library's own path. Nothing here is timed and no row is a performance claim.

1.1.1 *Established Results*

Deductive Results

Soundness. Any subset of a pattern's points is a necessary condition. No selection rule can lose a true occurrence. The proof uses neither order, dimension, alphabet nor interpretation of a symbol. In the engine it holds for legal probes over symbols, or over ranks numbered with the pattern in one population (`src/engine/nbody/orior/orior.h`).

No completeness. For any proper subset the converse is not entailed, in any domain and in any state. The exact compare is irreducible and no knowledge of the domain removes it.

Three corollaries. The selection rule is free. The alphabet need not be known. The index set need not be ordered.

Collision entropy is permutation invariant. It cannot detect an arrangement. H_2 is $-\log_2 \sum p^2$ and depends on the symbol counts alone. A shuffle preserves every count. A corpus and its own shuffle carry identical H_2, exactly and not approximately. No entropy of this order can separate a structured domain from a rearrangement of the same symbols, whatever the domain is. The measured agreement between English prose at 3.764 bits and a memoryless corpus at 3.779 is a weaker statement than this and follows from it.

What is derivable here and what is not falls exactly on the line between the two propositions. Every quantity this work derives without fitting is computed from the histogram: the uninformed anchor rate 2^{-H_2}, the contrapositive indication distance, the cascade depth $\log_2 N/H_2$, the cut per anchor 2^{H_2} and the branch point $(1+q)/(1-q)$ in units of 2^{H_2}. Every quantity it has failed to derive needed a model of the arrangement: the joint survival of a cascade, whether a regularity belongs to a language, and the shape of the advantage curve between the two arrangements.

Those two lists are the same division as the propositions. Histogram quantities are permutation invariant. They describe the maximum entropy case and are free. The arrangement remains after that, and Proposition 2 states that no knowledge of a domain removes the exact compare. The cost of an arrangement is made of checks that have to be performed. Deriving it would be obtaining the irreducible half without paying for it. Proposition 2 denies exactly that. The derivation failing is therefore entailed and not a shortfall in the attempt.

The practical consequence is that measuring the advantage curve is not a fallback. It is the only access to that half, and any curve so obtained is a fact about one domain and never a law.

The shuffle is the maximum entropy background and it cannot be wrong. Drawing uniformly from the arrangements of a fixed multiset is the least committal distribution consistent with the observed histogram. Every other background in this work is a model and can be false. This one is the data with one property deleted.

Exactly one distribution maximizes entropy under a given constraint, and that is where the finality comes from. Entropy is strictly concave, and a constraint fixing counts or marginals is linear and slices out a convex set. The constrained maximum exists and is attained at a single point. The background is therefore reached by solving a condition and never by searching. It carries neither seed, local optimum nor variation between runs. An infinite set of arrangements is reduced to one reference by that condition, and a departure measured against it is a number instead of the best figure found so far. The property has one stated way to fail: a constraint set that is not convex can admit more than one maximizer, and a constraint of any shape other than a count or a marginal has to be checked before this reasoning is used on it.

That reference is necessarily memoryless. The memoryless control returns 1.00 for that reason and is not a baseline anybody chose. Among all joint distributions with given marginals, the product distribution has the maximum entropy. So where the only constraints are single symbol frequencies, the maximizer factorizes: the positions are independent. Memoryless follows from the construction and is not an assumption added to it.

The controls in this document therefore are the reference, written out as data. Ten memoryless arms sit between 0.99 and 1.01 and never leave, and a memoryless corpus returns 1.00 because its distance from the reference is zero by construction. Reading those rows as a baseline that happens to sit near one understates what they are.

The corollary is the other half of a loop this document already writes from the far end. If the reference is independence, a departure from it is a dependence between positions, which is mutual information. The product rule entry measures exactly that and says so: anchors surviving together more often than the product of their rates is mutual information between their positions, at a median of 3.55 on English prose, which is 1.8 bits, and a maximum over 60 needles of 490336, which is 18.9. The deduction here and that measurement are one statement approached from two directions.

The null permutation and the memoryless control are not the same object, and the difference is finite sample. A shuffle draws uniformly from the arrangements of a fixed multiset, which preserves every count exactly and makes the positions exchangeable. A memoryless source draws independently with those marginals, which fixes the counts only in expectation. The two agree in the limit and differ at finite length. This document uses the first as its null and the second as its control without recording anywhere that they differ, and no row here is known to turn on the difference.

The field's name for the validity half is distribution-free, and that name bounds how far it has to be checked. A null permutation is drawn by relabeling the sample it was handed. The reference rests on exchangeability and takes nothing from whatever process produced the sample. The level of the test therefore holds for all source distributions at once, and the statistical term for that property is distribution-free, or nonparametric. The shuffle entry above reaches the same reference from the other side. A background asserting nothing past the observed counts holds against any source distribution.

The question of whether it works here, and then on all corpora of the same kind, has two halves, and this is the half with its answer already in hand. There is nothing to check per corpus, because the quantifier was discharged by proof before any corpus was read, and that proof is Fisher's from 1935, recorded below under the wider version of the same correction. A hundred agreeing corpora carry the weight of one, since the property was in doubt on none of them.

The property covers the source and stops at the partition. The reference is exact once the parts are fixed, and fixing them is a separate act with an entry of its own below, where a domain built to carry structure at one scale reads 0.117 at every width dividing that scale and 0.450 at every width coprime to it. Which of those two numbers a reader is holding is settled by the partition, and being distribution-free does not reach that choice.

The reach has no such proof, and closing it is years of work with no shortcut in it. Proposition 2 already sets the division. Soundness belongs to the construction and cost belongs to the domain's distribution. This document holds the rows where the construction is sound and reads nothing, a memoryless corpus at 1.00 and the digits of the square root of two at 1.00. A claim covering a whole kind of object is therefore empirical, quantified over a set nobody has enumerated, and no run of any length settles it.

The span table is ten measurements. An eleventh raises the count and moves the quantifier no distance, because each row is an instance of the same kind of evidence as the row above it. Getting from ten instances to a statement about a domain takes a controlled series inside that domain with an outside answer attached, which the doped lattice posit below sets out for one domain, where the arithmetic is free and obtaining the point cloud is the expense. Ten domains held to that standard is a program of years, and no arrangement of the existing rows shortens it.

How it reads is worth recording beside the arithmetic. Ten domains under one construction already strains a reader before they reach the caveats, and a row added at the current standard strains it further while supplying no more support than the last one did. The two halves therefore go over separately. The level is distribution-free and proved, and the reach is a list of measurements each carrying its own floor. Handed over together they read as a claim of universal yield, which the 1.00 rows in this document refute.

The maximum entropy domain is the best case for this filter. 2^{-H_2} per anchor is a floor. A filter that selects on symbol rarity reads only the marginals. An arrangement can add correlated

hits and cannot make the marginals more informative than independence already makes them. Structure therefore costs it and never pays it. Measured at matched first order statistics, English prose at 3.764 bits passes 15.9 alignments where a memoryless corpus at 3.779 bits passes 3.9, and C source at 3.627 passes 44.1. The first order quantity is the same to half a percent and the filtering is four times worse. All of the difference is arrangement.

This bounds the filter and not the search. Where an order exists a shift takes it, and Section 4.9 records the rare anchor arrangement losing to Horspool by 1.2 to 25 times for that reason. The structure that costs a marginal filter is the same structure a shift table consumes.

It is also the largest difference obtainable. A background that commits to any structure absorbs that much of the signal into itself. What remains measurable against it is smaller and mixed with the model's error. The maximum entropy background commits to nothing past the histogram. The residual against it is all of the arrangement information, and no other reference exposes more.

That is the same statement Section 7.2 records from the physics, arriving by a different route. Maximum entropy under a constraint is equilibrium, and a shuffle is the equilibrium state of a corpus, the measured departure is a free energy above it, and the maximum extractable is that departure and nothing beyond it. The correspondence is exact and not an analogy, since both quantities are a distance to the maximum entropy state under the same constraint. It does not revive the reference measure reading that Section 4.3.1 refuted, which concerned what an anchor measure is worth and failed because that quantity is not a relative entropy.

Those two together give the shape of every measurement here. The quantity that governs cost is computed from the histogram, the background is the maximum entropy arrangement under that same histogram, and structure is the residual between them. The residual is necessarily invisible to the statistic that defined the background. A second instrument is not a convenience and no refinement of the first one would have served.

Ownership of the three guarantees. Soundness belongs to the construction. Cost belongs to the domain's distribution. Existence of a match at all belongs to the source being stationary and ergodic, and the skip $\mathbb{E}[S] = 1/q$ is Kac's recurrence time.

What a probe set rejects is a coverage function. The placement descent is greedy submodular maximization and inherits its bound. Fix the corpus and the needle. A candidate probe p rejects a definite set of alignments R_p, the alignments whose byte at that displacement disagrees with the needle's. For a probe set P the rejected alignments are $\bigcup_{p \in P} R_p$ and the count $f(P)$ of them is a coverage function. Coverage functions are monotone, since adding a probe un-rejects nothing, and submodular, since an alignment already rejected contributes nothing when rejected again. Neither property assumes anything about the corpus. The descent in `orior_steer_spawn_coarms` scores every candidate by the survivors it would leave and keeps the smallest (`src/engine/nbody/orior/orior.c`), and fewest survivors left is largest marginal gain in f. That is the greedy algorithm on a monotone submodular objective under a cardinality constraint. By Nemhauser, Wolsey and Fisher (1978) it returns at least $1-1/e$ of what the best probe set of the same size rejects. The bound is on alignments rejected, holds only at a `sample_stride` of one (`src/engine/nbody/orior/orior.h`), and guarantees nothing for `orior_steer_plan_recursive`, which chooses among the caller's own offsets (`orior.c`).

Submodularity does NOT supply the destroy theorem's premise, and the claim that it does was written and then refuted. The claim was that diminishing returns gives the conclusion without the enumeration property: a marginal gain never increases as more is placed. If the best candidate at a level adds nothing, no candidate deeper can. Submodularity bounds the gain of a FIXED candidate as the placed set grows. It says nothing about a candidate absent from the earlier candidate set, because there is no earlier gain to bound it by.

The counterexample is three alignments. Let R_p be empty with candidate set $\{p\}$ at level 0: the best marginal gain is zero and the destroy rule fires. Let the candidate set at level 1 be $\{p, q\}$ with $R_q = \{1, 2, 3\}$. Nothing was placed at level 0. The population is unchanged and q has gain three over it. Stopping rejects none of the three and continuing rejects all three, while f is monotone and submodular at every step. The destroy theorem fails and submodularity did not notice.

The non-increasing enumeration premise is therefore load bearing and stays stated. What submodularity gives is the Nemhauser, Wolsey and Fisher bound and the identification of the stop test as greedy's zero marginal gain stop. What it does not give is the equivalence of stopping and continuing. Those are two theorems and only the first comes free. The premise has now moved three times and the condition has been correct since the second move; it should stop being relitigated.

The order within a placed set is free twice over, and the two arguments are independent. Soundness says the conjunction of necessary conditions is order independent. The count does not depend on sequence. Coverage says f is defined on sets. What a probe set rejects does not depend on sequence either. Placement and order are nulls for correctness by the first and for rejection by the second. Neither argument derives the other.

One read per alignment is a floor and it cannot be crossed by anything in this class. Let an engine decide each alignment using only reads taken at that alignment. Suppose it decides alignment a having read nothing at a. Its decision is then a function whose domain is the empty tuple. Its range holds one value and it answers identically whatever the corpus holds at a. An adversary edits the corpus at a and flips whether a is an occurrence; the engine observes nothing different and returns the same answer, which is wrong for one of the two corpora. At least one read therefore happens at every alignment classified, and total reads are at least N. The bound is attained by the empty probe set, which takes no probe reads and sends every alignment to a compare reading at least one byte. `test_steer` asserts both halves on all five fields.

The floor binds this class and says nothing about a skipping search. Boyer-Moore [3], Horspool, Sunday and the factor methods break the premise deliberately: a read at one alignment decides a range of others and most alignments are never visited. Their reads per alignment are taken over a sparse subset of A, which is a different denominator wearing the same name, and no ratio between the two quantities measures anything. This is recorded because the temptation to form that ratio has already been acted on once in this tree.

The entries from here through the one on Turing completeness costing the guarantee nothing are a conjecture about a composition that does not exist, and are kept as written. One descent is a finite automaton with data dependent control flow bounded above by a constant, and nothing in the engine grows while it runs (`docs/steering.md`). A reader the engine may call for more field is described in prose with no signature or contract, and nothing is being built toward it (`docs/findings-for-verification.md`). Whether a caller's loop over descents is Turing

complete is open: the target named is a tag system (Minsky 1961), and the encoding is owed (`docs/steering.md`).

The tape is the universe and the exam set is the surviving set, and the two together decide the classification. Three classifications were written and the first two were wrong, each because of what was taken to be the tape. Taking the corpus to be a fixed finite input gives a finite automaton: the survivor set is a subset of the alignments, at most $2^{|A|}$ states, and nothing grows. Taking the corpus to be the tape gives a linear bounded automaton, since an LBA is exactly a machine whose tape is bounded by a linear function of its input. Both readings assume the corpus IS the input, and it is not. It is a window.

The universe is the tape. It is unbounded and it is not a function of any input, because there is no fixed input to be a function of: the engine chooses what it reads next. The exam set is the surviving set, the part currently held under examination, and it is the read-write store. Its size grows as the engine extends its examination, and nothing bounds that. Where to look next is computed from the census and from what has already survived. The head movement is directed by the machine and not fixed by a caller. Unbounded tape, an unbounded read-write store over it, computed head movement, exact conditional branching on magnitudes, and unbounded iteration through self examination. The LBA objection fails because an LBA's bound is linear in an input this construction does not have.

The field supplies the transition function, and the construction is therefore not a curiosity. The engine holds a head and a store and imposes no rule of its own. The corpus at that position decides the survivors of a probe. The transition is read off the field and not computed against a model of it. That is the same statement the census entry makes, that the steering is the field's own and not a parameter, and the same one the maximum entropy background makes, that the reference is the data with one property deleted. Taken together with the tape being the universe it says where the program lives: the engine is a universal reader and it runs the field. Neither threshold, tuning constant nor fitted parameter enters anywhere, the property that would be lost the moment a rule was imposed instead of read.

The gap is in the interface and not in the theory. The entries take `const uint8_t *corpus` with a `corpus_len`, which is a window nailed down. The engine cannot ask for more universe. What is implemented examines a fixed region and the construction above cannot be reached from it. The change is one interface: a reader the engine may call for more, in place of a pointer and a length. Until that exists this remains the construction the implementation admits and not the implementation. That is the distinction every entry here is required to keep.

One invocation is total. The system is not, and an entry claiming otherwise was written and then withdrawn. Every loop inside one call is bounded by a quantity fixed before that call: the sweep to N, the descent to `ORIOR_STEER_ANCHORS`, the shape enumeration to `needle_len` and a caller's ceiling. That much is true and it settles nothing, because a system halting on every input decides its own halting. The question is whether the OUTER loop is bounded, and the outer loop is self examination, which nothing bounds. The withdrawn entry analyzed one invocation and stated a conclusion about the system.

Conjecture: the construction is a register machine over survivor magnitudes. The state is the truthy and falsy survivor vector over the alignments, whose length is the alignment count and is therefore unbounded with the corpus. Decisions are comparisons between magnitudes of that vector, and the descent makes them at every level and at its stopping test. Inside a descent

spawning masks the vector down and the destroy rule ends the descent and restores nothing: survivors are set standing once, at entry, and only cleared (`src/engine/nbody/orior/orior.c`). A descent therefore terminates, and $S(P)$ is non-increasing inside it. Survivors return only when a new descent starts, on the same field or another. Kept as a conjecture about a construction whose destroy step restores survivors: $S(P)$ is a non-increasing chain in a finite lattice and stabilizes while probes accumulate, that fails once a destroy restores survivors, and spawn alone terminates while spawn with destroy need not. The engine as built has no such destroy step, and whether a caller's loop over descents terminates is open. The magnitudes must be exact for a comparison to be a decision and not a decision about a rounding. The integer carry is there for that.

Turing completeness costs the guarantee nothing, the point of drawing the boundary where it is drawn. Soundness quantifies over the probe set placed at this instant and never mentions how it was reached or whether the process reaching it halts. It therefore holds along every trajectory, non-monotone and non-terminating included, and every intermediate state is an exact answer awaiting a compare. The anytime property is not a convenience attached to a terminating computation; it makes a non-terminating one useful.

The engine's implementation is bounded and the difference is not glossed. `ORIOR_EXACT_LIMBS` is 128 by default, which is 4096 bits, and a build selects any power of two from one limb up; each build's width is still fixed (`src/engine/base/no_rounding/exact_integer.h`). The multiply climbs a ladder, long multiplication, then Karatsuba from 32 limbs, then the Schönhage-Strassen transform from 8,192 limbs, and division is Knuth's long division, then Newton's reciprocal from 8,192 limbs (`exact_integer.h`); `test/engine/exact_transform_test` runs 4 checks with 0 failed at each width (`workbooks/engine/ledger.md`). A fixed width counter is a finite state machine at any width. The implementation admits the construction above and does not yet carry it out.

Empirical Results: Supported

result	figure	evidence
invariant over byte strings	9,396,207 occurrences, 0 refused, 582 rows	3 measures, 1 to 6 anchors, 5 corpora, lengths 1 to 2048
invariant off the line	213,840 occurrences, 0 refused, 252 rows	13 geometries, dimensions 1 to 8, rotated point set, complex alphabet
rotation is invisible	scatter and turned agree at every anchor count	same 8 points a quarter turn apart
uninformed anchor rate	2^{-H_2}, within 4%	4 corpora, H_2 from 0 to 7.83
contrapositive indication distance	$m(1-2^{-H_2})$, within 0.7%	84 rows carrying a ratio, 3 measures
what a measure can be worth	$\sum p^2/\mathrm{E}[\min_m]$, exact at a point mass	ceiling 1.000 measured 1.00
the ceiling condition	equals 1 exactly when p is uniform on its support	entropy does not order the corpora, this does
bias correction size	$(1-C)/((N-1)C\ln 2)$ bits	9 sources, four orders of magnitude
the correction at a point mass	0.0000 bits, orders 2 to 5, lengths 256 to 16384	16 rows, exactly zero
product rule failure	up to 4 orders of magnitude, always under-provisioning	6 anchors, structured C
the estimator's identity	probe with self hit is the plug in, without it is the U statistic	bias tracks to 0.02 bits
the estimator's cost	1.4 to 21.7 times the unbiased estimator, falling as $1/\sqrt{k}$	budget sweep 16 to 4096
product objective	11% over Horspool on C source, exactly 1.0000 on uniform	reads the histogram, and that makes it a ceiling
distance field with triangulated credit	13.2% over Horspool, beats the histogram oracle on every structured row	no alphabet model, $O(m)$ state
probes chosen from the answers so far	within 2% to 20% of the irreducible floor, exact on all 35 rows	no preprocessing, no carried state
the inverse multiply check	residual exactly 0 on all 35 rows	survivors reconstruct every observation
searching with no pattern	recovered a multiple of the record period from 512 reads	92 shifts against 0 on a shuffle of the same bytes
finding the boundary marker	space in English at a 5.58 gap, newline in records at exactly 16.00, semicolon in C	gap dispersion against a shuffle, no pattern given
boundaries locating on their own	English unique from $m = 64$, a 2726 fold cut using its most common symbol	perfectly periodic boundaries cap at the period, exactly 16
the same regularities across languages	space found in 5 of 6 at a gap of 4.70 to 5.40, Zipf slope -0.72 to -1.05, brevity negative in all	Spanish 1605 to French 1862, three families
description packed per unit	Finnish carries 11854 types and 11.9 bits per word against English at 5691 and 9.8, at the same budget	bits per byte spreads 23% where bits per word spreads 33%
subject words from connectives	gap dispersion above one returns begat, Sancho, aventura; the, and, of never appear	no stopword list, no grammar, no dictionary
the regularities are not alphabetic	the same text in Morse, four marks and no letters, Zipf -1.185 and brevity -0.652	both present with no letters in the corpus

Zipf is a property of the message	-1.185 against -1.183 under a permuted code table	relabeling cannot change how often a word recurs
brevity is mostly not a property of the code	-0.652 against -0.550 under a permuted table	at most 0.10 of it is Morse's assignment, the rest is structural
what a designed code table buys	22.4% fewer bytes than a permutation of its own codes	the design shows in length, not in the correlation
what a greedy in-order shift collects	$1/((1+m2^{-H_2})(1-2^{-H_2}))$, 47% to 91% at m=16, 1% to 5% at m=256	derived from rows already taken, no fitting
where the best anchor sits	advance saturates at 2^{H_2}, reaching 95% near $a = 3 \cdot 2^{H_2}$	predicts the chosen offset on four corpora
a controlled series with an outside answer on every row	383 of 383 readable games return their exact Grundy period	385 subtraction games, periods 2 to 45, worst margin 16.0 times the tallest shuffled margin
one loss per line in Nim	{1} and not a range, on every axis line	64, 4096 and 32768 lines, 2 to 4 heaps, Bouton 1901
Wythoff's positions off a played grid	equals the 1907 closed form exactly, a_n/n to 1.0×10^{-3} of φ	mex recursion holding no formula, sides 200, 400, 800
a board returns its own width	8 across the rows and down the columns, both arms	margin 7 to 15 times a scatter of the same pieces
a game's rules leave a departure and its players do not	1.225 against the null, 25 batch standard deviations; correlation with win share 0.067	9 pairings, 3 arms ordered by played win rate

Collision entropy is close to a constant for a language. The quantities derived from it are too. Ten English texts from 1609 to 1861, covering scripture, drama, epic verse, political argument, folk narrative, a novel, scientific writing and a cookbook, give H_2 between 3.693 and 3.891 with a mean of 3.800, a spread of 2.6%. The subject, the century and the author do not move it.

Two of twelve sit at 3.417 and 3.420 and both are layout. Their commonest symbol holds 0.241 and 0.254 of the corpus against 0.168 to 0.195 for the others, and line endings are folded to spaces before measuring, and a short lined file gains spaces and concentrates its distribution. One of the two is set one verse to a line. Neither is a property of the writing.

Every quantity in this work is denominated in 2^{H_2}: the cut per anchor is 2^{H_2}, the cascade depth is $\log_2 N/H_2$, the branch point is $2^{H_2}(1+q)/(1-q)$ and the uninformed rate is 2^{-H_2}. A constant H_2 therefore makes all of them constants for that language, where they had been measurements of a corpus. For English at 3.800 bits: a cut of 13.93 per anchor, a branch at a needle length near 15.0, and $\lceil \log_2 N/3.80 \rceil$ anchors. Whether to ship such a table instead of a histogram pass is a design decision and is not one recorded here.

It does not survive a change of source. It is not a property of the language. All ten texts above came from Project Gutenberg. Measured on Common Corpus instead, French reads 3.948 against 3.818 for Gutenberg French and Spanish reads 3.933 against 3.694. One language measured from two sources moves by 0.13 and 0.24 bits. The whole spread the constant was built on was 0.198 bits across ten texts. The between source gap is as large as the within source spread.

The cause is the character inventory. Gutenberg texts here carry 73 to 110 distinct symbols and the Common Corpus texts carry 143 to 229, because the second preserves accents, punctuation and Unicode that the first flattens. A larger inventory admits a higher H_2. So the figure is a property of a language as one publisher encodes it, and a table of per language constants would be calibrated to that publisher's conventions. Whether a constant exists per language and encoding pair is not established here, and Greek measures 3.688 and C source 3.627 under the Gutenberg treatment.

Corruption is detectable from the histogram, and windowing the histogram locates it. Three detectors over an English corpus with a known share of its symbols replaced by uniform draws (`examples/any_corpus/4_measure/noise_detection.py`). Collision entropy over the whole corpus reads identically for corruption scattered across it and gathered into one block, 3.790 against 3.790 at one percent and 3.889 against 3.889 at five, which is permutation invariance measured. Collision entropy over the whole corpus is also blunt: the natural band for English is 3.693 to 3.891 bits, and a global figure does not leave that band until somewhere between five and ten percent.

The same quantity computed window by window regains the position it cannot hold globally, because a corruption that is a small share of a corpus is a large share of the window containing

it. At half a percent gathered in one block the global figure is 3.777 and says nothing, while the worst of the 4096 symbol windows reads 6.369 against a median of 3.798, eight standard deviations out, and names the window. That figure is $\log_2 82$. The block sits at maximum entropy over the alphabet. Uniform draws have to give that. A clean corpus returns its worst window 0.23 deviations above the median and is correctly reported as carrying nothing.

The same test runs at every window and does not stop where the estimator breaks. Halving the window from 4096 to 32 on a block at half a percent gives separations of 8.0, 7.6, 7.5, 7.3, 7.0, 6.5, 5.7 and 4.3 standard deviations. The alphabet holds 90 symbols. The last three windows cannot estimate it at all, and the block is still found at 4.3 deviations from a window of 32. Undersampling lowers the clean windows and the corrupt one together, the median falling from 3.798 to 3.415 while the worst falls from 6.369 to 4.913 and the deviation stays near 0.35. Differencing two equally poor estimates keeps the contrast. A window of 8192 gives only 4.3 because the block is smaller than the window and is diluted in it, and that makes the maximum at 4096 a matched length and not part of the decay.

An unnamed quantity in the halving ladder. Taken at full precision, and starting at 4096 because 8192 dilutes the block and is not part of the decay:

window	separation	loss	loss ratio
4096	8.03210		
2048	7.60885	0.42325	
1024	7.49435	0.11450	0.2705
512	7.30620	0.18815	1.6432
256	6.97637	0.32983	1.7530
128	6.54592	0.43045	1.3051
64	5.68814	0.85778	1.9928
32	4.25064	1.43750	1.6758
16	2.48424	1.76640	1.2288

The loss per halving grows and the ratio of successive losses is stable over the last six rows of this corpus at 1.600 with a standard error of 0.107, an interval that contains φ, $\pi/2$, $\sqrt{e}$ and $3/2$ and identifies none of them.

There is no quantity there. Four corpora gave 1.049, 1.388, 1.600 and 2.352, which ordered with the alphabet weight and was recorded here as a finding. Run over 29 corpora it scatters: at H_2 between 3.69 and 3.83 the values are 0.91, 1.05, 1.07, 1.10, 1.16, 1.16, 1.18, 1.20, 1.30, 1.33, 1.41, 1.42, 1.64, 1.99 and 2.04, with scientific writing at 2.04 and a memoryless control at 0.91 sitting at almost the same H_2.

The ratio is not stable on a single corpus either. The novel gives 1.600 under one window set and 1.410 under the same procedure with one window removed. A line fitted through 42 corpora gives an unweighted slope of -0.118 and a weighted slope of +0.045, which disagree in sign, and absorbs 1.1% and 3.5% of the variance.

Tested without assuming a distribution, the association with the alphabet weight is absent and a difference between families is present. Spearman's rank correlation between H_2 and the loss ratio over 42 corpora gives $\rho = +0.049$ at $p = 0.76$. A Wilcoxon rank sum between the 19 natural corpora and the 10 memoryless ones gives $p = 0.0009$, with medians of 1.195 and 0.942. The quantity separates structured sources from unstructured ones and does not vary with the alphabet weight. The four point ordering suggested the opposite.

The standard errors quoted alongside those figures, here and elsewhere in this section, are not valid. The loss ratio over 42 corpora has a skew of -4.20 and an excess kurtosis of +19.67, giving a Jarque-Bera statistic of 800.1 against a one percent point of 9.21. Normality is rejected by a factor of 87, and every t and every σ computed from these values assumes something the data denies.

The r^2 figures do not depend on that assumption and the absence of a trend stands. The claim that one corpus gave 1.600 with a standard error of 0.107, and the reading that the interval therefore contained several named constants, was arithmetic on a distribution that does not support it.

The ladder also runs one step further than the alphabet allows: a window of 16 symbols over 90 distinct ones still separates the block at 2.48 deviations.

Dispersion against a null permutation fires lower, at one part in a thousand, and it is not position blind: scattered corruption moves it toward 1.00 by destroying clustering and a contiguous block moves it toward 0 because the block is itself a cluster. Its sign separates the two shapes. Reading the arrangement puts it on the side of the line that has to be measured. The windowed histogram reaches within a factor of five of it and stays on the side that does not.

How many anchors to place is the log of the domain in the effective alphabet. Each anchor cuts the survivors by 2^{-H_2}, which leaves $N2^{-kH_2}$ after k of them, and the excess over the one true occurrence reaches zero at $k = \log_2 N/H_2$. On English at $N = 728751$ that predicts 4.9. Measured with jittered placement the survivors beyond the true occurrence run 63240.9, 3290.3, 351.0, 17.2 for one to four anchors, a cut of 9 to 20 times each, and reach 0.5 at six and 0.0 at seven. The row at five breaks the trend at 173.1, which is a mean over 20 trials on a heavy tailed quantity where a median is the right statistic.

The cut per anchor is the constant Section 4.4 names, checked without fitting. Collision entropy computed from the symbol histogram alone gives 3.764 bits on English. An anchor should cut by $2^{H_2} = 13.59$. Backing the same quantity out of the four anchor sweep gives 3.735 bits, which agrees to 0.8%. As a prediction of the survivors themselves, $N2^{-kH_2}$ gives 53625, 3946, 290 and 21.4 against 63241, 3290, 351 and 17.2 measured, and a single histogram number carries four orders of magnitude to within 21% with nothing fitted. The variation in the per step cuts is noise around the constant and not drift.

A language carries constants of its own, and the arrangement measure is not one of them. Four texts in each of seven alphabetic languages, measured at character width to make a logographic script comparable (`examples/language/4_measure/language_constant.py`, `src/engine/r/hypotheses/language_variance.R`). The mean distance between word boundaries separates languages at $F = 13.21$ and $p = 3.5 \times 10^{-6}$, and collision entropy at $F = 9.02$ and $p = 6.2 \times 10^{-5}$, with the spread between languages exceeding the spread within one in both cases. A language does have figures belonging to it.

The rare half against a null permutation does not. It gives $F = 0.66$ at $p = 0.68$, and its between language spread of 0.0342 is smaller than its within language spread of 0.0731. Two novels in one language differ on it more than two languages do. The measure carries no information about which language it is reading. A universal has to look exactly like that.

Chinese settles it. Measured in within language standard deviations from the alphabetic mean, it stands 66.0 away on collision entropy and 22.1 away on the mean gap, carrying 3164 distinct symbols against 87 to 98, and 0.5 away on the rare half. A change of writing system moves everything structural by tens of deviations and leaves the arrangement measure where it was.

The cross language results hold across seven families and two scripts because the measure was never reading language identity.

That finding is published and belongs to Montemurro and Zanette. *Universal Entropy of Word Ordering Across Linguistic Families*, PLoS ONE 6(5):e19875, 2011. They shuffle the words to

build the null, the same construction used here, over 7077 texts in eight corpora covering Indo-European, Finno-Ugric, Austronesian, Afroasiatic and Sino-Tibetan families and the Sumerian isolate, and report the ordering quantity bounded near 3.3 bits per word with a relative variability of 0.07 against 0.23 for the entropy. Two hundred times the sample here, reaching Old Egyptian and Sumerian, and fourteen years earlier.

The statistic differs, since theirs is a Shannon entropy reduction per word and this is a dispersion of gaps over the rare half. Two unlike measures agreeing is worth recording. The conclusion is theirs. This work reached it independently and later, and found the precedent only when asked how such a thing could have gone unnoticed, and that order is backwards.

The rare half also departs from the null on a second source and in four more language families. Measured on Common Corpus instead of Project Gutenberg: Portuguese 0.32, Italian 0.36, Swedish 0.37, French 0.38, Turkish 0.46, Tamil 0.64 and Bengali 0.78, against a memoryless corpus at 1.00 on every one of ten parameter arms. With the earlier set that covers Germanic, Romance, Hellenic, Indo-Aryan, Uralic, Turkic and Dravidian over two independent sources and four scripts, and no natural corpus has yet returned 1.00.

The magnitudes are not comparable between the sources. Common Corpus rows are many documents joined, and joining distinct subjects drives this measure down on its own, from 0.828 to 0.626 for three works. The sign transfers between the sources and the size does not.

The published semantic primes are not the frequent half, and they divide along the same line it does. The Natural Semantic Metalanguage of Wierzbicka and Goddard proposes 65 words claimed to be undefinable and to have an exponent in every language, which resembles the finding above that the frequent half holds person reference, a conjunction and a copula or negation in every language measured. The two are not the same object. Ranked by frequency (`examples/language/4_measure/prime_ranks.py`), 26 of 66 primes fall in the top 100 of a novel and 23 of 68 in scientific prose, which is a third of them, with a median rank of 179 and 252 and one reaching 6443 in a vocabulary of 6730.

They split by category, and the split follows the head and the tail. In the novel the median rank is 39 for existence, 52 for the intensifier, 73 for the logical primes, 93 for time and 97 for the quantifiers, all inside the frequent half. The median rank is 449 for the evaluators, 493 for speech, 615 for the descriptors, 1055 for the actions and 1456 for life and death, all outside it. Being, negation, when, how many and very are machinery; good, say, big, do, live and die are content.

So the two schemes cut the same joint from different sides. One sorts by what cannot be defined and the other by where the information sits, and a prime lands in whichever half its category does. An earlier reading here treated the frequent half as the prime set and that was wrong.

The rare half departs from the null in every kind of writing, and the frequent half does not. Twelve corpora carrying six kinds of English writing, five languages, four centuries, a percussive re-encoding and a programming language all return a rare half between 0.48 and 0.76 against a memoryless corpus at 1.00, with no arm of the ten parameter sweep leaving 1.00. Scientific argument reads 0.48, economics 0.49, recipes 0.53, folk narrative 0.63, legal argument 0.64 and philosophical criticism 0.76. The kind of writing does not matter.

The frequent half is not invariant and part of the difference was an artifact. Its value tracked corpus size almost monotonically, from 1.01 at 106 KB to 0.72 at 2.4 MB, because a frequent symbol needs enough occurrences before a small real departure separates from the estimate. Truncating

every corpus to 400000 symbols moves economics from 0.72 to 0.87 and science from 0.88 to 0.96, while the rare half moves by at most 0.06 anywhere. So the rare half is robust to corpus size and the frequent half is not, and comparing frequent half values across corpora of different lengths is invalid.

Folding line endings is right for prose and wrong for source, and it moved one measure and not the other. A wrapped prose file carries line endings its publisher chose, and folding them to spaces lets the language show. A source file carries line endings and indentation its author chose, since the language ignores its own whitespace. Folding them discards the authored layer and keeps what the compiler reads. Measured both ways, the C sources give H_2 of 3.627 folded and 3.814 with the layout kept, a shift of 0.187 bits and the same size as the shift that made two Gutenberg texts look like outliers. The rare half measure is 0.50 either way and the frequent half moves from 0.59 to 0.60.

That is the division holding again: merging two frequent symbols changes the histogram and cannot change where the rare ones sit. Folding also removes an outlier. The figure 3.627 was cited here as a programming language sitting below the English band, and measured correctly it is 3.814, inside it.

In a formal language the human layer is measurable because it is definitionally what the compiler discards. Two channels carry it and neither changes what the program means. Whitespace is ignored by the language outright, and folding it moves H_2 by 0.187 bits as recorded above. Case is not ignored, but the convention that an upper snake name is a macro and a lower snake name is a variable is enforced by nobody: 42.2% of the uppercase runs in the C sources are three characters or longer, against 1.3% for a novel, 4.5% for scientific writing, 7.3% for the King James and 13.5% for political argument. The separation is clean and needs no identifier to be read.

The case channel is community convention and it does not separate formal languages from prose. Share of uppercase runs three characters or longer: makefile 0.544, shell 0.541, VHDL 0.495, Fortran 0.462, assembly 0.430, C 0.422, Scheme 0.280, Lisp 0.264, Perl 0.226, Ruby 0.213, R 0.210, Lua 0.159, Matlab 0.124, TypeScript 0.070, Modelica 0.060, Isabelle 0.025, Mathematica 0.024 and Lean 0.009, against 0.045 for scientific prose and 0.013 for a novel. An earlier version of this entry recorded a clean separation over the first ten languages, and four more break it: Lean sits below the novel, and Isabelle, Mathematica and Modelica sit inside the prose range. The measure orders communities that name things in upper case above ones that do not, and theorem provers name by mathematical convention so there is nothing to shout.

The language labels on that source are not reliable and the ones carrying findings were checked. A corpus labeled as a fabrication image format holds Ethereum contract code, on the evidence of `function`, `uint256`, `contract` and `pragma`, and was nearly reported here as a machine generated control. Checked by their distinctive keywords, VHDL carries `signal`, `port`, `attribute` and `generic`; Lean carries `theorem`, `lemma`, `have` and `calc`; Isabelle carries `shows`, `assumes`, `qed` and `proof`; Modelica carries `annotation`, `redeclare` and `extends`; and the makefile corpus carries `ifeq`, `endif` and `cflags`. Those five are what the table above rests on. A corpus labeled as Mathematica returns every token exactly once and is not trusted. This is the third mislabeled corpus in this work, after one labeled Hungarian that held German and one labeled Latin that held English.

VHDL is the strongest case of a convention with no machine content, because the language cannot see it. VHDL is case insensitive: `ENTITY` and `entity` are one token to the compiler and no program can distinguish them. Its share of long uppercase runs is 0.495, third of eighteen.

In C the case at least carries identity, and a convention there is a choice among distinguishable options. Here the options are not distinguishable at all and the convention is held anyway.

The obvious companion measure does not work. Folding the line endings shifts H_2 by 0.078 to 0.330 across those languages and by 0.110 to 0.139 for prose. The ranges overlap, and the shift is measuring line density and not an authored layer. Indentation spread does better, 2.28 to 9.65 for code against 0.10 and 1.98 for prose, with one exception: a makefile is flat, at 0.72, because its recipe lines carry one tab and do not nest.

The control that was missing all along: a structured domain with no author and no selection. Every corpus in this work that departs from a null permutation was made by a person. The measure detecting arrangement and the measure detecting human production were never separated. A genome will not serve, since the selection that shaped language shaped the organism, and all biology shares one machinery. Mathematics is under no selection at all. The digits of the square root of two return 1.00, at the null, as a number conjectured normal should. The gaps between primes return 0.93, outside the 0.99 to 1.01 band that every one of ten memoryless arms occupies.

So the strong reading is refuted. Nothing authored the primes and the measure sees them. It reads arrangement, and human production was never intrinsic to it. The weak reading holds and is quantitative: the primes depart by 0.07 where the human corpora depart by 0.22 to 0.68, three to ten times further, and the square root digits confirm that determinism alone is not what it responds to. The gaps were encoded as half the gap capped at one byte. Part of that 0.07 may be the encoding.

Nor is the signature human, on the fairest comparison available. Bird song, whale song, a wolf howl and two recordings of a person reading aloud were fetched and put through one pipeline: averaged to mono, decimated by block averaging to 8 kHz so nothing aliases, and quantized to one byte (`maint/data/fetch/vocalization_domain.py`). Whale song returns 0.25, a dawn chorus 0.30, human speech 0.42 and 0.43 on two independent recordings, and a wolf howl 0.46. The animals bracket the human, and two of the three depart further from the null than a person reading an encyclopedia does.

A dawn chorus is many birds at once. Comparing it against one person alone in a studio compares a concatenation against a single source, and joining sources moves this measure by itself. Matched by number of sources and trimmed to 110000 samples each, the single bird returns 0.23 against 0.32 and 0.45 for two people reading, and the dawn chorus returns 0.26 against 0.33, 0.49 and 0.71 for three recordings of crowds. Both matched comparisons put the birds ahead.

They are birds and not animals. The whale returns 0.37, inside the human range, and the wolf returns 0.62, less clustered than any human recording. Two of four animals depart further than the people do and two do not, with two or three recordings per cell.

The slice was wrong for a vocalization and correcting it changes the answer. A whale song unit runs one to three seconds and a wolf howl is seconds long, and a statistic over the gaps between rare amplitudes at 8 kHz reads inside a single call and never sees how calls are arranged. Taking the amplitude envelope at one symbol every 10 ms (`data/envelope_scale.py`) gives whale 0.44, dawn chorus 0.45 and wolf 0.49 against 0.56 and 0.72 for two human recordings, which separates without overlap where the sample scale had them interleaved. The envelope leaves only 7 to 40 distinct levels. The rare half is a handful of symbols and the figures are thin.

True unit scale is not reachable with these recordings. At one symbol every 50 ms a one minute clip gives 1053 symbols for the whale, 566 for the wolf and 281 for one bird, which is far too few. Measuring the arrangement of units needs recordings of tens of minutes.

The band where much animal communication happens is absent from the recordings by construction. Measured as a share of spectral energy, the wolf recording carries 0.0007 below 20 Hz, 0.0000 from 20 to 50 Hz and 0.0001 from 50 to 200 Hz, which puts 99.92% of it above 200 Hz with the low end filtered out. The whale carries the most low content at 0.0281 in the 20 to 50 Hz band. Sampling is not the cause, since 48 kHz resolves 20 Hz easily. Microphones and hydrophones roll off at the low end, and every psychoacoustic codec is built to discard what a person cannot hear. The encoding removes the infrasonic channel before the file exists. Any measurement taken from compressed archival audio is therefore blind to it.

Recorded for the band instead of despite it, the blue whale is the furthest from the null of anything measured here. Six uncompressed recordings from the NOAA PMEL Acoustics Program, each time compressed by ten to bring a call near 20 Hz into the audible band, with the speedup undone before measuring (`maint/data/fetch/infrasound_domain.py`). The band is present: 99.68% of one recording's energy lies below 20 Hz, against 0.0007 for the archival wolf and 0.0127 for archival human speech, a factor of a thousand.

Four blue whale recordings from four ocean basins return 0.25, 0.27, 0.36 and 0.43 at one symbol per 100 ms of real time, against 0.44 to 0.49 for animals in the audible band and 0.56 and 0.72 for human speech at the same scale. A fin whale returns 0.74 and the 52 Hz whale returns 0.80.

Between 4 and 16 symbols cleared the occurrence floor on those rows. The envelope had to be coarsened to 32 levels to give each level enough occurrences, and the rare half is then a handful of symbols. The recordings are 148 to 264 seconds of true time where the statistic wants tens of minutes. The agreement of four independent populations is the strongest thing about the result and the symbol count is the weakest; the fin whale row rests on four symbols and is not quotable.

A mechanism unrelated to communication also fits. A bird call is discrete and separated by silence while a person reading aloud emits a continuous signal. Clustered rare amplitudes follow from the shape of the emission and not from what it carries. Amplitude quantization cannot separate those. Doing so needs the measure applied to segmented calls instead of to samples, which is not built.

On a continuous irregular domain the construction as implemented finds nothing, and the matcher is why. Protein structures were read as what Section 4.2 covers: a cloud of points in three dimensions carrying values, with a pattern given as displacements and an alignment surviving where every displacement lands on a point holding the value asked for. The null shuffles the values among the points and leaves every coordinate where it is. The shape of the cloud is untouched and only the value to position association is destroyed.

At one anchor a 8015 point structure gives 0.023 live against 0.024 null, and a 9364 point structure gives 0.023 against 0.022. At two anchors both arms are zero and the ratio has no content. At three anchors the cascade clears the whole cloud, which is worth its own note: in text three anchors leave five to fifteen survivors, and in three dimensions they leave none, because a displacement must agree on three axes at once.

The cause is exact matching on a continuous domain. Coordinates were voxelized at two angstroms and the displacements were required to agree exactly, and two occurrences of one structural motif never land on identical voxel offsets. The thirteen geometries in Section 4.2 are

generated lattices and rotated point sets, where exact repeats exist because they were built in. Nature does not supply them.

Proposition 1 survives a tolerance, since a point lying within some distance of every displacement is still a necessary condition and a subset of those conditions is still necessary. What fails is this implementation.

Corrected, it works, and the correction took two steps because the first null deleted the wrong property. With a tolerance of two voxels the cascade returns survivors, 6.510 per trial at two anchors, and against the value shuffling null it still shows nothing at a ratio of 1.066. The rules a protein obeys constrain where its points are, not which residue occupies each one, and a null that moves the values and leaves every coordinate in place preserves all of the structure and it cancels.

Scattering the positions over the same bounding box while keeping the composition gives 6.510 against 2.592, a ratio of 2.512. The same live arm, a background that deletes geometry instead of composition, and the departure appears. So the construction reaches a natural continuous domain in three dimensions, the structure there is geometric and not compositional, and both failures were in the instrument.

The reading of that result is wrong and the proof of it refuted the rule it produced. Four clouds were built where the answer is known by construction: a lattice or a scatter, carrying values determined by position or drawn at random. The prediction was that the value null fires on the two with a pattern and the position null on the two with a lattice. Measured, a lattice with a pattern fires on both as predicted and a scatter with random values fires on neither as predicted, and the other two are wrong. A lattice with random values fires on neither, and a scatter with a pattern fires on both.

Two things follow. The nulls are not independent, since moving a point destroys which values are near it as well as where it sits. The position null deletes the value coupling along with the geometry and is strictly the stronger of the two. And the measure cannot see geometric order by itself: a perfect lattice carrying random values returns 0.690 live against 0.684 and 0.553, because a survivor needs a displacement and a value together and random values supply no repeated pairs.

So the protein figure of 2.512 is not the fold. This measure cannot read backbone angles or secondary structure, and what a scatter changes besides the geometry is the local density, since a protein is packed in its core and a uniform scatter is Poisson. Local packing is the likeliest reading and it is not established here. The rule that a null must delete the property being asked about is refuted as stated, because on this domain the two nulls do not delete separable properties.

A picture read as a sequence returns its own width, and the construction was already written for it. Section 4.2 holds the invariant over thirteen geometries and dimensions one to eight. An image is the case this work covers and not an extension of it: a domain whose alphabet is a range of values and whose arrangement has two dimensions. Read row by row, the second dimension survives as a periodicity, since a pixel and the pixel below it lie one width apart in the sequence.

Three paintings fetched at a width of 960 return 960 as the strongest lag or the second strongest, with 959 and 961 beside it, the shape a real period takes (`data/image_domain.py`). Where 960 is second the winner is a lag of 1, from a painted surface being smooth. Nothing in the detector is told that the bytes are a picture.

The controls hold. Uniform noise returns unrelated lags at an agreement of 0.0050, which is 1/256 and therefore chance. A generated ramp returns 511 at 0.9980, and 511 is the correct answer where 512 is the width it was built with, because the ramp is (*row*+*column*) mod 256 and stepping down one row and back one column leaves the value alone. The detector found the structure of the object instead of the width it was built with.

The three separate by how strongly they agree, and that ordering is already recorded here: noise at 0.0050 has nothing to find, a machine made ramp at 0.9980 is perfectly regular and so trivial to detect and empty to use, and the paintings at 0.02 to 0.30 sit in the band where a periodicity is both findable and carries a position.

On a board layout the same residue is a grid setting, and it separates two tool communities. A coordinate in a layout may hold any value the format's precision allows, the fabricator accepts it, and physics does not prefer one to another. Designers place on a grid regardless. Measured as the share of decimal coordinates landing on a grid (`examples/source/4_measure/layout_grid.py`), a KiCad corpus gives 0.389 on 0.05 mm and 0.023 on 25 mil, and an EAGLE corpus gives 0.133 on 0.05 mm and 0.414 on 25 mil. One works metric and the other imperial, an eighteenfold difference on the imperial grid, with the same physics, the same format capability and the same copper.

That measurement also identifies its own corpus, since nothing but a board layout places 41% of its coordinates on 0.635 mm. Both labels are otherwise unverified, and the mislabeled corpus recorded above is why that matters.

Constructs also pair across a distance the slice cannot hold. In the C sources the median separation is 7 symbols for a bracket, 16 for the two halves of a conditional, 20 for a parenthesis and 114 for a brace, with a mean of 212 for the last. Prose carries 30 parentheses and 4 braces in 728751 bytes. That answers a question Section 7.4.4 raised and left open: the C sources return structure at every span from 8 upward while prose has almost none between a word and a passage, and the spans above are where it comes from.

A further caution on comparing the two at all. A byte of C can be a complete operation, where a byte of English is a letter inside a word. One symbol per byte places the two corpora at different levels of abstraction. Section 4.10 records the symbol width as an unjustified choice within a corpus, and this is the same choice failing between corpora.

After that correction the frequent half is reading formatting. At a common 400000 symbols only two corpora keep a frequent half signal: C source at 0.58 and a cookbook at 0.70. Everything else, narrative and argument and science and law, sits between 0.87 and 0.99. Both survivors carry a repeated skeleton, a language syntax in one and a fixed section structure per recipe in the other. The measure is reading a template and not a language.

The cell rarest rule holds across kinds of writing. Against the rule the library uses, on scientific argument, legal argument, folk narrative and recipes, the median sizing error moves from 1.72, 2.32, 2.14 and 2.70 to 0.94, 1.70, 1.16 and 1.16, and the alignments passed move from 12.4, 14.6, 10.8 and 19.3 to 6.0, 9.6, 3.5 and 16.4. The cell rarest rule is better on both counts on all four. The worst case improves on three of them, from 90328 to 60 on science and from 221767 to 188 on legal argument, and is 7.8 times worse on recipes.

The product rule error is a measurement and not only a defect. Anchors surviving together more often than the product of their rates is mutual information between their positions,

and the size of the departure is that information. On English prose with anchors chosen by rarity the median is 3.55, which is 1.8 bits, and the largest of 60 needles is 490336, which is 18.9. A typical search carries under two bits of it and a needle that recurs carries nineteen. Taking the rarest symbol in each cell instead brings the median to 1.11, which is 0.15 bits, without changing the maximum.

The best filter measured is the rarest symbol inside each cell, and it is not the rule in use. Three anchors over 60 needles: taking the rarest symbols in the needle passes a mean of 15.9 alignments on English prose and 44.1 on C source; dividing the needle into cells and taking the rarest symbol inside each one passes 5.0 and 25.4. That is 3.2 and 1.7 times fewer, against the rule the library uses. On a memoryless corpus it passes 6.4 against 3.9, which costs 1.6 times where there is no structure to exploit, because the cells prevent it from taking the three globally rarest symbols. The cell rule also brings the median sizing error to 1.11 from 3.55. On a typical search it is both the better filter and the better sized one.

On a periodic corpus the histogram misses by 4096 and the coherence is predictable in closed form. On a corpus of period sixteen the histogram predicts one alignment in 65536 survives four anchors. One in sixteen does. Measuring the corpus for coherence before the search predicts that factor in closed form as $2^{(k-1)H_2}$, from one pass over the lags. This entry was in the delta null abstract and is the only copy of the figure.

The product rule fails in the tail and not in the middle, and reporting it as a mean hid that. Over 60 needles the ratio of measured survivors to the product of the anchor rates has a median of 3.55 on English prose and a maximum of 490336, five orders of magnitude apart. On C source the median is 87.5 against a maximum of 175369. A mean over such a sample is mostly its largest draw: at 25 trials one needle contributed 96% of a reported mean of 19684. The original entry in this workbook said the failure reaches up to four orders of magnitude, which was correct, and several figures written here at one time turned that into a typical value by quoting means.

Sizing from the measured pair helps the tail and not the typical case. Kac's lemma makes the expected distance between occurrences the reciprocal of their frequency, and a count of surviving alignments is an observed joint frequency and not an estimate of one, and that count is already the intermediate result of the cascade. Substituting it for two of the three marginals moves the median from 3.55 to 2.18 on English prose and from 87.5 to 17.9 on C source. Where the anchors are the rarest in each cell it moves the median from 1.11 to 1.18, which is no improvement. The correction is worth taking because it costs nothing, and it is not worth the two to three orders of magnitude first claimed for it.

The sizing error distribution, by quantile, and what a buffer would have to hold. Over 200 needles with three anchors, the ratio of measured survivors to the product of the anchor rates:

corpus and rule	p50	p75	p90	p95	p99	max
English prose, rarest	1.87	10.81	445.59	6625.99	51073.10	490336.80
English prose, cell rarest	0.93	2.06	9.97	90.92	2382.88	490336.80
C source, rarest	79.59	1219.25	12552.23	36286.09	147227.55	175369.07
C source, cell rarest	7.40	184.70	1219.25	3169.17	18056.99	44068.51
memoryless, rarest	0.87	1.32	1.72	2.08	2.48	2.93

The memoryless corpus holds its whole distribution inside a factor of three. The tail belongs to the structure and not to the measurement. The distribution is heavy tailed without being a single power law: the slopes between successive quantiles in log-log run -2.53, -4.05, -3.90 and -1.27, steeper in the middle than at either end.

What cell selection buys has a peak and then gives it all back. Dividing the two English rows gives 2.0, 5.3, 44.7, 72.9, 21.4 and 1.00 across the quantiles above. The advantage rises to 73 times near the 95th percentile and reaches exactly one at the maximum, where both rules return 490336.80 to six figures. So placement is worth the most in the upper middle of the tail and worth nothing at the worst case. The C source does not show the same shape, running 10.8, 6.6, 10.3, 11.5, 8.2 and 4.0.

The tail is the needle recurring in the corpus, and placement does not reach it. The largest ratio over 60 needles is 490336 under both selection rules, identical to six figures. A needle that recurs as a unit matches every anchor inside it together wherever they are placed, which puts the correlation between the needle and the corpus and not between the anchors. The jittered rule appears to fix the sizing and does not: it selects common symbols whose chance hits swamp the recurrence, and it passes 337.2 alignments to do it.

There are two structure meters here operating at different spans, and they agree. The null permutation reads clustering above a span of 32 and returns 0.73 on English prose, 0.50 on C source and 1.00 on a memoryless corpus, ordering the C source as the more structured of the two. The product rule error reads correlation inside a needle of 24 and returns a median of 3.55, 87.5 and 0.86 on the same three, in the same order. An earlier version of this entry had the orderings opposite and built an argument on the difference, and that came from quoting means of a heavy tailed ratio. Section 7.4.4 separately stated the block sweep as prose having almost nothing below a sentence, which was a claim about the clustering measure written as a claim about the text.

Collision entropy cannot see structure, and the null had to exist for that reason. English prose measures 3.764 bits and the memoryless control measures 3.779. The two are indistinguishable on the quantity that governs the cost. One carries four centuries of arrangement and the other carries none, and 2^{-H_2} predicts the anchor rate correctly on both. Cost belongs to the distribution and structure does not live in the distribution at all.

Predicting the aggregate rate and predicting one search are different problems. The uninformed rate 2^{-H_2} carries the survivors across the sweep to within 21%. Multiplying the probabilities of the particular symbols a needle put under the anchors is wrong by 2 to 14 times over the same rows. Sizing a system is solved and predicting a single query is not.

The product rule is accurate where it does not matter and wrong where it is needed. Over the same sweep its error runs 1.00, 1.02, 2.09, 2.28, 3.04, 13.89 for one to six anchors. At one and two anchors it is exact and tens of thousands of alignments survive. By the depth where the filter is worth using it is off by 2 to 14 times, and the error grows with every anchor added.

Where the anchors are placed changes the sizing more than which symbols they are. Three rules over the same corpora and the same needles: taking the rarest symbols in the needle gives a sizing error of 20992 times on English prose and 1494 on C source; spacing the offsets evenly with no reference to the needle gives 2.02 and 3.76; drawing one offset inside each evenly sized cell gives 1.84 and 1.97. The last is better than even spacing on both corpora and passes fewer alignments than it on English, 337.2 against 345.8. Even spacing is a comb and shares a period with whatever the domain carries; one draw per cell keeps the spread and gives the anchor set no period of its own.

On a perfectly periodic domain the placement rule stops mattering. A counter of period 90 returns a sizing error near 8099 and 8096 survivors under all three rules, identically. Every

alignment at the right phase survives whatever the anchors are. Testing placement against periodicity needs a periodic domain carrying noise, which is not built.

How far a vocabulary spreads across subjects is measurable, and this scale is the only one here with resolved bands. Scoring each word for clustering against a null permutation and averaging gives 0.758 to 0.833 for single works over four languages and three centuries, 0.509 to 0.667 for collections, and 1.010 for a memoryless corpus, with no overlap. C source falls with the collections at 0.533, which is forty files each concerning a different module. Hugo is the lowest single work at 0.758, and *Les Misérables* turns aside into Waterloo, the sewers and a convent. It is the most varied single work in the set.

The reading is causal and not observed. Joining English single works with the total length held near 900000 characters walks the score from 0.828 to 0.685 to 0.626 as works are added. Cutting the 66 book collection into equal pieces walks it back up, 0.509 whole, then 0.612, 0.648, 0.681 and 0.708 as the pieces fall from 8.2 books each to 1.0, with the highest single piece reaching 0.847. The reading is not a count alone: two English works joined give 0.685 against 0.667 for 37 plays, since a novel and an epic poem stand further apart than two plays of one period.

A cipher cannot remove what this measure reads unless it spends key equal to the message. A permutation of the symbols reproduces the boundary dispersion to four decimals at 0.2815. A repeating key of length 8 leaves a corpus that looks memoryless read whole, and splitting it at stride 8 and averaging all eight cosets returns 0.896 against 0.896 for the plaintext, exactly, since each coset was enciphered by one substitution. The key length falls out of a stride scan with nothing told to it: strides coprime to it sit at the memoryless value and strides dividing it fall, with the minimum on the true period, checked at key lengths 3 and 8. A pseudorandom addend as long as the message reads 1.004 whole and 1.004 under every coset scan, and is the only transform that erases anything.

Subsampling attenuates by a known factor. Scoring cosets at stride s with the length held fixed gives a departure from the null of $1 - c/\sqrt{s}$ with c between 0.237 and 0.277 at eight of ten strides. Cluster detection is a counting problem and a larger stride puts $1/s$ as many of a cluster's points in the coset. The signal falls as the square root of that. Structure is not lost at a stride, it is attenuated, and a longer corpus buys any stride back at quadratic cost.

One quantity does three jobs. 2^{-H_2} sets the candidate rate, the contrapositive indication distance, and the size of the bias correction. None was derived from another and all three were measured independently.

Five instruments agree on the zero information object, and that is one assumption held five times. Collision entropy 0.000, ceiling 1.000, refutation 0.000, correction 0.0000 at every order, and 2032 of 2033 positions surviving six anchors. All five read the corpus eight bits at a time. The flat corpus is `0x41` repeated, whose bit pattern has period eight. At any width from six bits up it takes exactly eight values and carries $\log_2 8 = 3$ bits. Measured 3.0000 at widths 6, 8, 12 and 16. The object is flat only at the byte boundary.

The symbol width was never justified and it is not free. $H_2(w)/w$, the discrimination per bit read, falls with width on every corpus: English gives 0.984 at one bit and 0.856 at eight. The byte slice forfeits 15%, and sixteen bit symbols forfeit 40%. Soundness is indifferent to the width, since Proposition 1 never mentions it. Which width is best depends on whether a read costs per bit, per access, or per machine word, and this work counts reads under the third without having said so.

A confident period reading is not evidence of a period, and no margin test catches it. Handed the gap word of Wythoff's losing positions, which is Sturmian and has no period because φ is irrational, `sequence_period` returns 8, then 13, then 21, then 34, then 55 as the window widens from 16 to 128. Every one of those rows clears its own shuffle floor, and across nine irrational slopes and seven windows the margin runs 3.4 to 29 times that floor. Nothing about the readings is weak.

They are not noise either. The reported number is the denominator of a continued fraction convergent, or of an intermediate fraction, of the word's own slope, on 61 of the 63 rows. A rotation by an irrational agrees with itself at the denominator of any good rational approximation to it. The detector is finding a real agreement at a real lag and reporting it correctly. Only the name on the output is false. The two rows the mechanism does not account for are the weakest in the table, at rank 0 and rank 8 of 63 by margin. The shape of the movement follows the slope: π, whose continued fraction has a 292 in it early, holds 7 across six consecutive windows, where φ moves at every step.

What separates the two costs one call per window and uses neither oracle, null nor second instrument. A period is a property of a sequence; an approximation is a property of a sequence and a window together. 115 of 115 subtraction games with a proved period give one answer across five windows, and 0 of 9 aperiodic words give one across seven. `measure.periodicity.stable_period` is that test. `sequence_period` is unchanged, since this is a second question and not a correction to it.

The crystallography rows stand, because a lattice has a period for the reported number to be. The general license to read a high margin as a period is withdrawn. Any period in this work taken at a single window is unverified until it is read at a second.

A flattened grid carries the repeat along each of its axes at once, and a single reported period is one of them. Three hundred and fifteen two-heap subtraction grids at five widths: the answer is the laid-out width on 108, the game's own Grundy period on 96, the width off by one on 68, and a combination on 43. The same grids shuffled return the width on 5 of 100, the chance rate. None of the four groups is noise. Where a grid's content is a function of the sum or difference of its coordinates the shortest flattened repeat is at $W - 1$, and the detector is right while the ground truth of W was the wrong question. Reversi returns 8 because nothing repeats along a Reversi row, which is a property of that corpus. Read the axes separately where the caller knows there are two.

The departure reads what the rules did to a board and not who chose the move. Reversi, 200 games per pairing: live clumping 0.6555 against a null of 0.5350, a ratio of 1.225, where six disjoint batches spread by 0.0089. A control game on the same board with the same squares filled and the flip rule removed reads 0.998. So the departure is the flip rule and a board being a board contributes nothing. Three arms of different strength in nine pairings order themselves cleanly by played win share, 0.645 for corner against random and 0.630 for greedy against random, and not at all by departure, which runs 1.184 to 1.255 at a correlation of 0.067. Every legal move turns over a bracketed run whoever picked it. Strength changes where the trace goes. It does not change what the trace is. Reversi is the only corpus in this work where the rules and the play can be varied separately, and the evidence is one game with three weak arms.

An earlier version of the flip control read 0.971 and the miss was ours. It kept Reversi's four fixed center pieces, laid out black over white and white over black. Four of a hundred and twelve

neighboring pairs therefore disagree by construction, where a draw would have half agree. That is 0.018 of the reading and all of the gap. The same four squares sit under every Reversi row above and push the same way. The ratio of 1.225 is understated by about that much and never overstated.

A domain that hands out its own answers costs a second a row, and the workbook's estimate of years was about obtaining corpora. The reach entry above says that getting from instances to a statement about a domain takes a controlled series inside that domain with an outside answer attached, and prices it at years. That price is a fact about crystals, proteins and held papers, where the reading is cheap and the row is dear. A subtraction game inverts it: the rules are four lines, the position space is generated by them, the Grundy sequence is provably eventually periodic, and the period is computed by a dull comparison routine sharing no code with the detector. Three hundred and eighty five games took under a minute. The series it makes still covers only subtraction games, and their sequences are exactly periodic over a small alphabet with no noise. That is the easiest case a period detector can be shown.

Empirical Results: Refuted

Recorded because each was built, measured on the case it was designed for, and kept.

mechanism	result
rare anchor as a whole search	loses to Horspool by 1.2 to 25 times, mean and worst case
salted anchor	loses on both as well
rare byte as a filter over Horspool	0.0% to 2.6%
discard the field on divergence	inert, never fires on 15 of 28 rows, at most 1% where it does
pull the anchor to the field's center of mass	2% to 42% worse than taking the largest value
three term control on the field	mean 1.005 against a plain lag, worse on 24 of 35 rows
deadband against the ambient level, amplify, settle	mean 1.014, better on 2 of 35 rows

No drift rate was obtained, over three designs that each lost to a different confound. Pairs matched for form lost to genre: 53 years of epic verse preserves more of the top 100 words, 0.5625, than 12 years across a change of form, 0.5038. A series held against one 1611 reference lost to genre as well, and its only genre matched interval is also its only undisturbed one. The disruption term and the genre term take the same value on the same rows. Three translations of one book lost to translator lineage: two years across a change of translator costs 0.1481 of the top 100 band where 389 years costs 0.0662, and the content band agrees better across four centuries, 0.6243, than between two contemporaneous translations, 0.5440.

What that work did establish is a floor and a ceiling with nothing between them. Within one language across 389 years the top 100 overlap is 0.7857. Across two languages it is 0.0204. Nothing measured falls between. The measure has a slow regime and a total regime and no resolved middle. A civilization ending sits in that gap, because a population leaving substitutes one vocabulary for another instead of drifting, and changes then arrive in groups with no rate having to vary.

The pattern across all seven is one sentence. Extracting more information per observation pays. Reacting more cleverly to information already extracted does not. Triangulating one answer across all m offsets wins 13.2%. Every refinement of the response is neutral or negative.

Said as state: one entry is a table walk and six are memory. The counterfactual credit computes what every offset would have earned from the symbol just read, by one backward pass over the needle, holding nothing. That is the 13.2%. Crediting only the offset that asked, carrying the field between searches, resetting on divergence, selecting by center of mass, adding accumulated and trend terms, and gating on a deadband are all memory, and together they are worth

nothing. The table is also reversible at the cost of a lookup: forward it maps a symbol to an advance, backward it maps one read to the set of alignments that read kills. Section 4.9.4 of the paper says an in-order walk collects between 3% and 99% of that set. The reverse direction is where the uncollected refutation is, and nothing here has walked it.

Why the center of mass row loses, and it is the equilibrium argument arriving from the other side. The center of mass of the field is the constant the arrangement varies about. Pulling the anchor toward it moves the anchor toward the quantity that carries no arrangement and away from the only part that does. The mechanism was built to reduce variance and what it reduces is the signal. It is 2% to 42% worse and never better.

That is the same statement this document already makes about the background. Maximum entropy under a constraint is equilibrium, a shuffle is the equilibrium state of a corpus, and the measured departure from it is all of the arrangement information. The center of mass is that equilibrium computed per field instead of per corpus. Reorienting toward it is therefore reorienting toward the null, and a mechanism that moves toward its own null has nothing left to read.

The equilibrium argument also predicts the two results that already hold. The largest value wins because it is the furthest point from the center, meaning the largest departure. The counterfactual credit wins at 13.2% because it computes what every offset would have earned, giving the distribution of departures instead of their mean. Both are readings of the residual; every mechanism in the losing table is a reading of the center or a memory of it.

The corpus work reached the same shape independently. Comparing whole distributions between two Lushootseed dialects failed at 1469 and 2768 bytes against a method resolving at 6707, and flattening the pooled counts to maximum entropy and testing one run at a time found the border at ə, southern 45 against northern 6, beaten by 1 of 200 random borders. Flattening the pooled counts is subtracting the constant. What was left was the deviation, and the deviation was the dialect.

This is an explanation of measured rows and not a new measurement. What it predicts and nobody has run: a mechanism scoring an anchor by its distance from the field's center of mass, instead of by its raw value, should track the largest-value rule and should beat it wherever the field's center drifts.

Where the constant is given, the method works immediately, and that separates chemistry from language here. A domain that states its own rules hands over the constant. Bond lengths, valence and angles are the invariant a molecule is held to. The deviation is available without having to be constructed, and a null can delete one property while holding the others. A domain that states nothing gives no such reference, and the constant has to be built out of the data before anything can be measured against it. Flattening the pooled counts to maximum entropy is that construction, and the Lushootseed border came out of it.

That is also the missing piece under the protein row. This document records the rule that a null must delete the property being asked about, and records it as refuted, because scattering the positions destroyed the geometry and the composition together and the two nulls were not separable. They are not separable without the rules. Knowing them makes a null that deletes one thing constructible. The rule is not refuted in general; it is refuted wherever the constraints are unknown, which is where it was tested.

Where the substrate is indifferent, all of the departure from maximum entropy is intention, and three measurements in this document are that one measurement. This is

the sharpest case of the argument above, because the constant does not have to be estimated. A substrate that accepts every option equally is already at maximum entropy by construction. The flat reference is exact and free, and a person put everything measured above it there deliberately.

VHDL is the cleanest instance and this document already holds it. The language is case insensitive. `ENTITY` and `entity` are one token and no program can distinguish them. The substrate is not merely permissive, it is blind. The share of long uppercase runs is 0.495 all the same, third of eighteen languages. Nothing in the machine can read that 0.495. All of it is a community holding a convention for other people.

A board layout is the same measurement in another domain. A coordinate may hold any value the format's precision allows, the fabricator accepts it and physics prefers none. The reference is flat by physical law. Designers place on a grid regardless: KiCad returns 0.389 on 0.05 mm and EAGLE 0.414 on 25 mil, an eighteenfold difference on the imperial grid with the same physics, the same format capability and the same copper. With the circuits held the same, the departure can only be a fact about the designers.

Whitespace in C is the third. The language ignores its own whitespace outright, and folding it moves H_2 by 0.187 bits. That 0.187 is the authored layer, measured as the difference between what the machine reads and what the author wrote.

The subject-word result is the same shape with an estimated reference instead of a given one. Gap dispersion above one returns `begat`, `Sancho`, `aventura` and never `the`, `and`, `of`, without a stopword list, a grammar or a dictionary. The grammar is the constant every sentence in the language varies about, and flattening it out leaves what the text is about. Where the substrate hands over its own indifference the reference is exact; where it does not, the pooled distribution has to stand in for it. The Lushootseed border needed exactly that, and it is the harder result for the same reason.

So the recipe is one sentence and it has four instances here. Find what the substrate cannot see or does not care about, take that as flat, and read the deviation. The deviation carries what somebody meant.

Said in one line: ask a question loaded with what you are certain of, and read what comes back. The certainty is the constraint, and it is the only thing supplied. The answer is the residual after that constraint is subtracted. Every row above measures that residual under a different name for the constraint: a shuffle of the same multiset, the pooled counts of two dialects, a compiler that cannot see case, a fabricator that accepts any coordinate, a published cell edge. Nothing is fitted and nothing is assumed past the certainty that was put in. The formal statement of the same thing is the two propositions below.

What is universal here is the soundness, and it is universal because it is deductive. What is not universal is the reading, and that is Proposition 2. The two halves get claimed together and they are not the same claim.

Proposition 1 has no parts and no domain in it. Any subset of a pattern's points is a necessary condition. No selection rule can lose a true occurrence, and the proof uses neither order, dimension, alphabet nor interpretation of a symbol. A mineral, a fold, a picture, a cipher and a sentence are therefore one object to it. Nothing was ported between those rows; there was nothing to port.

The physical anchor is real and this document already states it exactly. Maximum entropy under a constraint is equilibrium, and a shuffle is the equilibrium state of a corpus, the measured departure is a free energy above it, and the maximum extractable is that departure and nothing

beyond it. The correspondence is exact and not an analogy, since both quantities are a distance to the maximum entropy state under the same constraint. In that sense the reference is not chosen and not modeled: it is the data with one property deleted, computable from the sample, and it would be the same reference for anyone.

What does not carry is the reading. Proposition 2 says the exact compare is irreducible and no knowledge of a domain removes it. Soundness belongs to the construction while cost belongs to the domain's distribution. This document has the failures that prove the distinction is not decorative: a memoryless corpus returns 1.00 and there is nothing to read, a perfect lattice carrying random values returns 0.690 against 0.684 because a survivor needs a displacement and a value together, the digits of the square root of two sit at the null as a normal number should, and the protein nulls do not delete separable properties on that domain. In each the construction is sound and finds nothing, correctly.

So the model applies everywhere and reports something only where the domain put something there. A universal claim about applicability is supported and is deductive. A universal claim about yield is refuted by rows already in this file.

The scope stated as limits, the form a reader should be handed. Four of them and no others. The construction needs an instantiated object: a sample, a record or a measurement, and never a thing suspected of existing. It needs that object partitioned into parts, by a partition stipulated, estimated or supplied. It reads one quantity, the distance from the constrained maximum entropy reference. And it adds nothing, since the arithmetic supplies no knowledge and the constraint carries all of it.

The rows that found nothing sit inside those limits and not outside them. A memoryless corpus at 1.00, the digits of the square root of two at 1.00, a lattice carrying random values at 0.690 against 0.684 and 0.553, and the protein nulls that do not delete separable properties are four objects that existed, were partitioned, and were correctly reported as carrying nothing this instrument can see.

What the limits do not exclude is an absence. Reporting that something expected is missing needs no new object, because a hole is a departure from an occupancy the constraint already states. The oracle check's second direction is that reading: it asks which tokens a paper prints that no row holds, and finds what a person skipped by knowing what should have been there. The regularity makes the hole visible, and a domain with no stated regularity has no holes to find.

A thermodynamic argument for universal yield is available and it is about the wrong system. The argument runs that measurement presupposes a gradient, since recording a result dissipates energy and Landauer's bound puts a floor of $kT \ln 2$ on erasing one bit. Nothing at equilibrium can be observed at all and every object anyone measures is away from its own maximum entropy. The first half holds and constrains the observer. The second half does not follow, because an object at its own maximum entropy is measurable by an observer who is not at theirs. This document has the cases: the memoryless corpora return 1.00 on every one of ten parameter arms and the digits of the square root of two return 1.00, and each was measured by an apparatus with all the gradient it needed. The bound belongs to whoever takes the reading and carries nothing about what the reading finds.

The span across domains is the claim, and one construction carries it instead of a family of them. The pieces are individually old. A null permutation is standard, maximum entropy as a reference is standard, and correcting a damaged text against attested forms is standard. What

is not standard is that one construction, with neither order, dimension nor alphabet in it, reaches all of the following, and this document holds the measurement for each.

domain	what it read	figure	instrument
a crystal	the published cell edge, from tiled COD entries	3 of 3 exact, to 0.0006 Å	shift
an image read as a sequence	its own width, told to nothing	960 at a width of 960	shift
a Vigenère cipher	the key length, told to nothing	8 at a key of 8	shift
point clouds in 1 to 8 dimensions	the invariant, refusing nothing	213,840 occurrences, 0 refused	both
byte strings	rare half against a null permutation	0.48 to 0.76, memoryless 1.00	null
protein structure	departure against a position-scrambling null	6.510 against 2.592	null
a board layout	a design grid physics does not prefer	0.389 metric, 0.414 imperial	null
vocalization	source separation on the envelope	whale 0.44 to human 0.72	null
a dialect border	a published border the method never saw	ṡ, beaten by 1 of 200	null
a damaged digitization	a dropped mark, from the paper's own attestation	86 sites, chance 0.000 to 0.136	null

The last column is the distinction to hold onto when reading the rest of this. Three rows belong to the shift agreement detector and three of those three have ground truth somebody else published. The remaining rows belong to the null permutation identity, which carries most of this document and has no positive control with an external answer. Both are the same construction under Proposition 1, and they are not interchangeable as evidence.

On telling a human apart, the refuted claim and the supported one are not the same claim, and conflating them understates what is here. The refutation covers the implication that departing from the null means a person made it. The gaps between primes return 0.93, outside the band every memoryless arm occupies, and nothing authored the primes.

Discrimination against a calibrated non-human reference does survive, and that is a different question with two measurements behind it. Quantitatively the primes depart by 0.07 where the human corpora depart by 0.22 to 0.68, three to ten times further, which puts the two in separable bands. On vocalization at the slice that matches the unit, one symbol every 10 ms of envelope, whale 0.44, dawn chorus 0.45 and wolf 0.49 stand against 0.56 and 0.72 for two human recordings, which separates without overlap where the sample scale had them interleaved.

A non-human corpus makes that possible for the same reason everything else here works. It supplies the constant. Without one there is only distance from maximum entropy, which the primes also have. With one there is a second reference, and the human band is the departure from **that**, not from the flat null.

Which means the signature is read as an inverse, and reading it any other way is reading the center. A detector aimed at humanity directly is aimed at the central value itself, and this document records that attempt failing: departure from the null fires on primes, on whale song and on a wolf howl. Calibrating the non-human instead and taking what is left over is the same move as flattening the pooled counts before looking for a dialect, and as taking the largest value instead of pulling toward the field's center of mass. In every one of those the quantity wanted is the residual, and in every one the failed version went for the mean.

So the object being measured is not humanity. It is the distance from a non-human reference, and it carries a human reading only where that reference is built and stated. What makes the reading falsifiable is that the reference is a corpus somebody can disagree with and replace, which a flat null is not.

The weakness is sample size and it is the thing to fix instead of the logic. The vocalization separation rests on three animal recordings against two human ones, the envelope leaves only 7 to 40 distinct levels so the rare half is a handful of symbols, and this document already says those

figures are thin. Four blue whale recordings from four ocean basins agreeing at 0.25 to 0.43 is the strongest version of it and rests on 4 to 16 symbols clearing the occurrence floor.

A lattice is measured and a protein is measured, and they are not the same result. An earlier version of this entry said no crystal had been read, and that was wrong on this document's own rows. The invariant is measured over 13 geometries including generated lattices, 213,840 occurrences with 0 refused across dimensions 1 to 8 and a rotated point set. The four constructed clouds go further and separate the cases: a lattice carrying values determined by position fires on both nulls, and a lattice carrying random values fires on neither, at 0.690 live against 0.684 and 0.553, because a survivor needs a displacement and a value together and random values supply no repeated pairs.

Real crystals are read exactly, with published ground truth, and no row here is stronger. Published cells from the Crystallography Open Database, handed over with nothing told to the detector (`examples/crystallography/6_oracle/proof_positive_control.py`, catalog `CRY-6-001`), return the published edge on **453 of 453 axes over 151 structures**. Equal as integers, with no tolerance. Sites are carried as exact integers at 10^{-1024} angstroms. There is no tolerance to set and no error to average. It is the only positive control in this work whose answer came from outside it.

An earlier version of this entry reported the same 453 axes as landing *within one voxel* of 0.25 angstroms, at a mean absolute error of 0.0124 and a worst of 0.0554. That error was neither in the deposit nor in the detector. The reader put every site on a 0.25 angstrom grid before the detector saw it, and the figure was the grid. `examples/crystallography/2_partition` isolates it: over 111 axes the grid reading lands inside one voxel every time and lands on the published edge zero times.

A protein and a crystal are the same object to this construction, and the instrument produced the difference in the two results instead of the domain. Everything here arrives as points carrying values. Section 4.2 covers exactly that, and the invariant holds over 13 geometries and dimensions 1 to 8 without an alphabet. A mineral and a fold are both that. So the expectation that what works on one works on the other follows from the construction and is not a hope.

What differs is which of the two instruments was pointed at each. The crystal belongs to the shift agreement detector, the same one that returned an image width of 960 and a Vigenère key length of 8, and it now has external ground truth three times over. The protein figure of 2.512 belongs to the null permutation identity, which carries most of the findings in this document and has still only been shown not to invent structure on memoryless input. This document already records the two nulls on that domain failing to delete separable properties, and local packing as the likeliest reading of 2.512, and not established.

So the protein row is not weak because proteins are hard. It is weak because the instrument with no external positive control was pointed at it, and the instrument with three read the crystal. Pointing the shift detector at a fold is the obvious next measurement and has not been run.

The center-of-mass argument is why it crosses the physical and the conceptual, and it is the entire mechanism. The construction never reads a substrate. It reads the departure from whatever that substrate is indifferent to, and indifference is available in both kinds of domain for different reasons. Physics supplies it in the physical rows: a fabricator accepts any coordinate. The flat reference over a board is a physical law and the 0.389 above it is a person. Convention supplies it in the conceptual rows: a case-insensitive compiler cannot read `ENTITY` apart from `entity`. The

flat reference is the language definition and the 0.495 above it is a community. A protein's bonding rules and a grammar are the same kind of object to this construction, because both are constants that something varies about, and neither is ever read.

Nothing in the method therefore needs to know which kind of domain it is in. A physical constant and a conceptual one are both just the thing to subtract. The rows above are one measurement repeated under different names for the constant.

What the table does support is narrower than a theory of everything and still unusual: the same construction, unchanged, reading text, geometry, images, layouts, sound and language, with the domain supplying only its own indifference. Trimming the two unsupported claims makes the rest of it checkable, and a claim of priority over any row of it is a literature question this document has already been wrong about once, at Montemurro and Zanette.

Posit: the departure has a band, and a master works inside it. Zero deviation from the constant is a copy and carries nothing, because it is the constant. Maximum deviation is noise and carries nothing either, because it is the flat reference itself. Everything readable is between them, and a sculptor whose work holds a viewer is one holding the departure where recognition still fires and has not yet tipped into the uncanny. We see ourselves in it. That recognition is the constant, and the deviation beside it carries what somebody meant.

The measured analog is already in this document and is the image row. Uniform noise returns an agreement of 0.0050, which is 1/256 and therefore chance, and has nothing to find. A machine made ramp returns 0.9980, is perfectly regular, and is trivial to detect and empty to use. Three paintings return 0.02 to 0.30 and sit in the band where a periodicity is both findable and carries a position. The ordering of those three was recorded as an observation about detectability. Under this posit it is the same statement about where information lives, and the paintings are in the band for the reason the sculpture would be.

A sculptor found it without any of this, and the absence of theory is the part worth recording. A sculptor holding that band derived nothing. The instrument was perception with a feedback loop: they could see when it was wrong and adjust, and the optimum is reachable that way by anyone who can see. The theory explains what they found and did not guide them to it.

That pattern is already twice in this document. A VHDL community holds a case convention the compiler cannot read, at 0.495, and computed no entropy to decide it. Designers place 41 percent of their coordinates on a 0.635 mm grid that physics does not prefer, and derived nothing either. In both, a population converged on a departure from an indifferent substrate with no access to the quantity being departed from. Montemurro and Zanette's word ordering figure is the same thing at the largest scale available: bounded near 3.3 bits per word with a relative variability of 0.07 across 7077 texts in eight corpora and six language families, held there by speakers who know nothing of entropy.

A quantity that people converge on without being able to state it is evidence the quantity is real and not an artifact of the instrument reading it. That is the argument the sculpture case would extend, and it is why the case is worth testing instead of admiring.

Nothing here has measured a sculpture, or a face, or the uncanny valley, and the posit is recorded as a posit. What would test it is a departure scale over one form with a known constant, faces being the obvious one, and the prediction is a peak in readable structure strictly between the copy and the noise instead of a monotone rise toward either.

A symbol a bad digitization dropped is recoverable from the distribution, and the document supplies its own answer key. `maint/data/salishan/orior_algorithmic_extraction/symbol_sift.py`. Some of these PDFs drop a combining mark and leave behind the space the typesetter made room for. One word arrives as two tokens. A paper long enough to print a word twice prints it once whole and once broken, and that pair is a labeled example nobody had to supply: Davis and Mellesmoen sets `xʷəlp-í<p>l̓əx` in section 2.2.2 and `xʷəlp-í<p>l əx` in example (23).

Scoring is the radix again. Run counts over the tokens that carry no break are flattened to maximum entropy, and a candidate restoration is scored on how far its runs sit from where a flat distribution would put them. A run common everywhere contributes nothing, and the remainder is the part belonging to that restoration. On this paper the correct Salish restoration scores 160.5 and the one English case, `S kwxwú7mesh` to `Skwxwú7mesh`, scores -20.9, against a rate of 0.120 for a mark drawn at random from the paper's own inventory landing on an attested form.

Inconsistent damage corrects itself and systematic damage does not. That difference decides when this works. A channel that fails sometimes leaves its own answer key, because the word it broke on one page it set whole on another. A channel that fails every time leaves nothing to check against. Davis and Mellesmoen drops a mark on some printings and not others. Robertson collapses every labialization in the paper. The same tool run over it finds six sites and none of them a labialization, and reports that instead of guessing. The negative result is the confirmation: this document says a collapse destroys the distinction instead of disguising it, and here is the instrument agreeing on a paper where the collapse is total.

So the rate of a fault decides whether it is repairable from the text. A fault at rate 1.0 needs the page. A fault at any rate below it is a labeled sample of itself.

Two of that paper's roughly twenty-five damaged sites are readable this way and the tool reports the bound instead of hiding it. A word printed once and broken once has no second printing to check against. It is not recoverable by attestation and still needs the page. The method is exact where it applies and silent where it does not, the same shape as Proposition 1.

The precedent is already here. `font_substitution.py` scores a candidate mapping by how many mutated tokens become forms attested in a clean paper, and it moved attested tokens from 1 of 3599 to 811 on one paper and 2 of 4332 to 965 on the other. That is the same instrument with a coarser score.

Two things that follow and that nobody has run. Both are predictions and neither is measured.

Recovering a dropped symbol means reading the channel that dropped it. A channel with a per-source signature is therefore identifiable from the same residual, and handwriting is such a channel: a scanned manuscript is a distortion of a known alphabet, the distortion is systematic per writer, and a writer's hand is the deviation from the pooled letter distribution. The corpus half of this document already separates the message from its code, at Zipf -1.185 against -1.183 under a permuted code table where relabeling cannot change how often a word recurs, and brevity at -0.652 against -0.550 where at most 0.10 of it is the code's assignment. A writer is a code and the text is a message. This would rest on that separation.

Prosody is the same argument on the sound side, and the field is built. `src/engine/python/representation/sound/perceived_sound.py` keeps prosody separate from the segment field precisely because a field made shift invariant would put a stressed form and an unstressed one at one address, and it holds loudness, pitch against the speaker's own median, how that pitch is moving,

and voicing. Pitch against the speaker's own median is a per-speaker constant subtracted, the center of mass again. What remains in that field is the residual. Whether that residual identifies the speaker has not been asked. The measurement shows the field separating sources at all: whale 0.44, dawn chorus 0.45 and wolf 0.49 against 0.56 and 0.72 for two human recordings, without overlap.

Partial knowledge of the boundaries is enough, and that is Proposition 1 and not a new claim. Knowing that a thing exists and knowing some of its boundaries, not necessarily all of them, is sufficient to discriminate. Proposition 1 says any subset of a pattern's points is a necessary condition, with neither order, dimension nor alphabet. A partial constraint set is therefore still sound and loses no true occurrence. What incomplete knowledge costs is selectivity and never correctness. For rank probes that holds only where both sides are numbered in one population (`src/engine/nbody/orior/orior.h`). This document draws that same division everywhere else. Soundness belongs to the construction and cost belongs to the domain.

That division is why the corpus work can proceed without knowing the phonology. The mark set for a paper is a partial boundary, established one paper at a time and never complete, and it discriminates anyway. A narrower set reads fewer tokens and reports nothing false about the ones it reads, and being wrong in that direction is the safe way to be wrong. Each paper declares its own for that reason.

Where the one winning mechanism breaks, and why it is the same mechanism. The counterfactual credit gives offset a' the advance that the symbol in cell $s + a$ would have bought, while the cell a' would have read is $s + a'$. That is sound only where the symbol distribution does not depend on position. `periodic16` is the only corpus where the field is worse than Horspool across a run of lengths, and the loss peaks at needle lengths 16 and 32, the record period and twice it, falling away on either side. Reading the second cell instead of inferring it removes the assumption and costs the read the counterfactual exists to save. Not measured.

Where to branch between the two arrangements is derivable and lands at the effective alphabet. The greedy in-order shift collects $1/((1 + mq)(1 - q))$ of the available refutation with $q = 2^{-H_2}$. Setting that to a half gives $m = (1 + q)/(q(1 - q))$. The shift stops collecting most of the available refutation at a needle length of

$$\frac{m}{2^{H_2}} = \frac{1+q}{1-q}$$

The right side is the effective alphabet size times a factor constant to the extent q is small. Over the corpora measured: economics 12.89 against 2^{H_2} of 10.68, C source 14.54 against 12.36, Greek 15.05 against 12.89, English prose 15.74 against 13.59 and science 16.38 against 14.23. The ratio runs 1.151 to 1.207 and its variation is $(1 + q)/(1 - q)$ evaluated at each corpus instead of scatter.

The derivation agrees with the measurements already taken. Free order loses at every needle length of 16 or below on English and reaches 1.47 times the in-order reads at 128, against a predicted crossing at 15.74. The same 2^{H_2} sets how many anchors to place, at $\log_2 N/H_2$. One scale governs both the depth of the cascade and the length at which discarding the order stops costing more than it returns.

That makes the branch decidable at run time from two cheap numbers. H_2 is one pass over a histogram of the domain and m is known when the call is made, and a caller can choose between the two arrangements by arithmetic, without running either one, without a benchmark and without reading a symbol for its value. Whether to spend that is a design decision and not recorded here as one.

The advantage between the two arrangements has a minimum and a maximum, both measured. On English at $N = 2789$ the ratio of in-order reads to free-order reads runs 0.962, 0.878, 0.849, 1.028, 1.380, 1.467 and 1.329 at needle lengths 4 to 256. The worst point is at $m = 16$, the effective alphabet 2^{H_2}, and the best is at $m = 128$. An earlier entry recorded 1.47 at 128 as the best result, and it is the best of the lengths tested and not a peak that was located.

An attempt to derive the peak gives $m = r + \sqrt{r^2 + Nr}$ with $r = 1/(1-q)$, which is 56 here. The lengths are sampled at powers of two, which puts 56 one sample below the observed maximum, and the spacing cannot separate a correct prediction from one wrong by a factor of two. The same expression predicts a peak advantage of 31 against 1.47 measured. The cost model behind it is wrong somewhere regardless.

The needle is drawn from the corpus. The benchmark answers an easier question than the one posed. Every search therefore finds a guaranteed occurrence and pays the m read verification floor, which is already recorded here as its own finding. That floor brings the two arrangements together as m grows, and the convergence is the falling side of the curve above. The shape may belong to the needle selection and not to the algorithms. The construction is specified against a pattern that is unknown and of arbitrary width, and such a pattern is almost never present in a given domain. With no occurrence there is neither verification, floor nor convergence. The curve for the stated problem has not been measured. Also, N is 2789 bytes. At $m = 256$ the needle is 9% of the corpus and no asymptotic reading of any of these rows is available.

Correctness is measured in thirteen geometries and speed in one, and that one is the dominated case. The invariant rows cover dimensions 1 to 8, a rotated point set, a complex alphabet and an unordered index set. Every performance row, every comparison against Horspool, every advance and stride measurement, is a one dimensional byte string. That is the single domain where an order exists for a shift to take, which is where this construction is worst positioned, and the domain where it is the only available arrangement has never been timed against anything.

The cost of discarding order, dimension and alphabet is paid on every domain and the benefit arrives only where those were absent to begin with. On a line the cost is paid for nothing. Off the line nothing is given up, because there was nothing there to give. The measurements are arranged so that only the first case has a number attached.

Why the rare anchor loses on a line. An anchor at offset a advances by at most $a + 1$, and that makes $m - 1$ optimal for advance by construction, and advance governs search cost. Where an order exists it is most of the available structure and a shift takes it. Where no order exists there is no shift, and rejection is the only lever there is.

Design-Relevant Experiments

The vectorized cost model, measured and much smaller than derived. The model in Section 4.9.5 predicted 11 to 26 times fewer reads at $m = 256$. Measured, free order runs between 0.27 and 1.47 times the in-order reads, best at 1.47 on English at $m = 128$, and it loses at every needle length of 16 or below. Dependency depth is 2 on every row, as the construction requires.

A long search is verification bound. Every needle is drawn from its corpus. Every search confirms one genuine occurrence at a cost of m reads, and that term was missing from the derivation. At $m = 256$ the floor is 256 reads for any algorithm, Horspool runs at 1.10 to 1.57 times it

and free order at 1.04 to 1.16. An order of magnitude was never available to anyone. Adding the missing term accounts for the measurements to within 3%.

Both propositions are now measurements and not only arguments. An arm that probes and declares without confirming is 6.6 to 20.5 times cheaper and wrong on 32 to 64 of 64 searches on every row where the stride exceeds one. Across all 35 rows the number of searches that over-report equals the number that are wrong: no search in any corpus, at any needle length, at any stride, ever reported fewer occurrences than exist. That is Proposition 1 visible in data. The persistent over-report is Proposition 2 in the same column.

The one directional error is a control signal. A miss being impossible means any discrepancy is an over-count. It is detectable without knowing the answer and it means exactly one thing: the stride is too coarse. The boundary is sharp. English at $m = 4$ moves from stride 1 to stride 2 and from 0 to 61 searches wrong of 64.

A threshold on shift agreement needs a null permutation and not a formula. Tests of one shift share read positions, and a standard error computed as though they were independent is too small. The first run reported twelve findings on SHA-256 output. A shuffle of the same bytes holds the histogram and destroys the positions, and it reports sixteen. The raw count carried no information at all. Only `periodic16` survives the subtraction, at 92 against 0.

The Finnish outlier was a genre confound of mine instead of agglutination. The *Kalevala* is verse in strict meter. Its lines are more regular than its words and the detector finds the line. Finnish prose from 1870 finds the space and returns a brevity of -0.298 against English prose at -0.301. An earlier entry here blamed agglutinative morphology and that was wrong. Agglutination does show, in a mean word of 6.82 bytes against 4.94 and in 0.37 distinct forms per token against 0.11, and neither impairs the detector.

The symbol width was chosen by habit, and a Greek text is where that showed. Reading UTF-8's own framing out of the bytes gives 71.3% two byte sequences against 99.4% one byte for English. The byte level detector was measuring half a letter. Re-seating the 144 distinct Greek symbols one to a byte recovers `U+0020` as the boundary and puts Greek inside the group on every universal. The same re-slicing moves English by 0.001. It is not doing the work itself.

Vocabulary compression separates a translation from a composition. The 1611 KJV and the 1623 Shakespeare are twelve years apart in one language, both capped at 1048576 symbols. Shakespeare uses fewer tokens, 188474 against 199878, and 92% more distinct types, 25031 against 13058. The KJV's small vocabulary belongs to the translation and not to the era.

Exact matching cannot see an inflected formula. Epic register went undetected. The Iliad returned 108 formulas against Shakespeare's 142 and Austen's 128, which is no signature at all. Greek carries 59% more surface forms per token than English at comparable corpus size. Inflection does that to a repeated phrase. Testing the register needs lemmatization, which is not built.

The most regular thing in a text file is almost never the language. Four times the detector returned the format: the line ending of a wrapped file, Gutenberg's own markup, a ruled separator at a gap of 2 and a dispersion of 0.005, and verse numbering at a gap of 816. Every language result required stripping a layer of the file first, and all four were caught by reading the output.

A boundary has to be regular enough to find and irregular enough to locate with. The record corpus reports the tightest spacing measured anywhere, a dispersion of exactly zero, and its boundaries reduce the search by exactly its period and by nothing beyond it at any needle

length. English reports 0.162 and its boundaries locate uniquely from a needle length of 64. The variation in the spacing carries the position. The two ends of regularity fail in opposite ways: perfect regularity is trivial to detect and empty to use, and no regularity cannot be detected at all.

The pattern free detector finds periodicity and no other structure. English prose and C source report nothing above their own shuffles, and English peaks below its shuffle. Both have structure; neither has agreement with itself at a fixed offset. The shift formulation tests that hypothesis and no other.

A stride calibrated on one known pattern does not transfer. The over-count is monotone in the stride. One pattern with a known occurrence count can bisect the largest stride that reproduces it. Applied to 63 unseen needles the error rate runs 0 to 62 of 63. Every row with no errors is one where calibration drove the stride to one or two, which is slower than an in-order search. The calibrated and derived strides agree closely, 98 against 91 and 75 against 84. It is not a mis-set constant. The safe stride belongs to the pattern and the corpus together, and one measurement bounds only the pattern it was taken on.

The failure was not where it was flagged. The section carried a caveat naming the independence assumption, which Section 4.6 measures failing worst on `structured`. Independence held: predicted survivors 1.25 to 3.84 against measured 1 to 11. The omitted term had nothing probabilistic in it.

The accumulation is still a bitwise OR over an alignment bit vector, the survivors are still its complement, and the mask is still m bits wide whatever the alphabet and whatever the dimension. Those are properties of the construction and the measurement does not touch them.

Method: take the better fix

Take the better fix, always. Measure it. Record it. Record the rationale. Douglas's instruction, and it is a rule about deciding and not about any one defect. The opposite can look reasonable at the moment it is chosen.

What the smaller fix was, and why it was tempting. The projection groups a field into classes and labels them rarest first, and the label is a byte. A field with more than 255 classes cannot have them all named. What the code did with the excess was measured: it capped the class count during DISCOVERY, before the rarity ordering ran. The merged set was chosen by arrival order. That direction is the harmful one. A class occurring once has one chance to arrive early and a class occurring nine times has nine. The rarest classes arrive last and were merged first. Two fields with identical frequency multisets and opposite arrangements merged sets whose mean occupancies were 1.06 and 9.00, which no histogram can separate. The rarest class is the best probe the steering has. The code was spending exactly what the projection exists to find, and worst on the most natural arrangement.

Two repairs were available. Refuse such a field outright, which is four lines and fails closed. Or carry the classes in caller supplied buffers so the count runs unbounded and the clamp applies at relabel time, which changes the signature and needs the caller to find memory.

The smaller one was taken first and three arguments supported it. It fails closed, which this engine does everywhere. It is small, and the signature had already changed twice that day in the middle of a verification pass. And it enforced advice already given in three separate places, where every field was the shape that would have been refused.

Every one of those arguments is about the cost of the change and none is about the result. Refusing leaves a capability unbuilt: a field of a thousand classes could keep its 255 rarest apart and get the fast path, and under the refusal it gets nothing. The rule names that directly. A fix chosen because it is cheap to make is chosen on the wrong axis, and the axis it is chosen on is invisible afterwards, because what ships looks deliberate either way.

The two halves after "always" are what make it checkable. Measure it: the better fix is asserted at 400 classes counted and not capped, and at 256 ranks used with the rarest kept apart. The smaller fix would have failed the second of those assertions. Record it: both the defect and the discarded alternative are in `docs/findings-for-verification.md`. A reader meeting the buffers knows what they buy. Without the measurement "better" is a preference; without the record the next person re-derives the same tradeoff and may take the cheap side again.

It also settles a conflict this workbook had no rule for. There was a sound argument for leaving the signature closed, since it had just been verified. A preference about how to work is information, and the process rule outranks it. The rule was followed, and this entry fixes that order in advance.

A rule with no counterweight would be a license to rewrite everything, and the counterweight is already here. Better means better as a result instead of larger. The fix that did NOT get taken at the same time is the evidence: a proposed change from comparing against one representative per class to comparing every pair was measured first, and it was taken only because the representative form was outright wrong on a predicate that is not transitive, not because it compared more pairs. And a separate rework offered in the same hour was declined, because nothing measured showed the current form losing anything.

Criteria for Result Validity

Every quantity here is a departure from a background and none is a value. A dispersion of 0.28 says nothing; the same dispersion against a shuffle of the same bytes is 2.91 and says the positions carry something. A Zipf slope of -0.988 is inside the natural range and is also what independent draws give.

The results that held were measured against a background built out of the data by removing one property: a shuffle that keeps every symbol count and destroys every position, a corpus truncated to a common length, a generator swept over ten parameter settings, a total length held fixed while subjects are added, an observed pair count substituted for two assumed marginals. Such a background cannot be wrong about the property it removes, because it is the same data with that property gone.

The results that failed were measured against a background that was assumed. The product rule assumes independence. The Zipf reading assumed a memoryless process would not reproduce it. The frequent half comparison assumed corpus length did not enter. A mean assumed the ratio was not heavy tailed. The scale test assumed the statistic conserved, and the four order discontinuity it produced was the assumption failing and not the corpus changing.

That also states why the method reaches an unknown domain. A background cannot be assumed for something that is never interpreted. It is constructed from the domain itself and only the difference is read.

Analytical Corrections

Each was invisible until another instrument disagreed, and the count is not kept because it keeps growing. A predicted rate including the needle finding itself. A self similar corpus producing a false trend. A 1.24 control read as a finding when its *z* said noise. The corpus matched table prediction, refuted. A control summarized as a ratio at counts too small to carry one. A control generator changed while its comment still described the old one. A dimension prediction omitting the guaranteed self match, the first entry made a third time. An interpretation naming entropy as what governs a measure's worth, which the corpus ordering contradicts. And a prior art claim asserted from search snippets before either source had been read.

Then, in one stretch of the corpus work: a mean of a heavy tailed ratio quoted as a typical value, which put figures three orders too large into several entries here. An ordering of two structure meters built on those figures, which reversed once the median was used. An entry stating the measurements cannot read a multibyte script, left standing twenty lines below the entry recording the re-slicing that fixed it. Inflection offered as the reason epic register went undetected, refuted by two English epics failing the same way. A claim that prose carries almost nothing below a sentence, which was a claim about the clustering measure written as a claim about the text. Frequent half values compared across corpora of different lengths, which is a comparison the statistic does not support. And a chained prediction reported as recovering two to three orders of magnitude, which was one outlier moving.

Then, in the work that folded the engine into one translation unit, three more.

The dispatch rule's cost was quoted at three and a half times better than it measures, and the figure came from no run. The kernel header stated that the rule "gives up around one percent of the cycles the worst rule gives up". `bench_dispatch` had never executed on Windows, because its cycle counter gate tested `__x86_64__`, a predefine GCC and Clang set and MSVC does not, sending every MSVC build to a POSIX `clock_gettime` that MSVC does not ship. The target failed to compile and nobody read the failure. With the gate carrying the MSVC definition the driver runs, and over 42 rows it names the faster engine 39 times while giving up 9131790 cycles, which is 0.035 of what the worst rule on the board gives up. The one percent corresponds to no run recoverable from this tree. Both the header and `README.md` now carry 0.035 with its conditions.

The steering was reported from one corpus, and the corpus chosen was the weakest of the family. `docs/steering.md` led with 1.072 falling to 1.066 reads per alignment, measured on the AGPL license text, 24654 bytes, one needle length. Swept across the corpus family the same routes give 1.003 to 1.003 on uniform, 1.562 to 1.001 on the synthetic skewed field, 1.877 to 1.067 on skewed and 1.187 to an exact 1.000 on period 16. The license text is the smallest effect in the set, and quoting it alone understated the mechanism on three of the other four fields. It also concealed a distinction: on the skewed field the recursive reorder moves 1.877 to 1.875 while spawning coarms moves it to 1.067. What pays there is the spawning and the ordering does almost nothing. One field cannot separate the two because on the license text they move together.

A read floor row was printed in the route table's column and is not in the route table's units. The empty probe set was briefly reported at 1.000 reads per alignment beside routes whose figures count probe reads only. Its total counts probe reads plus compares, charging one byte per compare. The floor argument requires that charge, and it is not a cost: with no probes every alignment takes a full compare of up to `needle_len` bytes, making it the most expensive route

available. Read down one column it invited the conclusion that not steering matches the best steering. It is reported outside the table now.

Unverified Hypotheses

Each of these was produced by failures recorded above and each would have caught them earlier. None has been tested as a claim. They are written here as posits with the evidence that produced them and what would settle them, because a rule inferred from the cases that suggested it remains a hypothesis until it is tested.

Conjecture: the universe as an unbounded carrier. Let Ω be a carrier assumed to exist and to be knowable only through finite observations. Every subconfiguration of Ω then exists as a member of the carrier's configuration family, but that existence does not by itself provide a finite encoding or an observable boundary. An unbounded n-dimensional object may be queried by another unbounded n-dimensional object only after the relation between their carriers has been specified.

The associated posit is that an approximate representation can conserve a chosen information invariant without bounding the underlying variability. A bound imposed for convenience may inject information from the observer and alter the residual, while an unbounded representation preserves the question's open support. This is not yet a theorem: the invariant, the approximation map, the carrier topology, and the meaning of "conserve" must be stated before the conjecture can be tested. The phrase *consummatio omnino pulchra est* is retained as a motto for the conjecture instead of as evidence for it.

A constructed null deletes one property and it has to be the one being asked about. Proofed and refuted. It came from the protein structures, where shuffling the values among fixed points gave 1.066 and scattering the points gave 2.512 on the same live arm. Four clouds were then built with the answer known by construction, a lattice or a scatter carrying values set by position or drawn at random (`evidence/proofs/posits/proof_null_property.py`). Two of the four predictions hold and two fail: a lattice with random values fires on neither null where the position null was predicted, and a scatter with a pattern fires on both where only the value null was.

The posit assumed the two nulls delete separable properties and they do not. Moving a point destroys which values sit near it as well as where it sits. The position null removes the value coupling along with the geometry and is strictly stronger. The measure also cannot see geometric order alone, since a survivor needs a displacement and a value together and a lattice with random values supplies no repeated pairs.

Something narrower survives and is worth keeping. A null can delete more than one property at once, and a result read as evidence about one of them is unsupported until a null exists that deletes only that one. That is the form the protein reading violated.

Two instruments are in use here and they keep being reported as one. The dispersion of gaps against a null permutation covers text, source code, recorded sound, prime gaps, the memoryless controls and the protein structures. The shift agreement detector covers pictures, crystal lattices, ciphers and the fixed width records. They have different nulls, different validation and different scopes, and three separate claims in this work merged them: a positive control was reported for the measure that did not receive one, a cross media claim was written for a measure that

had not been run on two of the media, and an ordering of structure meters was drawn between them. Each was corrected after the fact. The pattern is worth its own entry because it recurred three times without being noticed once.

One representation for every corpus, reduced by a sum over vectors. Built, run over 273 corpora, and it does not replace the shift detector. Every corpus is written as points carrying values, and each point takes the displacement to the nearest point holding the same value. The lengths of those displacements give a spread, and their directions are summed as outer products, which survive the sign flip that makes a plain sum cancel. On a line the reduction separates: the ten monkey corpora sit at 1.009 against a null of 1.000, and the arranged families sit below it, source at 0.523 and the recordings at 0.420.

Two things are wrong with it and both were measured. Across 273 corpora the length channel tracks the size of the alphabet at rho -0.530, and a cross corpus row in that table is comparing symbol inventories as much as corpora, and none of those rows should be quoted until the scored symbols are matched by occurrence count instead of by rank. Worse, on the pictures the reduction is wrong where the shift detector is right. All three pictures are 960 wide. The shift detector returns 960 for all three with the 1920 harmonic second every time. The orientation channel returns 1324, 760 and 458, and those are the heights: 662 doubled, 760 exactly, 1374 divided by three. Reshaping a picture at a wrong width shears the rows into diagonals, and an orientation tensor scores a shear highest. The channel finds its own reshaping artifact before it finds the picture. The two instruments do not agree, the one that was supposed to subsume the other fails the case the other passes, and the picture width result stays with the shift detector.

The volume has no dimension to assign, and assigning one was the defect behind the picture failure. Giving each domain a dimension meant choosing a picture's width and then measuring the choice. Built again in bit space instead, where nothing is chosen: each symbol Gray coded to make a step in value one bit of distance, the bits laid end to end, and a window of n bits read as a point in binary n space. The reduction is a sum of outer products of those windows, the correlation of the n bit positions, and the statistic is how far its eigenvalue spectrum sits from the even one against the permuted null.

The sum over n does not converge, and the failure to converge is the result. Every arranged corpus is still climbing at 64 bits, the largest width swept: the picture reads 0.0066 at 8 bits, 0.2535 at 32 and 0.3677 at 64, with the peak sitting on the last width measured. A ceiling would therefore decide the total. The quantity that does not depend on it is the exponent of the growth, near 1.0 to 1.2 for the picture, the byte corpora, source and recorded sound. The memoryless corpora are the control and they stay flat, and that separates real correlation at length from the fact that an estimated correlation matrix grows lopsided with its size on its own. That bias would lift every corpus alike and it lifts none of them.

The numbering given to the symbols was suspected and cleared, except on pictures where it is the point. The corpora reading strongest were the ones whose bytes carry a real order. The reading was expected to be an artifact of numbering the rest here. Renumbering the symbols at random, which leaves every frequency and position untouched, refuted it: source, recorded sound, English and Greek all sit inside the spread of their own renumberings. Only the picture moves, 0.2535 against 0.0246 renumbered and 20.6 standard deviations out.

The two instruments are not measuring one quantity, and that part is settled by construction and not by a number. The gap measure reads only where each symbol falls. Renaming the symbols cannot move it, and it returned identical values to four decimals over six corpora under renumbering, and that check confirms the script is sound. The bit volume moves. One instrument invariant to a change the other is not cannot be a replacement for it. The attempt to have the volume subsume the shift detector and the gap measure was the wrong shape from the start.

The explanation offered for that split was wrong and its own test refuted it. The account was that the volume reads structure carried as nearness in value. Any domain with ordered values would move under renumbering while symbolic domains held. Recorded sound is as ordered as a grayscale level and it does not move, sitting 1.5 standard deviations from its own renumberings against the picture's 20.6. The picture is the only corpus measured here whose reading depends on the numbering, and what singles it out is not known. The neat account was written before the sound column was read.

What singles those corpora out is how long they stay near a value, not how small their steps are. Widening the set put all three pictures and humpback song on the moving side, at 278.2, 20.6, 16.0 and 8.1 standard deviations, with human speech, source and text holding. Nearness between neighbors does not separate that: speech sits at 0.54 and whale song at 0.48, both far closer than chance, and only one of them moves. Counting instead how long a corpus holds inside a band around where a run started, with the band set as a share of each corpus's own spread, does separate it. Vermeer reads 4.11 times the shuffled figure, Hokusai 2.56, whale song 1.77 and Starry Night 1.56, then a gap, then speech at 1.38, source at 1.11, Greek at 1.02, English at 1.01 and the memoryless control at 1.00. Every corpus above the gap depends on its numbering and every corpus below it does not.

Speech is the case that decides it, and the reason is physiological. Articulation never stops moving. Speech has the small steps and none of the stays, while a held whale call and a flat region of a painting both stay. The measure predicts which side a corpus falls on and not how far it goes: Starry Night has the largest dependence in the set and the smallest stay of the four that move.

Two defects were found in the measuring and are recorded because both produced confident wrong numbers. A band fixed at eight levels is wide for a corpus using few of them and narrow for one using many, which read Starry Night as having no stays while it has the strongest dependence, and read a corpus of eight symbols as one run covering all 120000 positions. Separately, a distance in standard deviations is meaningless where the spread is near zero, and that accounts for all of the memoryless control's 3.3 and part of Starry Night's 278.2.

Abruptness orders the magnitude inside art and inverts outside it, at rho 0.786 over seven paintings, which is not enough paintings. Stays say which side a corpus falls on and Starry Night showed they do not say how far, having the shortest stay of the movers and the largest dependence. Sharpness of the crossings between stays was the candidate, and four more paintings were fetched to test it before the numbers were seen. Two predictions held: the pointillist surface is the most abrupt of the representative paintings at 0.1095 and the most dependent at 302.9, and the flat regions with hard edges give by far the longest stay at 20.43 with the lowest abruptness at 0.0064. Two failed, and they failed because the prediction came from art history instead of from what a digitized picture holds. A dark canvas covered in small bright sparks was

called the smooth extreme and measured as the most abrupt thing in the set at 0.1709, since a dark photographed surface carries noise in its shadows and noise is abrupt.

The relation survived the predictions that did not, at rho 0.786, which for seven points sits on the conventional threshold and settles nothing on its own. What is clear without any correlation is that the quantity does not cross families: English text is the most abrupt corpus measured anywhere at 0.4903 and does not depend on its numbering at all. Abruptness orders magnitude only among corpora already inside the smooth regime, and every painting measured lies between 0.0064 and 0.1709 while speech is 0.2236 and text is 0.4903. Great contrast in a painting is still smooth against a page of prose.

The high frequency part of a painting was called camera noise and it is not, which reverses the conclusion drawn from it. Separating the two by a median of every three raised the ranking from 0.786 to 0.821 and left a quantity that ranked it better still at 0.929, and that quantity was reported as the grain of the photograph, making the whole ordering an artifact of how each canvas was shot. Averaging is the test that settles it. Anything uncorrelated falls as one over the square root of the block, and not one of seven paintings falls that way. Starry Night reads 12.11 at block one and 13.66 at block sixteen where noise would give 3.03, Vermeer goes 0.61 to 1.55 against 0.15, and every other painting holds flat or rises. The high frequency part has power at every scale. A painted surface and a natural scene have that and a sensor's noise does not.

The exponent of the power against frequency is the first quantity here that holds still, and every earlier statistic had a scale inside it. A window width was chosen, a ceiling was put on an unbounded sum, a band was fixed in levels, and a distance in standard deviations was read as an effect size while its denominator narrowed with every extra byte read. Data carrying structure at every scale has no characteristic scale for any of those to sit at, and each one had to be corrected once its scale showed. What does not belong to a scale is how the quantity changes with scale, and over crops holding each picture's proportions the exponent moves by 0.03 to 0.07 across a change of two and a half times in area. Measured over the plane it fits a power law at 0.97 to 0.99, against 0.42 to 0.67 when the same picture was read as one long line.

The values place the paintings where the literature on natural scenes places them. Starry Night 1.94, Whistler 2.00, Hokusai 2.07, Seurat 2.04 and Turner 2.21 sit at the value a natural scene gives, while Vermeer 2.87 and Mondrian 3.41 sit above it, where flat regions with little high frequency content must land. Text is 0.34, source 0.31 and the memoryless control 0.00, the white noise value that control has to return.

A plane goes through one dimension when the coordinates are interleaved, and the exponent halves. Every measurement on a picture here has been handed a width, and the single instrument failed on pictures in the first place because a file stored row by row puts vertical neighbors a whole width apart, and a reader that is not told the width cannot see the second dimension. Interleaving the bits of the column and the row removes the need to tell it: neighbors in the plane become neighbors in the index, and the index spends alternate bits on each axis. Read along that index the exponent comes back at 0.475 of the plane's, with a spread of 0.064 over seven paintings, matching the one half a curve filling a plane has to give, since a distance along it goes as the square of a distance across the picture. Doubling it recovers the plane to 0.04 on Seurat, 0.03 on Starry Night and 0.08 on Whistler.

The squashing works at any number of dimensions, and it was checked against fields whose exponent was put in by hand. Every other reading behind it is of a painting, where the

plane is the only authority on what the answer should be and there is nothing independent to catch a wrong one. Shaping white noise in the frequency domain gives a field built to a chosen exponent. The answer exists before the measurement. Asked for 2.00 the reading over the set returns 2.044, 2.005 and 1.958 in two, three and four dimensions, and asked for 3.00 it returns 2.968, 2.974 and 2.970. That is a positive control with a known answer for the spectral reading, which had none before, and it says nothing about the gap measure, which still has none.

Read along the interleaved index the exponent comes back at 0.469, 0.285 and 0.185 of the set's, against the one half, one third and one quarter that a curve filling n dimensions predicts. The form is right and every reading is low, with the shortfall growing as the dimensions do at 0.031, 0.048 and 0.065. Multiplying back by n undershoots by 0.335 on average and by 0.507 at worst. Storage order gives 0.429, 0.209 and 0.170 and holds no such relation, the control failing as it should.

One cause covers the bias and the pictures at once. The curve jumps whenever it crosses a boundary between blocks, the share of those jumps grows with the dimensions, and quantizing to eight bits lays a white floor over everything. Both add power at high frequencies, both flatten the slope that is read, and both cost more where the slope is steep. The fields asked for 3.00 therefore miss by more than the fields asked for 2.00 at every dimension.

Under half of that shortfall belongs to the curve, and one dimension cannot hold the rest. The two separate because a Hilbert curve never jumps, its consecutive positions always being neighbors in the set, and interleaving lacks that property. Measured along both on the same fields the shortfall falls from 0.110 to 0.063. The jumps account for 0.429 of it, and the remaining 0.571 survives a curve with no jumps to blame. That residue is the cost of the folding itself. It grows with the dimensions, 0.037 in two and 0.071 in three at the same exponent, and it grows with smoothness, 0.037 to 0.065 as the field steepens from 2.00 to 3.00, and a smoother set in more dimensions loses more of itself on the way down to a line.

A line through a set carries how many dimensions the set has, and it took two failures to measure it. The count sits in the index because interleaving hands bit zero to the first axis, bit one to the second and bit n back to the first, and a step of exactly two to the k crosses a bit belonging to axis k modulo n and the roughness at that step belongs to that axis. There is one magnitude per dimension and they exist only at the exact powers of two.

The first attempt looked for a repeat spaced n doublings apart and sampled eight times per doubling, which sampled away the only steps that carry anything, and it returned the same period for every field at every dimension because that period was the lowest the search allowed. The second read the powers of two correctly and scored them by how consistently they grouped, which returned the largest candidate offered in all twelve readings, since splitting few readings into many groups raises that score on its own.

Under both of those sat a worse error. The fields were built isotropic. Every axis had identical statistics, and there was one magnitude repeated n times instead of n magnitudes. No line can count what was never made different, and the first two attempts were asked to do exactly that. Rebuilt with a different correlation length along each axis, and scored against shuffles of the same readings so the group count cancels, eleven of twelve readings return the dimension count with nothing supplied but the line.

The exponent identifies a family and rebuilds nothing, which is this document's own first proposition turning up as a measurement. Every quantity refined here reduces a set to a few numbers, and whether that is a dense representation or only a description turns on whether

the set comes back from it. A field built to a painting's own exponent alone correlates with that painting at 0.079, -0.058, 0.028, 0.126, -0.197, -0.044 and 0.073, which is one number against six hundred thousand and no picture at all. Keeping every amount in the transform and discarding where each sits does no better. Keeping where each sits and discarding the amounts returns 0.41 to 0.46 on the representational paintings, and costs as much as the set.

The split is the point. A cheap invariant is a necessary condition. It can reject quickly and can never reconstruct, and the irreducible exact compare of the second proposition is here measured in numbers held: one identifies the family, the whole set is needed for the member, and nothing was found in between. A summary that could do both would refute the propositions this work rests on.

Saying that where each frequency sits costs as much as the set was too strong, and it confused needing every angle with needing every angle exactly. An angle is knowable and can be held coarsely. Rounded to eight steps around the circle, which is three bits, and with the amounts kept, every painting comes back at 0.97 to 0.99, and four bits reaches 0.99 and above. Nothing here needs an angle at full precision.

The accounting still lands where the propositions put it, for a reason the second table gives. Three bits of angle with the amounts flattened to one number reaches only 0.43. The amounts are carrying real weight and have to be stored too. A picture of N samples at eight bits has N halved independent coefficients. Three bits of angle each is one and a half bits per sample and cheap, while the amounts beside them bring the total back near what the picture cost to begin with. The result rests on the scaling: one number rebuilds nothing, the whole set rebuilds it, and nothing in between was found. What does not survive is the claim that the angles are the incompressible part, since they need three bits and not thirty two.

Each rebuild is a single draw and the amounts column carries one reading of 0.585 that should not be trusted without repeats. Flattening the amounts to their mean is a harsher test than the published one, which judged whether a picture stayed recognizable. The positions column understates what positions carry.

Text has no approximate representation at all, and the three bit result on pictures does not carry to it. Held to three steps of angle a painting comes back at 0.97 and is the painting. Text at the same three bits comes back at 0.975 and returns 9.7 percent of its characters in Austen, 5.6 in the Iliad and 9.1 in source. Every symbol needs eight bits of angle, which is no saving over the text itself.

The correlation is worthless here and the two columns show why. At each depth it reads 0.636, 0.900, 0.975, 0.994 and 1.000 for every corpus measured including the memoryless control, because it describes the rounding and not the corpus. A sample of a picture landing two levels off cannot be seen. A letter landing one code point off is a different letter. The share of symbols returned exactly is the only measure that means anything and it is far behind.

Quoting the coefficient in levels compares alphabets and not languages, and dividing by the range each alphabet occupies renders them comparable. There is no fixed width to measure a spread against. An alphabet of 3150 symbols needs about 11.6 bits and one of 90 needs 6.5. The same count of levels means different things in the two. In levels, 21 languages run from 5.21 to 149.32 with the spread between languages 17.83 times the spread inside one, and Chinese sits 150.2 deviations outside. As a share of the positions each alphabet occupies the same 21 run from 0.0507 to 0.0790, the ratio falls to 1.36, and Chinese sits 3.2 deviations out at 0.0855.

Japanese is the case that shows the normalizing is doing real work and not flattening everything. It carries 2364 symbols in a mixed logographic and syllabic script, reads 149.32 in levels, and lands at 0.0630 as a share, which is inside the alphabetic band between Hebrew at 0.0507 and Catalan at 0.0790. Chinese carries 3150 symbols and lands at 0.0855, above every alphabetic language but only 8 percent above the highest of them.

They come close and they do not converge. A ratio of 1.36 means the spread between languages still exceeds the spread inside one. A language keeps a value of its own, and 21 languages across Germanic, Romance, Slavic, Uralic, Celtic, Hellenic, Semitic and Japonic with one constructed language among them is enough to say that much.

Which symbol follows which identifies a language where every scalar failed, and its mistakes are the language families. The scalars were the wrong shape of thing to test: each summarizes the alphabet and discards the context, and the context is which symbol follows which, which is a square over the alphabet and not one number. Taking that square, with positions given by frequency rank so the first position is whatever a text uses most and a Greek square is comparable to a Japanese one, and holding each text out in turn: 52.8 percent over 31 languages against 3.2 for guessing, and against 24.5 for the best scalar. Eight ranks already reach 42.5 and the gain from there to sixty four is small. Most of it sits in how the commonest handful of symbols follow one another.

Whether the information in a text closes at a finite order is not settled here, and the reading that appeared to settle it was measuring its own sample size. Written as the decomposition it names, the question is exact: the entropy of a block of n symbols is the sum of n conditional entropies by the chain rule, each one what that order predicts that the orders below it could not, and the object closes if those increments reach zero.

Measured over 400000 characters against a shuffle of the same text, the increments rose to order 4 and fell, and that was recorded as closure at a finite depth. Measured again over six million characters they rise to order 5, and the value there is larger than order 4 held before. The peak moved with the amount of data. A measurement running out of samples does that and a text does not. Orders 7 and 8 return values like -17.9 and 13.6 and one outright nan. Nothing past order 6 is being measured at all.

Counting blocks cannot answer this at any size reachable here, since the possible blocks grow as the alphabet raised to the order while a text grows in a line. The instrument for it estimates an entropy rate without counting blocks, as the compression based estimators do, and that has not been run. The claim that the information is held at low order is withdrawn, and nothing replaces it.

What a text gains at each distance describes the text and not the language, and it is twice as much a property of where the text came from. [The measurement is sound and the wording is not. It shows the curve carries the source more than the language, not that it carries nothing of the language, and a ratio of 2.064 with matching at chance is consistent with a language signal buried under a larger one.] Since no single window holds a text, the sweep gives a curve instead of a number, and that curve was worth testing as a description of a language. On the question that gave the character web 0.926, one language from two places against two languages from one place, the curve gives 2.064, and it matches its own language elsewhere 2 times in 67 against a chance of 3.4 percent, which is chance.

The curve carries nothing about language and it is not directionless. A ratio above two says it belongs to the source, twice over. It reads how repetitive a text is, and that is subject matter: scripture built on refrains compresses far better than a novel does in the same language. The curve gathers one work's translations together across every language they are written in. The character web holds where this fails because how characters follow one another belongs to a writing system and stays put across whatever the texts are about.

The two instruments have separated by what they read and not by how good they are. One reads the language, the other reads the text. That was not the expectation going in, which was that a reading with more in it would be better at everything.

The encyclopedia texts fell out of this comparison entirely, since the shortest of them is under the 300000 characters the sweep needs. The comparison ran on books against the two translated works only.

Read by matching instead of by counting, the text has not closed at a quarter of a million characters. Counting blocks cannot see past order 4 because the possible blocks grow as the alphabet raised to the order. Matching has no such wall: at each position the shortest string that the window behind it has not already seen is found, and a dependency of any length shows up as a long match and nothing needs to be seen many times for the reading to work. Sweeping the window makes how far the text reaches the result and not an assumption.

In all six texts the distance between the text and its own shuffle grows at every window out to 262144 characters. Vietnamese goes from 2.19 to 2.78 bits, Russian from 1.49 to 2.37, Urdu from 1.24 to 1.99, Turkish from 2.06 to 2.59, Indonesian from 1.57 to 2.28 and the Arabic from 3.09 to 3.41. Nothing settles. Where counting could see four symbols, this sees a quarter of a million and is still finding something.

The shuffled rows are why that can be said. They rise with the window, from 3.71 to 4.19 bits, and a shuffle holds nothing beyond one symbol. The estimator produced that rise and the text did not. Both arms carry it equally, which spoils the absolute rates and leaves the widening distance between them standing.

That null was the wrong one for the question and the right one gives the same answer. Shuffling a whole text destroys sequential dependency and also destroys the fact that a book is not one distribution, since chapters change subject and characters speak unlike the narrator. Entropy is concave. One pooled distribution over a whole book holds more than its parts hold on average, and a matcher finding long matches near a position could be finding that a page shares vocabulary with the page before it, which is a book changing subject and not a dependency reaching across it.

Scrambling the letters inside blocks keeps every block's composition exactly and destroys only the order. Dependency comes through that, and it survives entirely. Against a full shuffle at 4.1848 bits, Vietnamese reads 4.0782 scrambled inside 256 characters, 4.1712 inside 2048 and 4.1043 inside 16384, while the text itself reads 1.3812, and every other text does the same.

Why it does not bite is worth keeping, because it is a fact about the unit and not about the argument. Letter frequencies barely move within a book: a chapter about a ship and a chapter about a trial use nearly the same alphabet in nearly the same proportions. Word frequencies move a great deal. The concern is sound and its size at this unit is nothing, and at word level it would not be.

The Arabic original reads 0.24 to 0.35 bits per symbol against 1.2 to 1.9 for every translation beside it, the reading a text built on refrains and repeated constructions gives to a reader looking for repeats.

None of this is a separate technique from compression. The entropy rate of a source is the rate an ideal compressor reaches on it. How much structure a text holds and how well it compresses are one quantity under two names. The instrument that compresses sees what the instrument that counts cannot.

Where the reading stops working is now measured instead of guessed, because the shuffle has an answer that is arithmetic. Its symbols are independent by construction, and a block of n of them holds exactly n times the entropy of one, and the gap between that and what the estimator returns is the error itself, exactly, at every order. Divided by the bins seen over the samples taken, an error made only of undersampling has to be one curve for every text.

It is one curve, and then it is not. At order 3 the eight texts read 0.93, 0.96, 1.02, 1.02, 0.93, 0.94, 0.99 and 1.01, and at order 4 they hold the same, matching the analytic prediction to within a few percent. From order 5 they leave it together, reaching 1.15 to 1.90, then 1.94 to 2.98, then 3.29 to 4.43. Thai leaves earliest and Thai has the least text, 1.1 million characters against six million, and undersampling behaves that way.

So the estimator is sound through order 4 and outside its range from order 5, and that boundary belongs to the arithmetic and not to any language. Inside it the increments are still rising, 0.52 then 0.86 then 1.07 for Vietnamese and 0.49 then 0.86 then 1.12 for Urdu, monotone with no turn. Both earlier peaks, at order 4 and then at order 5, sat outside the range where the reading means anything. What can be said is that the information has not closed by order 4 and nothing here sees past it.

This also replaces the correction that was being applied. Subtracting a shuffled text's increments from the real ones mixes the error with whatever the shuffle's own estimate did, and subtracting a value known by arithmetic does not.

A second reading pointed the same way and is just as weak: identification peaked at order 2 and fell off through order 7, which was read as the language living at low order. Distances between longer readings degrade as the count of numbers grows whatever those numbers hold, and that was not controlled for.

A dirty corpus does not give a slightly wrong answer, it gives a confidently inverted one, and it has happened twice here. A Greek to English lexicon, 21.2 percent Greek letters and 78.8 percent Latin, was one file of four and stood inside every Greek reading in this work until a first line was read by accident. Then seven Dravidian and Indo Aryan files were fetched carrying untranslated English in shares from 1.8 to 30.0 percent, and a family came out inverted: the pair that separated most recently read as the widest apart of seven, the family sat further from itself than from its neighbors, and three of four languages had a language outside the family nearest them.

Thirty percent contamination did not give a thirty percent error. It gave the opposite of the answer, and then three explanations were built and tested on top of it, none of which could have worked. Cutting each file to its own writing put the family back with every other setting unchanged.

The check that catches this was written earlier in this work, after the lexicon, and was recorded as worth having. It was not run on the next seven corpora fetched. So the lesson is not that corpora

should be audited, which was already known and written down here, but that a check one has to remember is a check that does not run. The check belongs where a corpus is loaded and not in a tool of its own.

The best test here was assembled out of the failures before it and could not have been designed first. Zulu against Xhosa closes every escape: a close pair, one script, one text, one set of translators, and a failure would have meant the reading does not follow language at all. Every one of those constraints came from something that had already gone wrong. Tamil and Malayalam supplied the need for one script. Uralic supplied the need for a shallow relationship instead of a deep one. The Chinese pairs supplied the need for one collection. The test is the leftover after the exclusions, and there was no way to reach it except through them.

Whether a failure teaches anything depends on what it removes, and each of these removed something the next one needed. That is worth stating because the same sequence read as flailing while it was happening and reads as a search afterward, and only the second reading is right.

What improved alongside it was the working method and the two should not be confused. The first three explanations offered for the Dravidian inversion were all wrong, and one of them was a confound the instrument is built to be immune to, proposed without checking. The predictions after that held, and what changed between them was not only that the space had narrowed: the files were being looked at before theories were built on them.

Every bound put on a quantity in this work has had to be taken back off, and the reading improved each time it was. This is not a mood about the work, it is the same defect found ten separate times in one sitting, each with a number attached.

A dimension was assigned to each domain and the picture reader returned heights instead of widths, and the cure was to assign none. A sum over window widths was stopped at 24 and the excess turned out to still be climbing at 64. The total was a property of the ceiling. A band was fixed at eight levels and read a picture spread over 160 of them as having no structure, and a corpus of eight symbols as one run covering everything, and scaling the band to each corpus fixed both. A spread was quoted in levels and put Chinese 565.9 deviations outside every alphabetic language, and quoting it as a share of the range brought it to 3.2. One code width was chosen for the binary reading and reading every width at once took it from 40.0 to 45.7 percent. A picture was measured at one resolution and the ordering moved with the resolution until crops holding the proportions settled it to within 0.03. A distance in standard deviations moved one painting from 20.6 to 122.8 on nothing but how much of the file was read, and a ratio with no narrowing denominator held still. A period was looked for at powers of two when the period was three, which no data could have satisfied. A split at 1810 was chosen by hand and the result rests on five books against nine. English was written into a corpus fetch and the resulting narrow range of dates was then reported as a property of cookery.

The pattern is the same every time. A bound is chosen because something has to be chosen, the measurement returns a number, and the number describes the choice. What works is not choosing better. It is sweeping the quantity and letting the data say where it stops mattering, which also turns a wrong ceiling into a visible result instead of a silent one: the excess that would not stop climbing is how the divergence was found.

The reading belongs to the language by a margin of about seven percent, and this bounds every language result in this document. Several languages are now held from four unrelated places: books, encyclopedia articles, and two separate works each translated into many

languages, sharing no subject, century, translator or kind of writing. That makes the question answerable without argument. One language read from two of those places sits 0.0867 apart. Two languages read from the same place sit 0.0936 apart. The language wins, and it wins narrowly.

Named directly, 44 of 100 readings find their own language somewhere else before they find any other language anywhere, which is seventeen times what guessing gives and is real. Some of the misses are relationships and not errors, Malay matching Indonesian being two names for nearly one language, and others, Chinese matching Hindi, are nothing.

Every earlier figure in this section was measured inside one source, and that arrangement hides this. Reading a language from four books of the same catalog and calling the agreement a property of the language assumes what this measures, and what it measures is that where a text came from carries almost as much as what language it is in.

Being a procedure leaves no mark here that being formally written does not already leave. [Overclaimed as a negative, and doubly so. The reading is a character web, which is a statement about neighboring letters being asked whether a text is instruction, and a reader tells a recipe from a novel at a glance. What was shown is that this reading does not separate them, on two texts of source code.] Four bands were gathered to test it: source code, which is a procedure carried out exactly; instruction written for rulers, which is procedure at the scale of a campaign and reaches back to Xenophon and Caesar; cookery, which is procedure at the scale of a kitchen with its measurements no longer readable; and narrative, which is not procedure. If procedure marks a text on its own, the two informal bands sit between the code and the narrative.

They sit with the narrative. Instruction for rulers falls 0.75 of the way from code toward narrative and cookery 0.66, and more decisively the bands do not separate at all: books of one band scatter among themselves by 0.0783 and 0.0818 while the centers of those bands sit 0.0321 and 0.0379 from narrative. Only source code is its own band, scattering by 0.0466 against 0.0722 to the nearest other, and the notation marks it and not the procedure, since the two informal procedures do not go with it.

Two limits on that. Source code is two texts here, which is too few for the one band that did separate. And the reading is a character web, which is a statistic about neighboring letters being asked a question about how a whole book is arranged, the same mismatch that made the earlier shuffle test return nothing.

Every negative result in this section is ambiguous in a way the positives are not, and they have all been written as though they were not. A reading that finds something has found it. A reading that finds nothing has established only that this reading did not find it, and that is compatible with the information being absent from the text and with the information being present and needing a competence the reading has none of.

The two are not distinguishable here and they have been reported as the first. Japanese authorship scores 19 percent by the square, and a Japanese reader is not in any doubt about which writer they are reading. Finnish and Hungarian read as unrelated and a historical linguist establishes the relationship from evidence a character stream does not carry. The participants in a conversation know exactly who is saying what, and this reading of the same words is clueless.

That bounds the phrase used throughout this work more tightly than the literacy limit does. What can be seen without context is that a text is language and roughly what kind of language. Almost everything a competent reader knows at once is opaque to it, and the failures above cannot be told from that opacity.

The oldest continuous language traditions on earth are outside this work entirely, and two separate things put them there. Everything here reads text. Aboriginal Australian languages carry tens of thousands of years of unbroken oral tradition and no writing. They are not underrepresented in the corpus, they are absent from the method. Whatever a reading of written language finds, it has never been asked of the longest continuous case there is.

The second reason is not about availability and should not be filed as though it were. In many of those traditions the language is sacred and inseparable from law and country, held under restriction by initiation, by gender or by clan. Taking such material because a repository serves it would breach protocols that exist for reasons unrelated to what a license says.

There is also a plain mismatch between the method and the object. This work reads text as pattern with the meaning stripped out, deliberately. That stripping let it treat an undeciphered script and a protein chain with the same instrument. Where a language is a religion, that operation does not miss part of the thing, it mistakes what the thing is, and the numbers would be about something else.

Everything measured in this work is the writing of people who were taught to write. A cookbook of 1390 is a king's cook's, a manual on campaigns is a general's, advice for princes was written for a prince, and the survival of any of it selects for the powerful before it selects for anything else. Common thinking rarely reached a book at all. Whatever these readings find, it is a property of literate production and not of people, and no measurement of a corpus can reach a majority that left nothing in it. This bounds every claim in this document that uses the word human.

Matched across two works that share nothing, the family comes through and the language does not. Every identification before this compared texts of one kind to each other, four books of a language or four pieces cut from one translation, and pieces of one translation resemble each other for reasons that are not the language. Two different works rendered into the same 22 languages remove that: a language's reading from one is matched against every language's reading from the other, with neither sentence, topic nor translator in common.

Read as characters it matches its own language 9 times in 22, which is 40.9 percent against 4.5 for guessing, and the binary alphabet manages 7. Setting each language's own reading aside, 10 of the 14 with a relative present sit nearest that relative, and the errors are why the two numbers differ so much: Dutch matches Norwegian, French matches Spanish, Czech matches Russian, Albanian matches Italian. The reading finds the right neighborhood and then picks wrongly inside it.

That inverts what one work gave, where identification ran high and the families came out weak, and this is the direction to trust. Within one work a book can carry a match. Across two there is nothing left to carry it except the language, and what carries across is family sized.

Every cross language reading in this work compared different books, and holding the content fixed turns out to cost the family result almost nothing. Greek was epic poetry, Dutch was a novel, several were encyclopedia articles and one Greek text was a dictionary, and a difference between two languages was also a difference between two books. One text translated into 43 languages removes that by construction, and it reaches Vietnamese, Tamil, Malayalam, Telugu, Hindi, Marathi, Nepali, Gujarati, Burmese, Thai, Korean, Armenian, Albanian, Turkish, Latvian and Estonian, which no book catalog here carried. Urdu is not in it.

Over the same 17 languages the books also hold, the fixed text puts 10 of them nearest a relative against 15 of 22 from the books, which at those counts is not a difference worth claiming. So the

families were not being carried by shared topic, and the earlier result survives the control that was expected to weaken it.

What does cost is breadth. Over all 33 languages with a relative present it is 18, and the misses stop being sensible: Afrikaans goes to Finnish, Hebrew to Hungarian, Italian to Slovenian. Telling Germanic from Romance among European languages that have borrowed from each other for a thousand years is an easier question than placing Dravidian against Indic against Tai, and most of those families arrive here with two members and no contact with the rest.

Identification reads 83.1 percent on the fixed text against 52.8 on the books, and that comparison is not available. One text per language has to be cut into pieces to give several samples, and pieces of one translation resemble each other far more than four separate books do. The number is inflated by the method and not by the corpus.

One binary alphabet carries every script at a small cost, once the codes line up with the characters. Taking positions by frequency rank makes alphabets comparable and still leaves each language in an alphabet of its own size, three thousand symbols for Chinese against eighty for Welsh. A rank divided by the size of its alphabet says how far through that alphabet a symbol sits, the same quantity everywhere, and held to a fixed count of bits it becomes a code from one shared alphabet with nothing dropped for falling past a cutoff.

The first attempt at it returned 12.4 percent and put 2 of 22 languages nearest a relative, against 52.8 and 15 for the characters read directly. The fault was not the idea. That attempt slid windows of three to six bits along a stream carrying ten bits per character. Every window mixed the end of one character's code with the start of the next and measured neither. One code to one character returns 40.0 percent at 32 codes and 13 of 22, and the confusions stay sensible, with Polish, Hungarian and Esperanto all going to Czech.

So the universal alphabet costs about 13 points of identification and two families against reading the characters as they come, and what it buys is that it needs nothing said about any script, drops no symbols, and puts a logographic language and a Celtic one in the same 32 codes.

Comparing the binary webs at one width compared different resolutions against each other, and reading every width at once recovers half of what that cost. A width means different things in different languages: 32 codes hold about a hundred Chinese characters each and about two and a half Welsh ones. Two languages read at one width are read at two resolutions and only one of them is fine. Reading every width and laying them end to end takes identification from 40.0 to 45.7 percent and the nearest eight from 73.3 to 76.2, closing the gap to the characters from 12.8 points to 7.1.

The levels are joined as they come. Each is already shares of one text and so sums to one before anything is joined, and scaling them to a common length first was tried and is wrong: it makes a coarse level weigh the same as a fine one when the difference between them is the whole reason several widths were read.

What it does not fix is the families, which stay at 8 of 22 against 15 for the characters. That points at where the binary alphabet actually loses. Telling two languages apart needs the shape of a distribution, which binning by rank keeps. Knowing that two languages are related rests on the particular letters they share, and binning three thousand characters or eighty into 32 codes merges exactly those. The universal alphabet keeps what separates languages and loses what connects them.

It was reached for as a first pass and it is not sound enough to be one. A cheap reading before an exact compare is only worth anything if it never discards a true match, which the first proposition here states outright, and a stage that drops truths silently makes every stage after it wrong. Measured as a filter over 31 languages it keeps the right one 40.0 percent of the time at the nearest, 53.3 at the nearest three and 73.3 at the nearest eight. Keeping a quarter of the field still throws away better than one true match in four. That is a classifier of middling accuracy and not a necessary condition, and the difference matters more than the accuracy does.

The two things it might be for do not peak together either. Five bits identifies best at 40.0 percent and puts 8 of 22 languages nearest a relative, while seven bits identifies worse at 36.2 and puts 13 of 22 nearest a relative. Naming a language and grouping languages are different objectives and a width chosen for one is not the width for the other.

Of what separates two German books, the author is worth 11 percent, the century 1 percent, and the book itself 88. Measured over 1244 texts by 608 authors, 268 of whom have more than one work here, with every text cut to one length. Two works by one author sit 0.0737 apart. Two by different authors of one century sit 0.0824. Two sharing neither sit 0.0833. So sharing a century is worth 0.0009 and sharing an author 0.0088, and the floor between two works by the same person is nearly the whole distance between two works by strangers.

That is the residue named in the diachronic result, opened and measured. What makes two books read differently is almost entirely that they are two books.

The reading is not one reading, because a character is a morpheme in some writing and a fragment of one in the rest. A Chinese character carries meaning and combines with other characters under rules. Which character follows which is which morpheme follows which, and that is semantic and syntactic structure read directly. A Latin letter carries no meaning. The same reading over an alphabet is orthography and phonotactics. Two different quantities have been carrying one name throughout this section.

That explains several things recorded separately. Chinese could not be read at byte width at all and had to be given character width, which was noted as a practical matter and is a difference in kind. It sat outside every band it was measured against. And reading the alphabetic languages at word width to reach the same unit made everything worse, because a word is not a morpheme and is several of them in an agglutinative language, which is where that test separated the languages by how much they inflect.

The asymmetry cannot be removed by choosing a unit. Chinese supplies morphemes in its writing and every alphabetic language needs segmenting to reach them, and a fair comparison across the two needs a morphological analyzer per language, which this work does not have.

Tone is the first thing measured here that clears the floor everything else sank under. Three tonal languages held here do not allow the tone to be taken out. Chinese fuses it into the character, Thai spreads it across marks and the class of the initial consonant together, and Japanese never writes its pitch accent at all. Vietnamese writes six tones as marks that are separate from the marks setting vowel quality. The tone can be deleted on its own.

Deleting it moves the reading 0.0949. Every other language here was stripped of all of its marks and none moved that far: Czech 0.0779, Romanian 0.0723, Turkish 0.0582, Spanish 0.0409, French 0.0369. Vietnamese carries 0.209 marks for every letter and that count is tone alone.

The control decides it. Two unrelated passages of Vietnamese, four hundred thousand characters each, sit 0.0874 apart. Vietnamese with its tone removed sits 0.0949 from itself. Deleting the tone makes a text less like itself than a different part of the same language is.

Everything else measured in this section sits under that floor. The writer is worth 7 to 11 percent, the century 1, four hundred years of changing orthography 5, and all of it is swamped by two books being two books. Tone is over the floor, and it is over it while every letter stays on the page.

Repeated outside the family, the decomposition holds. Japanese gives the writer 5 percent of the distance between two works and leaves 95, against 11 and 88 in German. Same order and same conclusion, in a language outside Indo-European that writes with three systems at once, marks no word boundaries, and never puts its pitch accent on the page. So the finding that two books are separated mostly by being two books is not a fact about Germanic.

The sizes are not comparable. Fifty two works by twenty three writers, thirteen of them with more than one work, giving 55 pairs by one writer against Germany's 1370.

Three numbers beat four thousand at finding a Japanese writer, by nearly five to one. How much Chinese-derived character a text uses against how much syllabary is a choice of register and period and person, with no Germanic equivalent, and it was written down as a prediction before the run. That ratio takes 24 percent of the distance where the character square takes 5.

Both figures moved when the corpus grew from forty works to fifty two, and they moved apart: the square fell from 7 percent to 5 and the ratio rose from 16 to 24. So the earlier run understated the gap, and this is the second time a small description has matched or beaten the square, after a twenty one bin histogram of word lengths equaled it on families.

Every structural result in this section was measured on one branch of one family. The breadth here looks wide: 26 languages outside Indo-European across Uralic, Turkic, Semitic, Sinitic, Japonic, Koreanic, Dravidian, Bantu, Cushitic, Austronesian and Kartvelian. But the questions that take the reading apart, what an author is worth, what a century is worth, what is left over, were all asked of English and German, which are two Germanic languages of one family.

So the figures above describe Germanic prose and are written here as though they described writing. That is not a small qualification: an interrelated family shares inherited structure, a shared alphabet, and a thousand years of borrowing from each other, and a constant found across its members can be a fact about the family and read as a fact about language. Correcting it needs a corpus outside the family with author and date metadata of the same quality. That metadata made the German result possible and every earlier attempt at this question lacked it.

Across three centuries of one language, with dates nobody here chose, time leaves a mark of about five percent. The German printed record from 1600 to 1899, prepared to one standard, each text carrying its year in its name, split into centuries by the archive. This is the question the cookery books could not answer: those were dated by the first four digit number printed in them, then by author lifetimes, and split at a boundary chosen so five books fell on one side.

A text lands in its own century 112 times in 210, against 33.3 percent for guessing. The graded form says more: mean distance climbs with the years between two texts, 0.0803 under 25 years, 0.0808, 0.0823, 0.0829, then 0.0820, then 0.0843 above 200. Monotone but for one dip, and the whole span of it is five percent.

German definition, capitalization and typesetting all moved a great deal over those centuries and the reading barely registers it. What separates two books written eighty years apart is mostly that they are two books.

One figure from that run should not be quoted. The correlation between years apart and distance apart is 0.076 and was computed over 21945 pairs, which come from 210 texts. Pairwise distances are not independent observations. The effective count comes from the texts and not from the pairs, and any significance read off that denominator is invented. The monotone ordering of the bins is the evidence here.

Dividing the frequencies out of the reading compresses every distance toward the average and does not separate script from structure. The reading holds two things: how often each character is used, which belongs to a script and its conventions, and what follows what given those frequencies, which is about how a language is built. Since every result in this section says the first dominates, dividing the joint counts by the product of the two marginals should leave the second alone.

Half of it came out as predicted. Tamil to Malayalam, the closest pair in its family and the widest distance measured, fell from 1.200 to 0.988 of the average. The two Chinese character sets fell from 1.188 to 0.988. Both of those distances are script and both closed up.

The other half is the answer. Zulu to Xhosa, the closest pair measured anywhere and the one result with no confound left in it, rose from 0.737 to 0.909. The control of one language in one character set rose from 0.498 to 0.700. Every distance above the average fell and every distance below it rose, and all eight landed near 1.0.

That is compression and not correction. The operation removes most of the variance, and what remains is distances that are all alike. The marginals were never only script: a language with many vowels or heavy affixation has different letter counts for reasons that are the language. Dividing them out discards that with the conventions. The ratio also weights rare pairs most, where the expected count is smallest and the estimate worst. Part of what replaces the removed signal is noise.

A writer is found at the word and not at the character, the same split the languages showed one level down. Seven writers, thirty one works between them, all English prose of nearly one period. The language, the script and the century are fixed and only the writer changes. Guessing gets one in seven.

Which character follows which gets 35.5 percent, only two and a half times chance, with a writer's own works sitting 0.0600 apart against 0.0680 for two writers. How often each common word is used gets 77.4 percent, with 0.0220 against 0.0331. The words carry the writer and the characters barely do.

The language work predicts exactly this, once the reading is understood as following surface. These seven share an alphabet and its conventions entirely, and a reading of which character follows which has almost nothing left to separate them, exactly as it had nothing to separate Zulu from Xhosa in the other direction and everything to separate Tamil from Malayalam. The misses say it plainly: Conrad taken for Eliot, Eliot for Conrad and for Stevenson, Doyle for Stevenson, which is period and register and not authorship.

The words are also where the objection about a text not being one distribution has force. That was raised against the whole approach and measured as negligible, because letter frequencies barely move within a book. Word frequencies move a great deal between chapters and speakers,

and the word reading here is built on exactly that quantity. Its 77.4 percent is carried partly by a drift that the character reading was immune to.

Every language carries more than one reading per written word, and a walk through a sentence multiplies possibilities and cannot collapse. Counted on treebanks, where each token carries its lemma and the case, number, gender and person it stands in, which counts readings sharing a surface form instead of estimating them. German 4.0936, English 3.8516, Polish 2.9505, Turkish 2.2818, Estonian 1.8068, Finnish 1.5965, Hungarian 1.5440, Vietnamese 1.4492.

That answers why a reading over a growing window kept climbing here and would not settle. The possibilities over n words go as the product of the readings each carries, and a product of numbers above one grows. The flattest language measured, which has no inflection at all, still stands at 1.4492 and reaches 41 over ten words. The steepest reaches 1321429. Nothing collapses on its own, and what collapses it in reading is context pruning the space faster than the walk opens it, and that pruning is parsing itself.

The prediction that went with this was backwards and the correction is the finding. Polish was expected highest for its case syncretism and English lowest for having almost no inflection. The order is close to the reverse. Agglutination reduces ambiguity instead of making it: a Finnish word defines its case in a morpheme of its own. The form states what it is, while English ambiguity is between parts of speech, run and that and set, and German adds case syncretism on top of the same. Many cases explicitly marked is the opposite of ambiguous, and the two were conflated here.

The corpora differ seventeen fold in size and a form counts as ambiguous only where the corpus used it both ways. The small ones understate. Hungarian and Vietnamese are the two smallest and among the lowest. The split holds among the large ones: Estonian is the largest at 344581 tokens and reads 1.81 against German at 263777 and 4.09.

Words for sounds follow the border and not the family. Hungarian is Uralic and Polish is Slavic and they have shared the Carpathian basin since the Magyars arrived into Slavic speaking country around 895. Words imitating sounds sit outside the core vocabulary that resists borrowing and attach to shared work, which puts them where a thousand years of the same fields would show. Taken from a dictionary that files them by what they are, stripped to bare letters so neither language is separated by how it decorates its vowels, and compared over equal sized draws since the lists run from twenty words to four hundred and fifty five.

Hungarian to Polish reads 0.5959 against 0.6530 for unrelated languages sharing no border. The claim holds. What is larger than the claim is the pattern behind it: every bordering pair beats every non-bordering pair whatever their families. Polish to Czech 0.5512, Finnish to Estonian 0.5748, Hungarian to Czech 0.5857 and Hungarian to Polish 0.5959 sit below Polish to Finnish 0.6167, Hungarian to Finnish 0.6320, Hungarian to Estonian 0.6883 and Polish to Estonian 0.6892, with no overlap between the two groups.

Hungarian sits closer to Czech, which it is not related to, than to Finnish, which it is. For this part of a vocabulary geography beats descent outright, reversing what the same kind of reading gives for ordinary words.

Estonian brings twenty words and holds both of the highest readings, though the pattern survives dropping it.

Read as sounds instead of letters the pattern holds and the effect halves. The dictionary keeps no transcriptions for these languages: Hungarian pages carry a template that builds the pronunciation from the definition when the page is drawn, and several Polish pages carry nothing. Transcriptions would therefore not be new evidence, and they would not be, since for five languages with phonemic orthographies the pronunciation is a function of the definition. What they give is the shared alphabet. That Hungarian sz and Polish sz stop counting as the same consonant when they are not.

Mapped that way, every bordering pair still sits below every non-bordering one: Polish to Czech 0.5919, Hungarian to Czech 0.5926, Finnish to Estonian 0.6374 and Hungarian to Polish 0.6448 against Hungarian to Finnish 0.6541, Polish to Finnish 0.6685, Hungarian to Estonian 0.7046 and Polish to Estonian 0.7094. The gap between the two groups falls from 0.0208 in letters to 0.0093 in sounds. About half of what the definition showed was neighbors sharing definition conventions and about half is the words sounding alike. Hungarian still sits nearer Czech than Finnish, the part the confound was most likely to have manufactured.

The mappings are written by hand here from the standard description of each orthography. An error in them is an error in the result. Every distance rises because the sound alphabet is larger than twenty six letters. The numbers move together for that reason and only their ordering can be read.

The families survive losing the alphabet entirely, and the claim below that they do not is wrong. Reducing every Latin written language to the same twenty six bare letters, with every accent, tail and stroke removed and nothing kept that any language holds and another lacks, gives 13 of 22 languages nearest a relative. Reading the same texts with every character they actually use gives 13 of 22. The alphabet is worth nothing to the family result.

Stripping it also corrects some pairings. Latvian moves from Czech to Lithuanian, which is Baltic finding Baltic. Polish moves from Slovenian to Czech, which is West Slavic finding West Slavic. Romanian moves to Portuguese, Spanish to Portuguese, Swedish to Norwegian. The definition was interfering with the family signal in those cases and removing it helped.

The bare letters still hold the sounds a language uses and the endings it inflects with, which sit in plain letters whatever a language does to its vowels. Latvian and Lithuanian pair on shared Baltic morphology as soon as their diacritics stop separating them.

Most of what this reading calls naming a language is matching an alphabet. [Overclaimed, and this is the correction. Removing the characters each language uses far more than the others moves fourteen of twenty to a different nearest, which is true, and it was read as the family signal being definition. Most of those move to another member of the same family. The fine grained question of which relative is nearest is carried by the alphabet and the family is not.] Hungarian is detected here while the field's own settings say Hungarian wants shorter runs than anything measured, the opposite of what its morphology needs, and both cannot be about the same thing. So the characters each language uses far more than the others were found and then removed, every language losing its own so none is handicapped.

Fourteen of twenty languages find a different nearest neighbor once those characters are gone. The ratios say why: Latvian uses its long a 28640000 times more than everything else combined, Turkish its dotless i 46996667 times, Vietnamese its barred d 18520000. Those are not characters a language favors, they are characters that occur in one language and nowhere else, and a reading that names Latvian by them knows nothing about Latvian.

The six that do not move are the informative ones and they are the real clusters: French to Italian, Italian to Spanish, Spanish to Italian and Romanian to Italian, which is Romance, together with Latvian to Czech and Polish to Slovenian. Those pairings survive every distinctive character being deleted.

Polish is the case where both hold at once. It is trivially identifiable by its alphabet, its slashed l alone appearing 22543333 times more than elsewhere, and its pairing with Slovenian holds after all of that is stripped out. Being unique in an inventory and being related to a neighbor are separate facts and this reading usually cannot tell them apart.

Mining the field's own settings for language constants does not work from abstracts, and the first result that looked like it worked was noise. The idea is sound: people have built language identifiers and authorship attributors for decades, language by language, and the run length each settled on is a statement about that language that nobody published as a finding. The whole bibliography of the field is served with abstracts, 131040 entries.

Three languages cleared a threshold of eight papers and fell in the predicted order: Chinese 2.35, English 2.53, Arabic 2.59, which is logographic lowest and root and pattern highest, exactly as a character carrying a whole morpheme against a morphology spanning a discontinuous pattern would give. Lowering the threshold to four papers brought in nine more languages and destroyed it. Hungarian reads 1.25, the lowest of twelve, and is the most strongly agglutinative language in the set, which should need the longest runs. German is highest at 3.00 and is fusional.

The method cannot work as built and more data would not save it. Counting how often an abstract says bigram or trigram measures what a paper discusses, and a paper reporting that trigrams failed mentions them as often as one reporting they won. The count reads the vocabulary of the field and not any property of a language.

The idea needs the results tables of full texts, where a tuned setting is stated as an outcome instead of named in passing. That was worth knowing before 42 megabytes were downloaded to find it out.

Run against every task here, the square wins two of four by margins the size of the noise, and the character frequencies alone come within three points of it. Six descriptions of the same texts were put to four questions. Which family a language belongs to: the square 50.0 percent, word lengths 44.1, the frequencies 38.2. Which century a German book is from: the square 58.5, the frequencies 55.6. Which English writer: the square 47.1 and the frequencies 47.1, exactly level. Which Japanese writer: the square 19.0 and the frequencies 35.7, which the square loses by seventeen points.

So the transitions buy six points on families, three on centuries, nothing on English authorship, and cost seventeen on Japanese, for sixty four to a hundred and ninety five times the numbers.

The wins do not survive being looked at closely. With 34 items scored, one text is three points, and the square returned 15 of 34 on families in an earlier run and 17 here on the same corpus with only the read length and the cleaning different. Its advantages are the size of the difference between two runs of itself.

The comparison establishes something smaller and firmer than the question that prompted it. Reading how often each character is used, with nothing at all about what follows what, is within three points of the whole apparatus on three of four questions and better on the fourth.

Twenty one numbers do what four thousand were doing. How the word lengths of a text are distributed puts 15 of 34 languages nearest a relative. Which character follows which, over a

square of four thousand ninety six, puts 15 of 34. Mean word length and its spread alone, which is two numbers, puts 12. So the square carries no more of a family than the lengths do, and most of the machinery in this section was holding a signal that a histogram holds.

They are not the same signal, since they fail on different languages. The lengths lose Estonian to Romanian and Finnish to Lithuanian. The square loses Afrikaans to Finnish and Dutch to Turkish. Partly complementary and neither better.

The prediction that went with this was wrong. Word length was expected to rescue the families the square lost, since Uralic and Dravidian are agglutinative and should have long words throughout. Estonian reads 4.08 and Finnish 6.30, which is a wide gap inside one family, and it is not noise: Estonian lost its final vowels and its words are shorter for that reason. Dravidian splits the same way, Malayalam at 8.84 and Telugu at 6.96. What failed the square fails the lengths, for reasons of its own.

Three languages cannot be measured this way at all. Chinese, Japanese and Thai mark no word boundaries. The descriptor that matches the whole apparatus does not exist for them, the plainest form of this reading not being one reading across writing systems.

Cut down to two symbols the relationship still shows, and word length is all that remains. The alphabet was coarsened by keeping only the commonest few symbols of each text and folding everything else into one, swept from thirty two down to two. At every level the close pairs stay closer than the far pairs, including at two, where the only distinction left is whether a character is the commonest one.

For most of these languages the commonest character is the space. At two symbols the text records word lengths alone. It survives translation, and the lengths line up with the pairings exactly: Zulu 5.70 with a spread of 3.75 against Xhosa 5.88 and 3.65, Spanish 4.41 and 2.59 against French 4.66 and 2.66, with Finnish at 6.41, far enough from Spanish to separate them.

at the bottom of the sweep the reading is a single magnitude for each language and not a relation between quantities. The reduction leaves a scalar.

One of the four comparisons is void at that level and it was not noticed until the commonest characters were listed. Shona and Somali have the letter a as their commonest instead of the space. Their two symbol reading records where a falls and not where words end. Any distance between them and a space keyed language at that level compares two different quantities, and the Zulu to Somali reading at two symbols is then meaningless. The other three hold.

The one clean success was challenged on its definition and survived. Nguni languages write their clicks with c, x and q, which is a use of those letters no other language here makes. The pairing of Zulu and Xhosa might have been two definition conventions agreeing and not two languages. It is not. Those two use the letters less than everything else measured, 6.00 and 11.29 per thousand against Vietnamese at 74.91, Romanian at 48.50 and Spanish at 45.32, because clicks are not frequent enough in running text to move a letter distribution while Romance languages lean on c constantly. Removing all three letters from every language moves the pair from 0.0756 to 0.0758.

The same run corrects how that result was described. Spanish to French reads 0.0763, the same distance. Zulu to Xhosa is not the uniquely tight pairing it was called. It is an ordinary close pair sharing a script, where the envelope says it should be, and that is a smaller claim than the one made for it.

Given a close pair with nothing in the way, the reading finds it, and that fixes how far it reaches. Every failure so far had somewhere to hide. Tamil and Malayalam are close and came out widest apart of seven, and their scripts encode different distinctions. Uralic lost its family, and its members have had a thousand years of unlike neighbors. The Chinese pairs showed the collection moving the reading more than the language does, and those collections were different works.

Zulu and Xhosa remove all of it. Both Nguni, close enough to be partly mutually intelligible, both in the Latin alphabet, both from the same translated work as the other 43 languages here. Content, register and translators are fixed together. They read 0.0756 apart, the closest pair in the matrix, each finds the other, and the nearest thing to either of them otherwise is 0.0845.

So the instrument reads surface closeness and the depth it reaches is now measured. A close pair sharing a script is found: Zulu with Xhosa, Danish with Norwegian, Serbian with Slovenian. A close pair whose scripts diverged is lost, which is Tamil and Malayalam. A deep relationship is lost whatever the script, which is Finnish and Hungarian across four thousand years. Shona is the confirming miss: Bantu like the other two and far enough from Nguni that it goes to Somali instead.

That partly restores the European result without restoring what was claimed for it. Danish and Norwegian are a genuine close pair sharing an alphabet. Finding them was finding something. What it was not is descent, and the same reading loses every family whose members have drifted far enough apart to need one.

With the language held fixed and only the characters changed, the reading moves by twice what changing nothing moves, and where a text came from moves it more than either. Every earlier test changed two things at once. Chinese holds one still: simplified and traditional are one language in two character sets, and both sides here come from the same software translations.

Simplified to traditional reads 0.1253. Traditional to Hong Kong, the same language in the same characters from the same collection, reads 0.0585. So changing the characters costs 2.1 times what changing nothing costs.

The comparison that should have been the scale for that turns out to be the largest effect in the table. The same language from another collection sits further away than a different language does: the simplified Mandarin of software translations sits 0.1554 from the Mandarin of a translated scripture and 0.1274 from Korean, and two Mandarin corpora from two collections sit 0.1140 apart. Ordered by how much they move the reading, it is the collection first, the characters second and the language third.

That is the source and language result again, measured a different way and coming out worse. There it was 0.0867 against 0.0936, a margin of seven percent in the language's favor. Here the source term is larger than both of the others outright.

The tool's own line saying the writing does not matter more than the language should be disregarded. That line sets the clean within collection script pair against cross collection language pairs. The comparison then has register on one side and characters on the other.

A family separated by an ocean is found, and the same test loses a relative three times older, which puts the depth limit on both sides of one run. Malagasy is Austronesian, spoken in Madagascar, with its nearest relatives across the Indian Ocean and everything it has borrowed from since being African and French. Read against its family and its neighbors, all from one

collection so the content is fixed, its nearest is Malay at 0.0770, then Indonesian at 0.0814, then French at 0.0834. Its family averages 0.0856 against 0.0972 for everything else. Descent wins.

Uralic answered the same question with contact and this does not overturn it, because the two differ in age and the answer follows the age. Malagasy left the Malay world about twelve hundred years ago and is found. Maori split from that branch some three thousand years back and reads 0.0983, further from Malagasy than French and Portuguese are, while being the only other language in the run from its own family. Finnish and Hungarian separated four thousand years ago or more and were lost. Found at a thousand, lost at three, in one family and one run.

Two faults in the run. Swahili is the African contact language nearest to Madagascar and this collection holds 1195 characters of it. Contact is represented by French alone, which is real and is colonial and recent. And Tagalog was dropped with a message reading that no script is known for it, the gate's note and not the reason: it was excluded for holding 104971 characters against the 133333 the run required. That is the same misleading label already fixed in the Uralic tool and not fixed here.

On the family built to tell descent from contact, the reading follows contact. [Overclaimed as a negative. This says the reading did not find Uralic, and it was written as though it said the descent is not there to be found. Historical linguists establish that family from evidence a character stream does not carry. The failure is compatible with the relationship being present and invisible to this reading. The European result is settled, since the reading follows contact where the two can be told apart. What is not settled is anything about how deep a relationship can be and still exist.] In Europe descent and contact travel together: every pair the reading grouped shares an alphabet and a thousand years of borrowing, and nothing there separates the two. Uralic separates them. Finnish and Estonian sit on the Baltic among Germanic and Baltic neighbors, Hungarian sits two thousand kilometers away surrounded by Slavic, German and Turkic and has for a millennium. One family by descent, almost nothing shared by contact, and the relationship took an argument to establish for that reason.

Read on files the gate passes, Uralic sits 0.0559 from itself and 0.0535 from the neighbors it borrowed from. Not one of the three has a relative nearest it. Finnish goes to Swedish, which ruled it for six centuries. Hungarian goes to Czech. Estonian goes to Romanian, which is neither and is the mark of a reading close to noise at this distance.

So what was recorded here as recovering language families was recovering contact. Danish and Norwegian, Polish and Czech, Spanish and Portuguese are all neighbors as much as they are relatives, and the reading cannot see which of those it is answering. The line written earlier, that the mistakes it makes are the families, should read that the mistakes it makes are the contact zones, and in Europe the contact zones are the families.

On a family whose branchings are ordered, the reading returns the opposite of the family, and that weakens the European result badly. Dravidian was chosen because its account is not much in dispute and it gives a graded prediction: Tamil and Malayalam nearest, having split around the ninth century, Kannada beside that pair, Telugu furthest as South Central. Four languages on one parallel text, every text cut to one length so nothing follows from how much arrived.

Tamil to Malayalam comes back at 0.1005, the widest distance in the whole matrix. The most recent split in the family reads as the most distant pair measured. The four Dravidian languages

sit 0.0873 from each other and 0.0853 from three Indo Aryan languages, and Telugu's nearest of everything is Bengali.

Each of those languages writes in a script of its own. The scripts were suspected and aligned by where each character sits inside its own Unicode block, which is a transliteration with no judgement in it since those blocks are laid out in parallel by design. The alignment moves nothing: Tamil to Malayalam goes to 0.0980 and all three tests fail exactly as before. The reason it could not have helped is that the reading ranks characters by frequency inside each text. Which codepoints a script uses never entered it. A confound was proposed that the instrument is built to be immune to.

The unit was suspected next, and with better reason. A letter sequence suits an alphabet because that is where an alphabet keeps its context, while an abugida keeps its context in the cluster: a consonant carries a vowel already, a dependent sign changes it, and a virama binds one consonant to the next, and a codepoint is a piece of a unit and not a unit. Read as clusters instead, it gets worse. Tamil to Malayalam goes to 0.1186 and two languages leave the family where one left before.

The inversion was untranslated English in the files, and cutting each to its own script brings the family back. Three explanations were built and tested before anyone looked at what the files held. They hold English in wildly uneven amounts: Telugu is 30.0 percent Latin characters and 47.8 percent Telugu, Hindi is 14.8 percent Latin, while Tamil is 2.1 and Marathi 1.8. These are subtitles from educational video and each translator left a different share untranslated. Telugu's nearest of everything was Bengali because a third of Telugu's file is not Telugu.

Cut to its own writing, Dravidian sits 0.0939 from itself and 0.0947 from Indo Aryan, and every one of the four has a Dravidian language nearest it, where three of four left before. The family is there and it was buried under the English.

A tool for exactly this was written earlier in this work, after a Greek to English lexicon was found standing in a Greek corpus, and it was recorded as worth having. It was not run on any of these seven files. Three theories about scripts and units were built and tested against a defect the existing check would have named first.

What remains after cleaning is real and is about the writing. The order still does not hold: Tamil to Malayalam is 0.0984 against 0.0869 for Kannada, and Tamil sits furthest from everything measured, 0.0984 to 0.1211. The inventories say why. In the same length of cleaned text Tamil uses 48 distinct symbols and every other language uses 69 to 85. Tamil script encodes fewer distinctions than its neighbors, the Sanskrit question again and not something the reading can be blamed for.

Folding the aspirate and voiced stop rows onto their plain forms, the distinction set Tamil keeps, was tried before the cleaning and did not help. It removed 67 of Malayalam's 524 symbols, an eighth. Whatever separates these inventories is not confined to those rows.

Tamil holds 149 distinct codepoints and 1008 clusters in the same length of text. Malayalam holds 524 and 2644. The two languages that separated most recently carry the most unlike writing inventories of the seven, by three and a half times, and the reason is Sanskrit. Malayalam took Sanskrit phonology into its script, the aspirates and voiced stops that the Grantha letters supply, and built a literary tradition on mixing the two languages. Tamil script does not encode those distinctions and Tamil has a long history of declining to adopt them.

So the reading fails in a particular direction and not at random. It is reading a writing culture, and these two languages are as far apart in writing culture as any pair here while being as close

in descent as any pair here. In Europe those two things travel together and here they are at right angles. The same reading recovered families there and inverts them here for that reason.

The remainder is worse for the European result than the Dravidian failure is on its own. Danish and Norwegian are nearly one written language, Polish and Czech share both conventions and vocabulary, and every European pair the reading grouped correctly shares an alphabet and centuries of borrowing. Descent and contact travel together there and nothing in that test separates them. Dravidian has the descent without the contact, and the reading finds nothing. So what was recorded as recovering families may be recovering shared surface, which in Europe tracks descent and here does not.

Grouping the languages by those squares returns the families, and the tree it is scored against is argued and not measured. Fifteen of the 22 languages with a relative present sit nearest a language in their own family, against roughly one in nine by chance, and the first pairs formed are Danish with Norwegian at 0.034, then Serbian with Slovenian, French with Romanian, and Portuguese with Spanish. Two of the misses are not misses: Finnish sits nearest Swedish, which is six centuries of one country and an official language and not a mistake, and Polish sits nearest Esperanto, whose author's first language was Polish.

Greek sitting nearest Hebrew was called a script artifact here and it is neither an artifact nor a miss. Writing both in Latin letters was the test, since a transliteration keeps every word and changes only the symbols, and Greek stays nearest Hebrew across it while Slavic holds together across it too. So the script is not what joins them. What joins them is that neither has a relative in this set: Hellenic and Semitic each hold exactly one language here. The nearest neighbor of either must fall outside its own family whatever it measures. They were scored as errors and they were never scoreable, which is also why the honest figure is 15 of the 22 languages with a relative present and not 15 of 28.

One of the four Greek texts is a Greek to English lexicon and had been standing in every Greek reading in this work. It is 21.2 percent Greek letters and 78.8 percent Latin. It was found by reading the first line of a file, which is not a method. Every language text was then scored by the share of its letters belonging to the script its language is written in and judged against the other texts of that language. One of 106 is out of line, and it is that one. Excluding it moved Hebrew off Greek, removed the pair from the identification errors, and brought out Afrikaans sitting nearest Dutch, which is correct since one descends from the other.

That agreement is weaker evidence than it appears. A family tree is a reconstruction argued from cognates and sound correspondences. This is agreement with a scholarly consensus and not a check against a fact, and where the two disagree nothing here can say whether the instrument or the reconstruction is wrong. That is a different kind of check from the protein bond lengths, where valence fixes the answer whatever anyone believes. Those are an oracle. This is a strong prior.

The generated corpora mostly stay outside the languages and the kinds are not cleanly separated. The sharpest control passes: the chain built with English letter frequencies, made to imitate a language, sits nearest another generated corpus. Eight of ten generated corpora do. But the mean distance among the languages is 0.06317, among the generated 0.09780, and between the two kinds 0.08574. The gap between kinds is smaller than the spread inside the generated set, and one generated corpus sits nearer Hungarian than most of its own siblings. The square reads

the structure of a written language, and a distribution chosen well enough reaches part of the way into it. Calling it a human signature is more than has been shown.

The errors are what makes it more than a classifier. Danish goes to Norwegian and Norwegian to Danish, Polish to Czech, Russian to Czech, Serbian to Slovenian, and Chinese to Japanese. Two Scandinavian standards that are nearly one written language, West Slavic neighbors, South Slavic neighbors, and a script Japanese took from Chinese. Esperanto goes to Polish, the first language of the man who constructed it. Nothing in the construction knows about language families, and the mistakes it makes are the families.

No quantity measured here identifies a language, and the coefficient is among the weaker ones at it. [Overclaimed as a negative. Four quantities failed to identify a language and that was written as though no quantity does. A reader of any of these texts identifies its language instantly. The information is in them and these four do not reach it.] That the spread between languages is 1.36 times the spread inside one says a language keeps a value of its own. Whether that value picks the language out from every other is a separate claim with a pass and a fail. It was tested that way: each text held out in turn, every language given a value from the texts that remain, and the held out text assigned to whichever language it lands nearest. Over 102 texts in 28 languages, guessing gets 3.6 percent. The coefficient gets 9.8, which is 10 hits against 3.7 expected and real at a binomial probability near 0.003, and which is not identification. Collision entropy is the best of four at 24.5 percent, the count of symbols gets 15.7, and the spread of gaps gets 5.9, which is barely above guessing. All four together get 19.6, below the best single one, the mark of fitting noise at under four texts per language.

Where the misses go says the same thing. Welsh is taken for Chinese, Greek for Finnish, Hungarian for Danish. A quantity carrying a language's identity does not confuse a Celtic language with a logographic one.

This refutes that any of these four is a constant identifying a language, and it does not refute that such a constant exists. Four texts is a thin basis for a language's own value. The shares above bound the quantity from below and leave the upper end open.

Reading the alphabetic languages at word width makes it worse, and what it separates them by is morphology. A Chinese character stands for a morpheme and a Latin letter for a piece of one. Comparing one symbol against one symbol compares unlike units, and words are the closest unit to a morpheme reachable without a morphological analyzer for each language. At word width the ratio rises from 1.36 to 1.52 and Chinese stays outside at 4.1 deviations. What the word width does instead is sort the languages by how much they inflect: Finnish at 0.1612, Hungarian at 0.1609, Polish at 0.1618 and Welsh at 0.1564 against Spanish at 0.1099, Norwegian at 0.1106 and Dutch at 0.1122. An agglutinative language defines many morphemes into one word. Word width overshoots exactly where it overshoots most. The morpheme claim is therefore untested and not refuted, since a word is not a morpheme and testing it needs segmentation this work does not have.

The coefficient only becomes a property of the corpus once it is taken at the numbering it converges to. Read at whatever numbering a file arrives in it belongs to the numbering: the Iliad returns the least of anything measured at 0.056 because polytonic Greek needs 141 code points laid wide, and the same poem renumbered returns four times as much. Minimizing the weighted spread over whole positions has a known answer, the commonest symbol taking

the middle and the rest going outward by frequency. The value is reachable and not merely approached. Greek moves from a spread of 31.44 to 7.83 and from 0.056 returned to 0.222, Finnish gains the most at 0.250, and English gains 0.194.

That the numbering was being read is measurable: the ordering as given and the ordering at the tightest numbering agree only at rho 0.825. The mechanism survives the change at rho -0.996 between the tightest spread and what it returns, while the count of symbols falls to -0.664. What packs the mass is the frequency distribution and not the size of the alphabet.

Under the tightest numbering the languages sit closer together than their alphabets do. Greek carries 141 symbols and German 73, and both land at a spread of 7.83 and 7.84 and return 0.222. The four natural languages span 4.76 to 7.84 where as given they spanned 15.61 to 31.44, and source code sits outside them at 9.49. Four languages cannot carry a claim about languages, and one near tie among fifteen corpora in a range from 4.76 to 17.01 is not unlikely on its own. This is recorded as something to test on many languages and not as a result.

The spread of the values carries the result, and the generating parameters reach it only through that one number. The generated corpora settle this because their alphabet and weighting were chosen when they were made: ten of them across alphabets of 8, 26 and 64, uniform and geometric weightings at two rates and an English weighting, at three chain depths, measured beside five real languages carrying 70 to 141 symbols. Ranked against how far the values spread, the share of symbols returned gives rho -1.000 over all fifteen. Ranked against the count of symbols it gives -0.911, and the cases that separate the two say which is doing the work: a generated corpus of 26 symbols with a spread of 3.12 returns 0.523 while one of 8 symbols with a spread of 2.49 returns 0.627, following the spread order and not the alphabet order.

Alphabet size, weighting and chain depth are three knobs feeding one quantity, and once the spread is known the parameters that produced it say nothing further about whether the corpus can be put back. The level is not predicted, only the order: taking the error as normal and asking how often it stays inside the half level between symbols overstates the share returned by as much as 0.238, predicting 0.860 where the corpus returns 0.627, the size of error a reconstruction correlated between positions and heavier tailed than normal puts on that estimate.

What sets the cost is how tightly the alphabet is packed in value, which is not a property of the writing. The memoryless corpus carries eight symbols spread across the whole byte range and returns 62.7 percent at three bits, since an error has to be large to reach the next symbol. English packs about 65 symbols into contiguous positions one apart. Anything above half a level is already a different letter. The exponent alone returns 1.5 percent, which is one in 65 and exactly chance.

A protein represented by its bonds in assembly order returns the chemistry, and two earlier representations of the same molecules returned nothing because each discarded the thing that carries it. Laying every atom into a grid discarded the order and left a scatter in a box that is mostly empty. Taking the step from one alpha carbon to the next kept the order and discarded the bonds, an alpha carbon being a summary of a residue and the step between two of them not a bond. Walking the backbone as it is built, nitrogen to alpha carbon to carbon to the next nitrogen, gives the bonds themselves, and their lengths come back at 1.47, 1.52 and 1.32 on ubiquitin, 1.47, 1.53 and 1.34 on hemoglobin, 1.45, 1.52 and 1.33 on the chaperonin and the spike, and 1.48, 1.52 and 1.34 on crambin. Chemistry gives 1.46, 1.52 and 1.33. Five structures from 137 to 24024 bonds agree to within 0.02 angstroms.

This is a positive control and not an instance of the signature this work is chasing. The three lengths are set by valence, the carbon to nitrogen bond being shortest because resonance gives it partial double bond character. The repeat is a physical fact about electrons and nobody wrote it, and an answer of that kind is worth having. The answer exists before the measurement and is not supplied by it, a condition every other reading here has lacked.

The reader used everywhere else in this work cannot see that period, and the reason is structural. It samples roughness at lags 1, 2, 4, 8 and 16 and sorts them by position modulo the candidate count, because it was built for an index whose period counts bit positions. A period of three in a sequence sits at lags 3, 6, 9 and 12, and no power of two is a multiple of three. No data can make that reader return three. It returned 2, 2, 4, 2 and 2 on series whose period is known. Two different quantities were both being called a period of n, the same conflation that reported one instrument's result as another's three times earlier in this work.

Read at every lag instead, all five returned a multiple of three, but only two returned three: the others gave six and twelve. A series repeating every three agrees with itself at every multiple of three. The tallest of those is settled by noise and taking it reports a harmonic as the period. Scoring a candidate together with all of its multiples returns three on five of five.

A phase error had to be found first and it is worth recording because it looked like a result. A backbone run holding a break is cut into pieces, and every piece after the first begins somewhere other than the first bond of a residue. Concatenating them mixed the three bond types together. The large structures read 1.43, 1.43 and 1.43, the average of all three reported three times, and it looks exactly like a molecule whose bonds are all the same length.

It does not work on proteins, and the prediction that it would was wrong. Every set behind the result above was built here with a correlation length placed along each axis by hand, the case most favorable to the method. Real structures from the protein data bank are the honest test: three dimensional, shaped by nobody here, and anisotropic because a folded chain is longer one way than another. The prediction written before the run was that every structure returns three. Three of ten readings did, and nearly every score is negative, meaning the real grouping is worse than its own shuffles and there is no signal to have.

The first run smoothed the deposited atoms by 1.6 voxels and put every structure at an exponent between 5.8 and 7.5, far outside the 2 to 3 the synthetic fields used, with every reading along the line pinned near 1.0, the curve's floor. That was worth ruling out and it was not the obstacle: sweeping the smoothing down to 0.3 brought several structures into range and the scores stayed negative.

What separates the one structure that does give signal is how much of its box it occupies. A spike glycoprotein of 22812 atoms returns three at every smoothing and scores 1.85 at the finest, decaying to -0.24 as it smooths. A chaperonin carries 58674 atoms and is a hollow barrel, and gives nothing. In empty space every axis is identical, and a line crossing mostly vacuum has no magnitude per dimension to carry, and a protein is a thin chain in a large box. The method needs the anisotropy spread through the set and not confined to a small occupied part of it, and that condition holds in the fields it was demonstrated on and fails in the first real case it met.

The interleaved curve loses the property that carries the count. Split by curve, interleaving gets six of six with scores reaching 9.28 standard deviations above its own null, while the Hilbert curve gets five of six at scores of -0.05, 0.06, 0.50 and -0.01, which is no signal and makes those picks chance. The Hilbert curve was the better of the two above, losing 0.063 against

0.110 because it never jumps. A jump is a block completing, and which block completes is which axis just turned over. Removing the jumps removes the dimension count with them. Fidelity and dimensional structure trade against each other here and a curve cannot be chosen for both.

This is the quantity that was assumed away when this construction was first described as carrying every corpus with only some information elided. The elided part is not a qualifier and it is not small. Under the best curve available it is 0.063 in exponent, more than half of everything the folding costs, and it is a property of reducing dimensions and not of any choice made here.

Reading along the curve recovers the steep ones badly, missing Mondrian by 0.99 and Vermeer by 0.34, and the reason is a limit of the method and not of those pictures. The curve has structure of its own at every scale, and that structure puts a floor under the exponent any reading along it can report, and a picture smoother than the curve cannot be measured through it. Read as stored the same pictures rank against the plane at rho 0.464, and read along the interleaved index at 0.679, the wrong statistic for a scaling law and recorded only because it was computed first.

The ordering is stable under a sixteenfold coarsening, at rho 0.893. It is not a reading of the sampling. The ladder was built to decide exactly that, since a quantity measured at one resolution is measured at whatever resolution the file happens to carry and the finest scale is where an artifact would sit. The magnitudes are not stable and should not be quoted: reading 600000 bytes of a file instead of 120000 moved one painting from 20.6 to 122.8. The order survives, the numbers do not, and every earlier figure in this entry was taken at the shorter read.

Print the content and not only the statistic. Proofed as an audit, upheld, and it demonstrated itself on its own output. Nine problems were found here by reading and none by a statistic leaving a range: six formats read as language and three corpora holding something other than their label. Written as a check and run over every corpus on disk (`evidence/proofs/posits/proof_corpus_audit.py`), it flags 65 of 185, catching line wrapping in eight Gutenberg texts, HTML markup and base64 blobs across many Common Corpus entries that had never been examined, and two synthetic corpora whose names claim a language they do not hold.

The audit also found what looked like a serious problem and was not. Three corpora matched a marker for publisher boilerplate, including one Finnish text that would then carry English, and a fourth matched HTML. Counting the occurrences shows one to three instances of a phrase in files of 459 KB to 1.26 MB, which is a stray mention inside the body and not a block of license text. Nothing measured is affected.

Two things follow. The check flags on presence where it should flag on volume, which is a defect in it and fixable with a threshold. And the alarm it raised was itself a statistic that reading the content corrected. The posit applied to its own output.

The quantities here are heavy tailed by default, and a mean and a deviation need the distribution checked first. Proofed on the core measure, upheld in practice and wrong in its description. It came from two derived figures, a loss ratio whose Jarque-Bera reached 800 against a one percent point of 9.21 and a product rule error where one draw in twenty-five carried 96% of a mean. Tested on the per symbol ratios those are built from (`data/dump_ratios.py`, outside the repository, and `src/engine/r/hypotheses/ratio_normality.R`), Shapiro-Wilk rejects normality at one percent for five of seven corpora. The advice holds and rests on seven cases.

The description does not. Skew runs in both directions, from +2.98 on prime gaps to −0.37 on C source, and the excess kurtosis is modest everywhere except the primes at +15.6. A logarithm

does not repair it and often makes it worse, reaching 5×10^{-12} on Greek. These are not log normal either. They are non normal in ways that differ by corpus.

The shape of that distribution is itself a second detector. The memoryless corpus passes, at $p = 0.891$, and every structured corpus fails, at p below 0.007. Prime gaps fail hardest at 2.2×10^{-8}, which agrees with their tail reading of 0.93. The two measures order the same corpora the same way. This fell out of a test run for another purpose and is not otherwise measured here.

A negative control cannot show that an instrument works. Proofed, and it passes for one of the two instruments here. Every control other than the protein structures is a memoryless process, which shows only that an instrument does not invent structure. A crystal supplies the other kind: the Crystallography Open Database publishes a cell edge for every entry. The answer is a number someone else measured.

Tiling published cells and asking the shift detector for the period (`examples/crystallography/6_oracle/proof_positive_control.py`, catalog `CRY-6-001`) returns the published edge on 453 of 453 axes over 151 structures, as an equality between integers. Nothing was told to the detector.

The quantization question is real on a voxel grid, and exact sites remove it. The reading above runs on a voxel grid of 0.25 angstroms, which puts none of those cells on a whole number: 10.1000 is 40.40 voxels, 8.6633 is 34.65 and 14.0574 is 56.23. Successive tiles then land alternately on the two lags straddling the true period, and the share of agreement at the upper one carries the fraction between them, giving 0.407 against a true 0.400, 0.618 against 0.653 and 0.227 against 0.230. The edges recovered that way are 10.1018, 8.6545 and 14.0568 angstroms.

That recovery works and still answers a question the reader created. The voxel came from this tree and not from the deposit, and removing it removes the fraction along with the error: sites are carried as exact integers at 10^{-1024} angstroms and a period is read off the whole difference set, with neither sweep, ceiling nor lag to straddle. Of the three cells named above, the fluorite entry at 8.6633 angstroms is in the present cache and returns 8.6633. The halite and pyrite entries quoted are not, since the sweep fetches by mineral name and drew different deposits; every fluorite, halite and pyrite entry it did draw is among the 453 read exactly.

The errors are 0.0018, 0.0088 and 0.0006 angstroms against a grid of 0.25, which is between fourteen and four hundred times finer than the slice. Sub voxel precision discarded at every point survives in the proportion between neighboring lags. No single measurement holds it and the relation between two of them does.

The scope is narrow and matters. That is the shift agreement detector, the one that also returned an image width of 960 and a Vigenère key length of 8, and it now has external ground truth three times over. It is not the null permutation identity. That measure failed on the protein structures and it carries most of the findings recorded here. That measure has still only been shown not to invent structure on memoryless input, and no positive control with published ground truth exists for it.

A quantity that fails to conserve under a change that should not move it indicates the instrument and not the domain. Proofed as a battery, and it earns its keep. It came from one case, where the same measure over a quarter, a half and the whole of a corpus gave 1.41, 1.25 and 19685 through a mean over a heavy tail. Run as a set of transformations with stated expectations (`evidence/proofs/posits/proof_conservation.py`), it produced the null's standard deviation, which the whole of this work had been compared against and never measured, and it found one defect and one wrong expectation.

The null's standard deviation. Reseeding the null twelve times on one corpus gives a mean of 0.7186 with a standard deviation of 0.0074, about one percent of the value. Every separation recorded here is far above it: a human corpus at 0.73 against a memoryless one at 1.00 is 36 standard deviations, and prime gaps at 0.93 are 9.

The defect. Relabelling the symbols cannot change the gaps between occurrences and has to conserve exactly, and it moves by 0.9 standard deviations. The ranking sorts symbols by count. Symbols sharing a count are ordered by their label, and relabelling changes which of them falls in the rare half. The defect is small and it is real.

The wrong expectation. Doubling a corpus was expected to conserve and moves by 9.9 standard deviations, to 0.6453. Concatenating a corpus with itself puts an exact copy of every symbol exactly one length away, which is periodic structure the operation introduced, and the measure is right to report it. Reading backwards moves 1.6 standard deviations for a similar reason: shuffling a reversed corpus with one seed is not the reverse of shuffling the original. The null is effectively reseeded. Truncation to either half moves 2.0 and 4.5 standard deviations, the two halves of a novel genuinely differing. Block shuffling and five percent noise move 4.9 and 34.2 standard deviations, as they should.

The symbol width has to match the scale of the structure being looked for. Proofed and refined, and it corrects two earlier diagnoses. A domain was built whose only arrangement is clustering of units four bytes wide (`evidence/proofs/posits/proof_symbol_width.py`). Swept across slices, widths of 1, 2 and 4 return 0.117, 0.119 and 0.117, and widths of 3, 5 and 6 return 0.450, 0.510 and 0.459.

The structure is visible at every width dividing the true scale and lost at every width coprime to it. A repeated unit of four bytes contains repeated subsequences of one and two bytes at fixed positions inside it, and a divisor keeps the signal. Phase matters once the width exceeds one: at the true width of four, the aligned slice gives 0.117 and the worst offset gives 0.309. Misalignment costs more than a wrong width. A width of one divides every scale and has no phase, and that makes the finest slice the one that cannot hide structure.

That refutes the reading given to two earlier failures. The Greek text and the vocalizations were described here as read too finely, and a finer slice cannot lose an arrangement. What those cases failed at was selection: the Greek detector ranked candidate boundaries and a code unit won, and the 8 kHz vocalization reading carried waveform smoothness alongside the call structure. Both are choices about what the slice lets one compare, not about whether the structure is present.

Derivative work should sit closer to the center than original work does, and that is a different claim from the human signature this document refuted. Posited and untested. Every earlier attempt here to read a human signature measured departure from a flat null, and departure from flat fires on prime gaps, on whale song and on a wolf howl. That reading is refuted above and stays refuted. This posit does not use it.

It uses the center of mass argument this document already makes. The center is the constant a domain varies about, it carries no arrangement information, and a mechanism that moves toward it has nothing left to read. A model fit to a corpus and asked for the next symbol is doing exactly that: it is built to sit where the corpus is densest. Its output is therefore predicted to fall nearer the center of its own source distribution than a person writing on the same subject would, and the quantity that reads that distance is already built.

The sculptor band posit above is the same statement from the other end. Zero deviation from the constant is a copy and carries nothing because it is the constant. The readable band lies between the copy and the noise, and a derivation is predicted to land below it.

If it holds, what follows is a detector for derivative work, and it would not need to tell a person from a machine. It would need to tell what was fit to a distribution from what was not.

Why the nearest measurement here does not settle it either way. Ten chains built to imitate a language put eight of ten nearest another generated corpus, and the mean distance among the languages is 0.06317 against 0.09780 among the generated and 0.08574 between the two kinds. That entry concludes calling it a human signature is more than has been shown, and it still is. But those chains scatter: 0.09780 among themselves is wider than the languages at 0.06317, the opposite of collapsing toward a center. A chain samples a distribution crudely and lands anywhere in it; a trained model is fit to it and lands at its mode. The two fail in opposite directions. The chain result is not evidence about the model case and cannot be quoted as either support or refutation.

The posit says nothing about text in particular. Nothing in the construction mentions a medium, and the prediction is about fitting a distribution and sampling from it, as a generative model does in any medium.

The reason a medium never enters is that every one of them arrives here the same way, as points carrying values. Section 4.2 covers exactly that, and the invariant holds over thirteen geometries and dimensions one to eight without an alphabet. A text, a painting, a recording and a mineral are one object to that. What a medium does not fix is the unit: the points can be taken at any scale, and until a scale is chosen there is no reading to have. That choice is a declaration, and this document has measured what it costs twice. The symbol width proof built a domain whose only structure sits at one scale and found it visible at every width dividing that scale and lost at every width coprime to it, 0.117 against 0.450. The picture failure was the same defect from the other side: assigning each domain a dimension meant choosing a width and then measuring the choice, and the reader returned heights instead of widths until the assignment was removed.

The unit is not free to choose either, because a medium supplies its own. Its frequency content, the orientations it holds and the range its amplitudes take are properties of the thing and not of the reader, and a representation inherits them by construction. Setting a boundary the medium did not set destroys what sat across it, and this document measured that ten separate times in one sitting: a dimension assigned per domain returned heights instead of widths, a sum stopped at 24 bits was still climbing at 64, a band fixed at eight levels read a picture spread over 160 of them as having no structure, and a spread quoted in levels put Chinese 565.9 deviations outside every alphabetic language where quoting it as a share of the range brought it to 3.2. The entry covering those ends by saying what works is sweeping the quantity and letting the data say where it stops mattering.

In a new medium, therefore, reach for the reading that carries no scale inside it. The spectral exponent is the first quantity here that held still, because it is how a quantity changes with scale instead of its value at a scale somebody picked, and it moves by 0.03 to 0.07 across a change of two and a half times in area.

A reading is undetermined until its partition is fixed, in any dimension and in any state, and a partition is fixed in one of three ways. The quantifier is Proposition 2's and it is here for Proposition 2's reason: nothing in the construction names a dimension. Nothing about it can turn on one. The three ways have settled names. A partition is **stipulated**, chosen in advance

and held. It is **estimated**, by sweeping the coarse-graining and taking the value where it stops moving. Or it is **supervised**, supplied as ground truth by something outside the sample.

They are not interchangeable, and they differ in what each one contributes. A stipulation is a modeling choice: it adds no information and fixes a frame, as `paper_config.py` does when it declares a mark set per paper. An estimate extracts what the sample already carries and can reach nothing past it, as the window ladder and the symbol width sweep do. Ground truth alone adds information the sample did not contain, and the two strongest results in this document are the supervised ones for that reason: the cell edges came from the Crystallography Open Database and the dialect labels came from Mellesmoen and Kye, and neither was derivable from the corpus each was measured against.

Only ground truth settles a collapse, and this document proves that the hard way. `repairs.py` states that a collapse destroys a distinction instead of disguising it. Neither a stipulation nor a sweep recovers page $\mathtt{k}^{\mathtt{w}}$ from an extracted `w`: the information is not in the sample to estimate. A person reading the rendered page supplies it. Four papers here are stuck at that point and the answer to each is supervision, the price those four carry.

Where this already exists, and Kolmogorov's paper has now been read. The dependence of an entropy on the partition it is taken over is the reason Kolmogorov-Sinai entropy is defined as a supremum over partitions instead of a value at one, which puts the observation above at 1958 and not here.

Kolmogorov, A new metric invariant of transitive dynamical systems and automorphisms of Lebesgue spaces, Doklady Akademii Nauk SSSR 119(5):861-864, 1958. Retrieved from Math-Net.Ru, the Academy's own open archive, and read in the Russian original at pages 861 and 862. The supremum is not a later convenience of the field: it is in the definition itself. Equation (3) on page 862 defines the conditional quantity of information as a supremum, and the line directly beneath it states what the supremum runs over: it is taken over all finite decompositions of the space into disjoint measurable pieces, written there as all finite decompositions $M = A_1 \cup A_2 \cup \cdots \cup A_n$ with the pieces disjoint and drawn from the subalgebra. Conditional entropy is then defined from that same quantity as $H(\mathfrak{A}|\mathfrak{C}) = I(\mathfrak{A}, \mathfrak{A}|\mathfrak{C})$. It inherits the supremum. Section 1 opens by constructing the partition $\xi_{\mathfrak{C}}$ a subalgebra generates, and the invariant h is defined in Section 2 on the back of it.

So the entry above is 1958 and it is not close. Kolmogorov did not observe that an entropy depends on its partition and leave it there; he removed the dependence by quantifying over every partition, the stronger move and the one this document did not make.

Sinai, On the concept of entropy of a dynamical system, Doklady Akademii Nauk SSSR, 1959, 768-771. MR 21 #2036a. Not retrieved. What is known about it here comes from Rokhlin's survey, which is read, and the entry says so. It does not imply the paper was opened.

The volume number is unsettled and is left out above. Rokhlin's bibliography prints 121; the common citation gives 124. The scan mangles digits elsewhere in the same reference list. Neither number is worth asserting from it. MR 21 #2036a identifies the paper without one.

Rokhlin, Lectures on the entropy theory of measure-preserving transformations, Russian Mathematical Surveys 22(5):1-52, 1967. Retrieved from Math-Net.Ru and read. It is a survey and it is used here as one: it reports what Kolmogorov and Sinai did, and it is not either of their papers.

Two things it settles. Its Theorem 6.6 states the supremum in English, that "the entropy of a measurable partition is the least upper bound of the entropies of finite measurable coarser partitions," the same construction Kolmogorov's equation (3) carries, and it confirms the reading of it above from a second source. And it says what Sinai's 1959 paper contributed: Kolmogorov's original invariant was defined through a generator, which left open which automorphisms have one, and "Sinai [30] proposed to change Kolmogorov's invariant to the now generally used entropy $h(T)$."

So the pairing this document has been citing as Kolmogorov-Sinai is two separate moves, and the entry above had them as one. Kolmogorov quantified over partitions. Sinai replaced the generator-bound invariant with the form the field now uses.

So the posit carries to any medium on the same terms it carries to text, and in each one the unit is read off the medium and not imposed on it. This document already holds the authored side in several of them: the three paintings at an agreement of 0.02 to 0.30 and seven at spectral exponents of 1.94 to 3.41 against the value a natural scene gives, two human recordings at 0.56 and 0.72 on the envelope, a board layout at 0.389 metric and 0.414 imperial where physics prefers neither, a VHDL community holding a case convention at 0.495 that its own compiler cannot read, and 0.187 bits of C whitespace the language ignores outright. Each of those is a measured human arm with no generated counterpart beside it, and adding one completes the experiment in that medium.

The paintings are the readiest, because the exponent already has a positive control with a known answer: fields built to a chosen exponent return it to within 0.05 in two, three and four dimensions. A generated image measured the same way lands somewhere against seven paintings whose values are already on the page.

The confound that would produce a false result, also from this document. Distance from a reference is not distance from a generator. This work measures the collection as the largest of the three terms it separated: the same language from another collection sits 0.1554 away where a different language sits 0.1274. Where something came from moves the reading more than what it is. Work by another person, from another source, would read as far from the reference as a derivation might, and a detector built without controlling that would report the provenance of its reference and call it a finding.

What would settle it, and the design is already in this document. Zulu against Xhosa leaves nothing to hide behind, because both are Nguni, both in the Latin alphabet, and both from the same translated work. Content, register and translators are fixed together, and only the language varies. This question takes the same shape with one term swapped: one subject, one register, one medium, made once by a person and once by a model. That the only thing varying between the two arms is the generator. Anything short of that measures a collection.

Nothing here has run that. No figure in this document bears on generated work, and an accuracy for it would be a number nothing measured.

A doped lattice is the case where every condition this construction wants is already supplied, and the row that found nothing is the argument for it. Posited and untested. The four constructed clouds include a perfect lattice carrying random values, and it fired on neither null, at 0.690 live against 0.684 and 0.553. The reading recorded there is that a survivor needs a displacement and a value together and random values supply no repeated pairs. That is a fact

about the instrument and not about the cloud: the geometry in that arm is perfect and the measure could not reach it, because the conjunction it tests carries a term that holds nothing.

Doping is that arm with the terms separated and both of them wanted. The host lattice is the geometry, which the shift agreement detector reads and has read three times against published cell edges. Which site holds which species is the composition, which the value null on fixed positions reads. The protein row is the argument for the second one: the value null returned 1.066 there because a protein is constrained in where its points are and not in which residue occupies each one, and doping is the exact inversion of that.

The reading inverts too, and anyone running it needs to be told so first. Correct substitutional doping is a random occupancy of host sites. Agreement with the null is the pass and departure from it is the defect. Clustering, precipitation and segregation are what a departure looks like, and they are what degrades carrier mobility. Everywhere else in this document a return of 1.00 means nothing was found. Here it means the material is within spec.

Why the candidate set is closed, which is where Proposition 2 stops binding. Substitutional sites are host lattice sites and interstitials are the tetrahedral and octahedral sites, both periodic and enumerable from the host structure, and chemistry bounds which species can sit where by radius, valence and coordination. A contaminant is therefore a labeling of a known finite site list and not a point in a continuum. That removes what defeated the protein arm outright: coordinates were voxelized at two angstroms and required to agree exactly, and two occurrences of one motif never land on identical offsets. On a lattice exact repeats exist without being built in, and nowhere else here do they.

The reference is not a shuffle, and chemistry already holds what it is. Dopants interact with the host and with each other. A correctly doped material therefore carries short range order, and a plain random labeling null would report that order as clustering. The reference wanted is the maximum entropy configuration under the known interaction energies, which is this document's own construction applied where the constraints are tabulated instead of estimated. The reference is the chemistry case again: a domain that states its own rules hands over the constant.

What argues against it. The field has its own version and it is not new here. Atom probe tomography cluster analysis uses random labeling tests, which shuffle species among measured positions, alongside Ripley's K and nearest neighbor distributions. How far that practice already uses an interaction aware null instead of a plain shuffle is not established in this document and should be read before any claim of improvement is made.

The order of work, which carries this past being an application. Published atom probe datasets carry their process conditions with them: implant dose, anneal temperature, the distribution the process was meant to produce. That is a corpus in the sense this work already uses the word, an outside body with the answer attached, standing to a doped lattice as the Salishan hand extractions stand to a paper. So the first run is not a prediction. It is a validation against material whose doping is already known, and only material that clears it earns a prediction on material that is not.

That order matters more here than anywhere else in this document, because of what the null permutation identity lacks. The two instruments entry records that the shift agreement detector has external ground truth three times over from published cell edges, while the null permutation

identity carries most of the findings and has only been shown not to invent structure on memoryless input. A published doping dataset would be its first positive control with an answer from outside. That is the largest single gap in this work, and this is a route to closing it.

What it costs, and the arithmetic is free. The method needs atomic positions carrying species labels, which means atom probe tomography or aberration corrected microscopy. That is expensive, partly destructive, and samples a small volume. Every figure above comes free once the point cloud exists, and obtaining the point cloud accounts for the entire expense.

An absence is a reading, which corrects a caveat written here first. A vacancy is a known site holding nothing, and an antisite is a known site holding the wrong species. Both are questions about occupancy over an enumerated list, the case this posit is strongest on. An earlier draft of the paragraph below grouped them with the defects that defeat it, and that was backwards: the regularity of the lattice renders a hole visible, because a hole is only a hole against a structure that says what should have been there.

The same move is already load bearing elsewhere in this document under another name. Direction two of the oracle check asks which tokens a paper prints that no row holds, and finds omissions by knowing what should be present. Absence detection is that check applied to a lattice.

Where the claim stops. Dislocation cores, grain boundaries and strain fields put atoms off the ideal lattice and into a continuum, and those are frequently the defects that matter most. The closed candidate set holds for occupancy of an enumerated site list, vacancies and antisites included, and fails for the structures that leave the list.

1.1.2 *Open Problems*

Open Design Questions

A spatially correlated domain of more than one dimension. Needed to test a centroid based anchor rule, since on an independent domain every point of a pattern is interchangeable and such a rule correctly shows nothing. That domain is the n-dimensional analog of the separation result on a line. Not built.

The mechanism of the product rule failure, since the obvious candidate is refuted. Selecting by rarity chooses the symbols the rare half of the distribution shows gathering into passages, which predicts correlated anchors and a product that under-counts. Measured, the error is 20992 times on English prose and 1494 times on C source, while the clustering of the anchors chosen runs the other way, 0.899 against 0.631. More clustering, less error. The scale is why: a needle of 24 symbols sits below the span of 32 where prose carries no structure at all. The gathering cannot be operating inside the window the cascade uses. Short range recurrence of the needle's own sequences is the remaining candidate and is not measured. A sweep of needle length from 8 past 128 would say whether the two come into agreement where their spans finally overlap.

Which anchor rule to prefer, as a curve and not two points. Selecting the rarest symbols in the needle gives a sizing error of 20992 times and passes 15.0 alignments. Selecting maximally separated offsets, which reads nothing of the needle, gives 2.02 times and passes 345.8. Neither dominates: rarity is the better filter with unusable sizing and separation is the honest sizer with a filter

23 times weaker. Which one is right depends on a term not yet in this work, whether survivors are verified and discarded or buffered and handed on. Under-provisioning is a performance question in the first case and a correctness question in the second, because the survivors have to be stored somewhere. Measured by `survivor_regimes.py` on main (ANY-5-002), the discard regime favors rarity, which leaves the fewest survivors to verify, and the buffered-and-handed-on regime favors separation, because a rarity buffer that drops no survivor must hold the worst-case survival ratio. This workbook measures that ratio at 490336 on English prose and 175369 on C source, against 22.6 and 1638 for the spread rule. Under-provisioning in that regime drops a real occurrence, the correctness failure the term names.

Whether the memoryless control is enough of a null for the heterogeneity result. The word burstiness bands rest on one memoryless corpus at 1.010. The Zipf claim was withdrawn because it was trusted on two arms, and the ten arm sweep exists already.

Whether subject heterogeneity predicts anything about anchor quality. The measure orders corpora by how far their vocabulary spreads across subjects and nothing has asked whether that ordering predicts the product rule error, the achievable advance, or the safe stride.

Open Paper Questions

Baeza-Yates and Régnier, average running time of Boyer-Moore-Horspool, Theoretical Computer Science 92(1):19-31, 1992. The one work most likely to contain the distributional results in Sections 4.3.1, 4.4 and 4.4.1. Three attempts across three paths have returned 403, the last of them against the publisher's own tokened link. This is a paywall, and retrying it will not open it. Until it is read, the distributional side of this work is unchecked. Navarro's survey, which was blocked the same way and is now read, cites it. What it contains can be approached from that side without claiming to have read the paper itself.

The filtration literature in approximate string matching, which was the closer prior art and is now read. Navarro's survey was retrieved on the fifth attempt from a course mirror. Four earlier attempts across two paths returned 403. What remains open here is narrower than the entry that stood here before: the survey covers filters that are necessary conditions on edit distance over strings, and whether the same construction stated without an order, a dimension or an alphabet is also old is a question it does not answer.

The approximate membership literature, for a contract this work shares and did not invent. A filter that never refuses a true occurrence, admits false ones and leaves confirmation to the caller is the Bloom filter contract and the quotient filter contract, and both predate this by decades. What differs is where the guarantee comes from: those construct it over a hash of a set held in advance, and Proposition 1 derives it from a subset of a necessary condition being a necessary condition, with no hash, no codebook and nothing built from the pattern. That is a positioning claim and not a novelty claim, and it goes in Section 7 only after both are read in full. Two claims of novelty in this work were already withdrawn after the source was read.

A structured system that no person produced. Every corpus scoring away from the null permutation was written by someone, prose and C source alike. The measure detecting structure and the measure detecting human production are not distinguished by anything measured. A genome or a protein sequence separates them, and Rao and colleagues use exactly those.

Lemmatized text. Two results are blocked on it. Exact matching cannot see an inflected formula. Epic register returned nothing across three languages. And no cross-linguistic test of a semantic universal is possible without both a translation and a lemmatizer.

A corpus series matched for form and densely sampled. Three drift designs each lost to a different confound. The design that would settle it holds one register, a court chronicle or a legal series or a newspaper run, sampled at close intervals across a known disruption and a known quiet stretch. The genre term stops taking the same value as the disruption term.

The site reports behind Section 4.13.08. Göbekli Tepe, the composite burials, the Clovis terminus and the community ending deposits are discussed from recall and cited nowhere. The section makes no archaeological claim as a result, the correct state until they are read.

Separating the count of subjects from the distance between them. Two joined English works score 0.685 and a corpus of 37 plays scores 0.667, since a novel and an epic poem stand further apart than two plays of one period. The measure responds to both and neither is isolated.

Section 7.2 and Section 7.3 are in the wrong order in the paper. Reordering means moving a block that has not been read in full, and renumbering would break this workbook's citation to Section 7.2, which is currently correct. Left as it is and recorded here.

Reviewed Literature

Vishkin's dueling paradigm and deterministic sampling, from Neuburger's survey. Order free elimination, alphabet independence, and periodicity as a stated precondition with a published repair are all there, and this work arrived at them independently. Recorded in Section 7.1 of the paper.

What that citation should not be stretched to cover. The deterministic sample is computed from the pattern before any text is read, by stacking it and analyzing its own self overlap. That preprocessing pass buys the worst case guarantee at $\log m$ probes, and it is the step an adaptive search declines. The two are different algorithms with different preconditions: a fixed probe set needs the per pattern analysis, and Section 4.9.6 measures that a stride calibrated on one pattern is wrong on up to 62 of 63 others. Skipping the analysis while keeping a fixed probe set is the arrangement that fails. An adaptive search has no guarantee at all, and has instead that the error is one directional. Accuracy is monotone in effort with no preprocessing and no bound.

Jaynes, Information Theory and Statistical Mechanics, Physical Review 106(4):620-630, May 15 1957. Retrieved and read to page 622, covering the abstract, the introduction and the opening of Section 2. The background this work uses is his principle, and his own wording is that the maximum-entropy estimate "is the least biased estimate possible on the given information; i.e., it is maximally noncommittal with regard to missing information." The shuffle entry above reaches that construction from the null permutation and cited nobody for it, the arrangement that already cost this document once at Montemurro and Zanette.

What the reading corrected. Jaynes credits Gibbs with the mathematical facts about maximizing entropy, writing on page 621 that they "were pointed out long ago by Gibbs," and credits Shannon with what was missing, which Appendix A sketches on page 622. That is uniqueness of H as a measure of uncertainty, and the entry above asserts something else, the uniqueness of the constrained maximizer, from strict concavity and from no source. The two claims are easy to run together.

Where the assertion actually sits. Page 623 does assert it: the maximum-entropy distribution "may be asserted for the positive reason that it is uniquely determined as the one which is maximally noncommittal with regard to missing information." So Jaynes states the uniqueness of the constrained maximizer one page past where the entry above stops reading, and states it without proving it there. The entry above is therefore in the same position he was: the claim is his as well as ours, and neither of us proved it in the place we asserted it.

How these quotations were checked. Both Jaynes papers on disk are page scans and yield one character of extractable text between them. A text search over them therefore reports every quotation as absent from a paper that contains it. The pages were rendered at 150 dpi and read. Every quotation attributed to Jaynes in this section was confirmed that way, against page 620 for the abstract, 621 for Gibbs and Shannon, 622 for Appendix A, and 623 for the uniqueness sentence. An automated check over these two files can only ever report that it read nothing, and a check that reads nothing is never a pass.

Jaynes, Information Theory and Statistical Mechanics II, Physical Review 108(2):171-190, October 15 1957. Retrieved and read at page 171. It restates the principle from the first paper. Take everything known as the constraint, and the distribution with maximum entropy under it is the most unbiased representation of that knowledge, "the broadest one compatible with the given information."

It adds a sentence bearing directly on the three ways a partition is fixed, above. Jaynes writes that information as to the state of a system can be lost down any number of routes and "the only way in which it can be gained is by carrying out further measurements." The entry above reaches the same division from this document's own rows, where a stipulation adds nothing and an estimate reaches only what the sample holds. Jaynes states it as the content of the second law in 1957, and the entry above now cites him for it instead of arguing it from the crystal and dialect rows alone.

Navarro, A Guided Tour to Approximate String Matching, ACM Computing Surveys 33(1):31-88, 2001. Retrieved and read at Section 8, the filtering literature this document calls its closest prior art. It is closer than that. Navarro defines a filter as an algorithm that discards text areas that cannot contain a match, states that one is "normally unable to discover the matching text positions by itself," and writes of his own worked example that "it is necessary that either sur or vey appear in an approximate occurrence, but it is not sufficient." He then states the requirement outright: "Any filtering algorithm must be coupled with a process that verifies all those text positions that could not be discarded by the filter."

That is Proposition 1 and Proposition 2 in one paragraph, published in 2001. The field also has a name for the quantity this document calls the candidate rate, which it calls filtration efficiency.

The lemma is older than the survey and the survey says so. Section 8.2 attributes the partitioning filter to Rivest 1976: cut a pattern into k+1 pieces and at least one must appear unchanged in an approximate occurrence, because k errors cannot alter k+1 pieces. Fifty years.

What the reading cost, and what came through it. No claim of novelty for the construction survives this. A necessary condition used to discard candidates, with a mandatory verification pass and no missed occurrence, is the filtering paradigm and it has a literature. What is not in this survey is the generality: every filter in it is a necessary condition on edit distance over strings, where the pieces are substrings, the errors are edit operations and the index set is a line. Proposition 1 as stated here carries neither order, dimension nor alphabet. It reaches a point cloud in

eight dimensions that a k+1-pieces lemma has nothing to say about. Whether that abstraction is also old is a question this survey does not answer and the approximate membership entry below still opens.

The wider version of the same correction. Each piece of this work is standard in the field that owns it. Filtering is standard in string matching and Navarro reports it dominating a large range of parameters. The null permutation is Fisher's, from 1935, and is standard in statistics. The maximum entropy reference is Jaynes's, from 1957, and is standard in physics. Three fields have each held one of these for decades. What none of them has published is one instrument, unchanged, run across a crystal, a dialect border, an image, a board layout and a corpus with the numbers in one table. The span table in this document is that. What stands is the bridge, and the span is not the claim.

Parrondo, Horowitz and Sagawa, Thermodynamics of information. The bound on work from measurement is mutual information, $W \geq -kTI(X;M)$, and the maximum extractable work from a non-equilibrium state is its free energy above equilibrium. The equilibrium reading in Sections 4.3 and 4.8 survives. The reference measure reading does not, because Section 4.3.1 shows the governing quantity is not a relative entropy. Recorded in Section 7.2.

The Sproat and Rao exchange on entropy and the Indus script, all four items in full. Sproat in Computational Linguistics 36(3):585, the replies from Rao and colleagues and from Lee and colleagues in 36(4), and Sproat's answer to both. The position argued is that these statistics do not separate a language from anything else, demonstrated by applying a published classifier to Mesopotamian deity symbols and to 75 texts made from dice throws, both of which it reports as writing. Rao's answer is that sufficiency was never claimed and that the evidence is a posterior over several properties at once. Their own description of the Zipf-Mandelbrot property is that it is necessary and not sufficient for language, the asymmetry between the propositions here reached independently.

What that reading cost. Three of the four regularities in Section 4.13 were withdrawn after building the counterexample class it describes. A memoryless process with a delimiter, swept over alphabet size 8 to 64, delimiter rate 0.06 to 0.35 and uniform, geometric and English weighted distributions, returns a Zipf slope of -0.978 to -1.332 against a natural range of -0.723 to -1.185, a brevity correlation of -0.124 to -0.401 where the strongest natural text reaches -0.364, and a type count of 5824 to 22160 against 4171 to 11854. The Zipf slope, the brevity correlation and the type count carry no evidence about production. Only dispersion against a null permutation survived, at 0.99 to 1.01 for every memoryless arm against 1.79 to 4.82 for natural text, and it establishes only that a corpus is not memoryless. C source departs from that null further than English prose does.

Montemurro and Zanette on the entropy of word ordering, PLoS ONE 6(5):e19875, 2011. Read for its method, its corpora and its numbers. It shuffles words to build a null and reports the ordering quantity near 3.3 bits per word over 7077 texts in eight corpora and six families, with a relative variability of 0.07 against 0.23 for the entropy, and calls it a statistical linguistic universal. What it cost: the cross-language invariance recorded here is a rediscovery of that result, found after the measurement instead of before it, and the entry above now says so. The argument for their specific value of 3.3 bits was not read and nothing here tests it.

The Natural Semantic Metalanguage prime inventory, read as the published chart of 65. Wierzbicka and Goddard hold that each is undefinable and has an exponent in every language.

The claim is semantic and carries no statement about frequency, and the measurement above tests that and confirms it. The tradition it comes from is the Lwów-Warsaw school. Reducing meaning to indefinables and composing everything else is Tarski's and Leśniewski's program applied to natural language. Only the prime inventory and the universality claim were read; the argument that each is genuinely undefinable was not, and nothing here tests it.

The published corpus of cryptographic oracles. The digest is held to 1554 vectors from two bodies. That is two bodies and not the field.

Comparison against more published entropy estimators. Two were used. Neither end of the working range of the candidate count estimator is characterized.

Claim Boundaries

A genuinely unenumerable alphabet. Cannot be performed. The construction's tolerance of one is supported by the core never reading a value, and not by a measurement. The implemented route is the descent's equality oracle (`src/engine/nbody/orior/orior.h`, `orior.c`), and a growing alphabet is measured on it: from $\sigma = 2^8$ to 2^{24} its memory stays at 131057 bytes while a table indexed by symbol grows (`docs/findings-for-verification.md`).

Whether the invariant results are evidence for anything. They are evidence that the implementation agrees with Proposition 1 over the ranges tested. They are not evidence for the proposition, which needs none.

What the surviving statistic actually detects. Three of the four regularities in Section 4.13 were withdrawn after a memoryless process reproduced them: the Zipf slope at -0.988 from independent draws, the brevity correlation reaching -0.401 which is stronger than any natural text here, and the type count overlapping at three arms. Dispersion against a null permutation is the measure left standing, and the C sources score further from that null than English prose does. The statistic orders corpora by how much structure they carry and identifies no language. Whether it identifies a person is the open question above.

What the window sweep separates a period from. One family. Every aperiodic word it was tested against is Sturmian, a rotation by an irrational. A sequence aperiodic in some other way could hold its answer across widening windows and defeat the test, and nothing rules that out. Nine slopes is nine, under this document's own rule that an eleventh instance moves a quantifier no distance.

What the game series covers. Subtraction games. Their Grundy sequences are exactly periodic after a short preperiod, over a small alphabet, with no noise, and the workbook's own division says a result obtained inside one distribution stays there. The 383 rows establish that the detector is right where an answer exists and is checkable. They establish nothing about a corpus that is noisy, approximately periodic, or long.

Whether the rules-not-play finding is about flip rules. Reversi is one game and the mechanism given for the finding is its flip rule specifically. A game that constrains where a piece is placed, and flips nothing, might leave a trace a stronger player deepens, and none has been tried. The three arms are also one line of heuristic each, and the 0.067 correlation is measured over a strength range a competent program would sit entirely above.

A game is not a natural object. A person put its regularity there deliberately. That makes it a positive control in the same class as a published cell edge, with less excuse, since nobody had to

go and measure a subtraction game. It supports a single claim a memoryless control cannot: the detector finds structure where a person put it.

What the $1 - 1/e$ bound covers, which is less than it appears to. It bounds alignments rejected by k probes against the best k probes. It does not bound reads. Rejecting an alignment early saves the reads a later probe would have spent on it. Two probe sets covering identical alignments can cost different numbers of reads, and every read figure in `docs/steering.md` is a measurement outside the bound. It further assumes the marginal gains are scored exactly, which holds only at `sample_stride` of one; above one the planner scores on a sample, the oracle becomes approximate, and the ratio does not hold as stated. The bound has never been checked against an optimal probe set here, because obtaining one means enumerating every set of size k. It is a proved floor and is nowhere reported as an observation.

The engine is not asymptotically distinguished on a byte corpus, and the regime where it would be has not been measured. The general alphabet model, in which an algorithm may only ask whether two symbols are equal, is where the construction's three corollaries would pay: no order, no alphabet table, no hash. Main and Lorentz proved $\Omega(n \log n)$ for repetition detection in that model, which reads as a barrier the engine sidesteps, and it is not. The bound is $\Theta(n \log \sigma)$ in the alphabet size σ, and their $\Omega(n \log n)$ form assumes $\Omega(n)$ distinct symbols. At $\sigma = 256$ the factor is 8 and the bound is linear. On the byte corpora every figure in this workbook was taken over, the engine beats nothing asymptotically and no claim of that kind is made. The separation, if there is one, lives at large or unenumerable σ, where a bad character table is σ entries and nothing on the engine's oracle path is indexed by symbol, while the byte path's census is a 256 entry table (`src/engine/nbody/orior/orior.h`); the descent's output is one entry per alignment and grows with the field. The entry above records that case as unperformable, and a growing σ is measured from 2^8 to 2^{24}, short of that limit (`docs/findings-for-verification.md`). The $\Omega(m)$ space reading is withdrawn (`:294-310`).

1.1.3 *Unresolved Decisions*

Making `ancorae` instantiable. A `src` change, now authorized. The justification is weaker than it was: under a scalar read model the fixed table costs 25 to 30 times when mismatched, but the whole rare anchor arrangement loses to a shift anyway. The vectorized measurement above would settle whether to build it.

The praet and DMA integration. Set aside mid-flight. `PraetScheduleContext` was renamed to `PraetOrdoContext` by a rename script that skipped its own protective pair, and that has not been ruled on.

Audience for the paper. A publishable version needs the shift literature read. An internal design record does not.

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2

PARTICLE PHYSICS: THE ENGINE ON THE ATOM AND THE STANDARD MODEL

This chapter is a case study, and the full treatment is a research paper of its own at `theory/theory/particle_physics/`. It carries the engine's method to a domain it was not built for, atomic and particle physics, and the point lies in what the method kept the same: an object written as points carrying values, read by equality and by census magnitude, in exact arithmetic, against a background drawn from the data, and checked by an answer from outside. The runs are under `examples/particle_physics/`.

Note. All 11 of those runs import from `src/engine/python/`, and the modules they import (`representation`, `reference`, `measure`) are byte-identical to the ones every figure in this chapter was read from. Nothing here is rerun against them.

2.1 THE ATOM AS AN ACCUMULATION

An element is the electrons of its neutral atom, each an electron a point at an integer quantum state, the spin doubled to keep it integer. Two electrons at one state is a Pauli violation, and `exact.contested`, the same domain-blind primitive that finds two elements on one crystallographic site, finds it. Over all 118 elements the exact reading places Z electrons at Z distinct states and none is contested. Exclusion holds by a count and not by an assertion.

The differentiating electron of an element carries a group signature, its subshell type and its subshell's population with the principal number dropped, and that signature recurs down the table. The recurrence is the periodic law, and it is measured against two backgrounds: a shuffle that keeps the counts falls to the coincidence floor, and a filling without exclusion collapses to a ladder with no structure at all. The shell closures fall out of the signatures at Z equal to 2, 10, 18, 36, 54, 86 and 118, the noble gases, and the gaps between them are the row lengths 8, 8, 18, 18, 32, 32, read with no row told to the reader.

2.2 THE TRAJECTORY, EXACT

The Bohr orbit of a one-electron ion is an exact rational: a radius n^2/Z in Bohr radii and an energy $-Z^2/n^2$ in Rydbergs, with the quantum number an exact integer and nothing rounded. Within a spectral series the physical constant cancels, and a line ratio is a pure rational the model predicts with no measurement in it, the Balmer ratios reading 20/27, 125/189 and 5/8 against the first line.

2.3 THE PARTICLES

The confirmed Standard Model particles, six quarks, six leptons and five force carriers, are written by exact integer quantum numbers, charge in thirds and spin doubled, with no mass in the reading. Three things emerge. The three generations recur with identical quantum numbers, one structure seen three times as a periodic group was one signature seen down a column. Each generation's electric charge, with every quark counted in its three colors, sums to exactly zero, the anomaly-free condition and the reason an atom is neutral. And the free particles, the leptons and the bosons, carry a charge that divides by three while the six quarks do not, the exact statement of why a quark is never seen alone.

2.4 THE ORACLE, AND THE DISCIPLINE

The ideal Madelung filling is a model, and it is held against the measured ground states from the NIST Atomic Spectra Database. The oracle covers all 118 elements, 108 measured through hassium and 10 predicted above it and marked so, and reads off the 20 aufbau exceptions where a real atom fills against the order, chromium and copper and the lanthanides and actinides among them, with most elements agreeing as the positive control.

The whole subject steers by census magnitude and by equality, and never by a bound. The sift narrows the candidate periods with the rarest anchor first, a lag is a difference between two agreeing places or it is not, and no mass and no tolerance enters. A bound on a mass or a distance was the one decision the domain refused, and refusing it kept every reading exact.

3

WANTS

A want is a posit, stated in the terms of the field it reaches into, beside the open question between it and a status a run could give it. Wanting it does not mean the work has it. The parts of these posits that a bench can decide are stated as hypotheses in the thought experiment [32], and the rest stand here.

3.1 ABSOLUTE ZERO AS THE NOISE FLOOR

Posit. Absolute zero is the noise floor of physical measurement: the reference state every structure is read against. It is not an empty state.

- Classical thermodynamics treats $T = 0$ K as the end of motion. In quantum field theory a system cooled to $T = 0$ keeps its state space and sits in its ground state, where zero-point fluctuations persist.
- Thermal entropy $S_{\text{thermal}} = k_B \ln \Omega$ goes to zero for a non-degenerate ground state, as the third law requires. The state space Σ stays fully populated: at every coordinate $g \in G$ the field operator keeps a nonzero expectation value $x(g)$. The posit holds that the vacuum's entanglement entropy across a bipartition is maximal.
- By the Nernst unattainability principle $T = 0$ is reached only as a limit, after infinitely many steps. Absolute zero is therefore a reference state, in which thermal uncertainty is replaced entirely by irreducible zero-point noise. It is not an informational vacuum.
- The posit maps the anchored search onto the two regimes:

search	thermal region, $T > 0$	absolute zero, $T = 0$
field state $x(g)$	thermal fluctuations and quantum noise	ground-state vacuum fluctuations
alphabet Σ	thermal occupation modes	zero-point vacuum modes
indicator δ_0	comparisons blurred by thermal noise	exact matching of quantum states
pruning ∂A	dissipative, Landauer cost above zero	pruning that preserves entanglement

- The domain map $x : G \to \Sigma$ stays defined at every coordinate. The elements of Σ go from thermal mixed states to pure ground-state field amplitudes.

- The ground state is the search's maximum-entropy baseline: the field without structure, before an anchor set A is applied to read local structural correlations.
- Lowering the temperature removes thermal noise without removing the background data. The anchors Anc_A and the field's correlation functions are then resolved by the equality operator δ_0. Absolute zero is the fundamental noise floor: the maximum-entropy background every physical measurement is taken against.

What stands open. The vacuum's entanglement entropy across a region follows an area law, set by the region's boundary, and is not maximal [33]. Nothing in this repository measures a physical ground state, and the mapping onto the anchored search is an analogy until a measurement of the search depends on it.

3.2 UNIFICATION AND THE CONTINUUM

Posit. The universe has no initial singularity. It passes through a smooth, non-singular bounce from a previous universe, and relativity follows from the density of information in a continuous medium. Its parts, in order:

A. No initial singularity. Run backward, general relativity takes the universe to a single point, where the theory breaks down. The posit holds that the energy cost of compressing information forbids that point. A Planck-epoch compression of everything into one point cannot happen, and the universe instead passes through a global phase inversion.

- The cost of packing information rises with its density, through the quantum noise between the groupings being packed (here called informational viscosity). Packing all of the universe's information into one sub-Planckian coordinate would need the quantum foam packed perfectly, and the cost of that diverges.
- Before the cost diverges, the medium reaches a maximum information capacity (here called the cosmic breaker). The collapse stops at that bound and does not reach a singularity. The bound comes from the same physics as everything else, and no separate rule is needed to exclude the singular state.

B. The bounce. The universe passes smoothly between two macroscopic states, and the information entropy is inverted between them.

- The previous universe expanded until its information entropy reached its maximum $H_{\max}$. It dissolved into unstructured sub-Planckian noise: a field that is uniform apart from its noise.
- At complete uniformity there are no distinguishable coordinates, distances or particles, and scale has no meaning. Uniform randomness at cosmic scale cannot be told from uniform randomness below the Planck scale.
- The previous universe's complete dispersion is the initial state of ours (here called the phoenix effect). The maximum entropy of the old universe is the maximum potential of the new. Matter was not compressed. The coordinate system inverted.

C. The low-entropy beginning. The past hypothesis asks why our universe began in a highly ordered, low-entropy state [26]. Standard cosmology takes it as an initial condition. The posit holds that it follows necessarily from continuum mechanics: our universe began ordered because it inherited a fully reset, uniform sub-Planckian state from the previous one.

D. The cosmology in four statements.

1. The medium is a smooth continuum below the Planck scale.
2. Information is a physical, fluid concentration moving through noise.
3. Physical law is computed locally, at every point.
4. Time is emergent. It is the direction in which information entropy increases, the thermodynamic arrow, as the global inversion plays out.

E. Relativity from information density. Gravity and velocity are both measures of how densely information is packed locally, and that density sets the rate at which the field updates its state. The field's parallel computation is very large even at one update a second, and it runs everywhere at once.

- Mass is information congestion. A particle or a mass is a localized knot or eddy, a dense concentration of the field's data. A dense concentration slows the rate at which information dissipates, as a heavy drag does in a fluid. That rate is time, and gravitational time dilation is the field taking longer to update a dense state.
- Velocity is structural compression. A body moving through the medium near the speed of light is compressed against the oncoming field, which raises its local information density and slows the rate at which its entropy increases. An outside observer sees time dilation and length contraction. Where relativity says space stretches, the posit says the field's data is compressed.
- The speed of light c is the saturation threshold of the medium: the maximum rate at which the field passes information from one point to the next. Past that density or rate the field closes into a black hole. A black hole is the ceiling of local information density.
- The posit introduces no constant without a mechanism. Each constant is a property the medium already has, and a derivation finds it instead of setting it.

What stands open. The posit gives no equation, and a test needs one first: the Lorentz transformations and the field equations of general relativity derived from information density. The step at maximum entropy, where scale loses its meaning and the end of one universe becomes the start of the next, is the premise of conformal cyclic cosmology [31]. Gravity derived from information and entropy is the program of Jacobson [28] and Verlinde [35]. A maximum information density, reached by a black hole, is the Bekenstein bound [27]. A maximum speed at which a local medium carries information is the Lieb–Robinson bound [29]. The universe as a computation whose rate is set by its energy is Lloyd's [30]. The posit has to be set against each of them.

3.3 SUPPOSITIONS, BY THE HOUR

The notebook is kept in the order the thinking happened. Entries 1 and 10 (tail error), 4 to 6 (doping), 7 (proteins) and 17 (cultural material) are stated as hypotheses or as a constraint in the thought experiment. Entry 13's demon is answered there by the delta null: the boundary refuses and does not predict. The other entries stand here as wants. An approximate representation of an object, written appxrep, is a representation of it as vectors and magnitudes, held without a bound on its order.

1. Tail error correction is working. Adding a periodic perturbation drives the joint error on an undisturbed co-arm toward zero by several orders of magnitude.

2. This correction is another appxrep, and like the others it should be left unbounded.

3. Crystallography. A grain boundary and a grain size are instances of the boundaries every natural process leaves. Each has its own boundary curve, shaped by the physical processes that bound it and by the conditions while it formed, and any gradient of scarcity in those conditions records where it formed.

4. Crystallography. A dopant in a host lattice plays the part a salt plays in a hash, above all a known dopant: its distribution has already been measured.

5. Crystallography. It follows that the complement of the set of measured arrangements is determined as well, and the difference between the two is a real quantity.

6. Crystallography. Posit: from measured dopant distributions, new allowed dopant arrangements can be predicted.

7. Biology. The Ramachandran plot gives the allowed backbone conformations of a protein, and any structure known to exist can be held as an appxrep. It follows that a protein held as vectors and magnitudes is a natural representation and keeps the order of assembly. Posit: if the known allowed structures exist, the complement of the allowed set exists and is a subset of the superset. If the posit holds, it predicts chirality and new allowed structures.

8. Salishan. The hand extraction comes first. After it, write specific hypotheses that ask the corpus narrower questions and exclude more of the unbounded set: tailored amplification and tuned nulls. There is a natural complex curve, and the closer it sits to the imaginary axis, the more information can be extracted. The curve is a function, and it cannot be defined until the extraction is finished.

9. Conjecture. The universe is a set that exists and can be known, and every subset of it necessarily exists. Any set can be represented approximately, with its information conserved. An unbounded, undefined set of n dimensions can be used to ask questions of other unbounded, undefined sets of n dimensions, and such an object encodes everything about itself in itself. Bounding the object under examination adds information that is not in it and destroys its unbounded power to discriminate. Consummatio omnino pulchra est.

10. Tail error correction works best with a random perturbation that has no affine structure. Error reduction against percent inclusion has not been plotted. It is a recursive function of the tail, and the prediction is a smooth natural curve. A straight line would show a branch that could reach a tail error of 10^{-26}.

11. Posit. A noise error-correcting code learned in one domain carries its affine regularity to another through field transforms. Seen from the data, the transforms are mutations. Operationally the code is a dimensionless field transform, and it fails under bounding. Every transform has to keep that property, so that no dimension is lost. The data is mutated, which conserves its unbounded variability, while a transformation aims at a goal other than asking the data a question, and that goal introduces information that is not in the data. A mutation has no goal. It is random, and random noise cancels itself on inversion.

12. Self-detection needs no reflexive loop and no separate layer. It is an appxrep difference composed with a null permutation. Treating self-observation as a dimensional transform avoids the paradoxes of self-reference: Gödel sentences, infinite regress and the halting problem.

    a) The null permutation π_0. A state transition applies a permutation π to the state, $S_{t+1} = \pi(S_t)$. The null permutation is the identity on the active transform axes. Executing it does no structural work on the data, and the operator still registers the boundary condition. Self-detection is a transition whose net permutation is zero on every operational dimension.

    b) The appxrep difference as a projection. The ground set Σ can discriminate without bound. Projecting a state onto a lower-dimensional appxrep therefore leaves a difference between the state and its projection, $\Delta_{\text{self}} = S \setminus \text{appxrep}(S)$. The differencing is applied across dimensions instead of across clock ticks, and it needs no external monitor and no stack of nested interpreters. Differencing across time measures how the state evolves, $S_t \to S_{t+1}$. Differencing across dimensions measures the difference between the unagglomerated superset noise $S_{\max}$ and the compressed projection of the state held in memory.

    c) Self-detection reduces to $\text{appxrep}(\Delta) \circ \pi_0$. The machine therefore uses the same code paths, registers and set operations for observing itself as for its primary computation.

13. Laplace's demon is reached through a topological boundary and Jaynes's proposition [8]. The proposition is a boundary condition present in Σ, and it realizes the holographic principle: the information of a three-dimensional volume can be encoded on its two-dimensional boundary [34]. The universe is a set Σ, the measurement is Δ_{null}, and the engine is the topological boundary.

14. The algorithm is about five days old at this entry, and nearly all of that time has gone into reading Salishan. The Salishan corpus still has a long way to go, and no error may be introduced into it. The hand verification passes are multiplied: each added pass returns less, and it is still worth its cost.

15. Performance. Tail accuracy is now the largest target, and it has to reach its floor much faster. Once its mathematics is settled, the engine is to be tuned for superscalar hosts, with

the assembly breakdown kept for the teaching material that comes with the engine. It is to be written four ways: full assembly; pure C23 SWAR at machine word width; fractional, with a branch that follows a baseline; and wider buses where the hardware accelerates them. This waits for the hand extraction of Salishan.

16. Causality bounds the sampling rate, and today the hardware sets that bound. Between samples the information varies without bound, as a continuum. Posit: the lower bound on the sampling interval is the Planck time, which can be approached and never met unless latency and propagation time are removed from the universe. If they are always present and predictable, materials science can speed up their physical function, and the time between samples can be filled with a delta, so that time itself is not needed. If the engine works as designed, it records every transient to a tail error of 10^{-26}. State information is encoded in the object's entropy, and conservation of energy requires that it is.

17. A cultural object cannot be reconstructed unless the whole living object is encoded in the object itself. Repeating patterns, counterexamples from teaching, and fragments of the same protoculture surviving at the edges of collapsed civilizations regularly help fill cultural gaps. Where something was never recorded, determining its meaning takes much more care, and meaning is never injected where there is none. That widens the base set to include the unknown: more is included, which means more data for each complete set. Storing unbounded variability is not possible with conventional methods, since every variable costs cycles, memory or both. The algorithm is to use the whole of Salishan, from the whole input corpus, to recover faded text, damaged art, and the distribution and concentration of populations, combining archaeological work with known measures of distance between languages. Archaeologists and linguists already derive these. Anthropology may have known it longest, since it is the field's own thought exercise: if A moved from B, knowing from archaeological evidence what lay between, would the move be possible under any human circumstance? Anthropology considers aspects of human movement other fields leave alone. A subject that was never written down lives only as an idea, mother to daughter, father to son, community to family, society to community, and if all of those are gone it cannot be recovered. For that reason the tool is not to be used to regenerate anything without a human representative of that culture. The information is not thereby gone. Gone is the ability to tell a true statement, a member of its set, from an untrue one. Representing something untrue about a culture as true would be a grave ethical violation, and it would destroy the culture itself. This proof therefore only examines what is already there and asks whether it recreated it. It does not ask where the edges of what is known lie. The tools for that exist. Their author built them and does not use them for that.

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